

32-Bit

Microcontroller

TC33xEXT

32-Bit Single-Chip Microcontroller
AA-Step

32-Bit Single-Chip Microcontroller

Data Sheet

V 1.1, 2021-03

Microcontrollers

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1 Summary of Features

The TC33xEXT product family has the following features:

- High Performance Microcontroller with two CPU cores
- Two 32-bit super-scalar TriCore CPUs (TC1.6.2P), each having the following features:
 - Superior real-time performance
 - Strong bit handling
 - Fully integrated DSP capabilities
 - Multiply-accumulate unit able to sustain 2 MAC operations per cycle
 - Fully pipelined Floating point unit (FPU)
 - up to 300 MHz operation at full temperature range
 - up to 192 Kbyte Data Scratch-Pad RAM (DSRP)
 - up to 64 Kbyte Instruction Scratch-Pad RAM (PSPR)
 - up to 64 Kbyte Data RAM (DLMU)
 - 32 Kbyte Instruction Cache (ICACHE)
 - 16 Kbyte Data Cache (DCACHE)
- Lockstepped shadow core for one TC1.6.2P
- Multiple on-chip memories
 - All embedded NVM and SRAM are ECC protected
 - up to 2 Mbyte Program Flash Memory (PFLASH)
 - up to 128 Kbyte Data Flash Memory (DFLASH 0) usable for EEPROM emulation
 - BootROM (BROM)
- 64-Channel DMA Controller with safe data transfer
- Sophisticated interrupt system (ECC protected)
- High performance on-chip bus structure
 - 64-bit Cross Bar Interconnect (SRI) giving fast parallel access between bus masters, CPUs and memories
 - 32-bit System Peripheral Bus (SPB) for on-chip peripheral and functional units
 - SRI to SPB bus bridges (SFI Bridge)
- Optional Hardware Security Module (HSM) on some variants
- Safety Management Unit (SMU) handling safety monitor alarms
- Memory Test Unit with ECC, Memory Initialization and MBIST functions (MTU)
- Versatile On-chip Peripheral Units
 - 6 Asynchronous/Synchronous Serial Channels (ASCLIN) with hardware LIN support (V1.3, V2.0, V2.1 and J2602) up to 50 MBaud
 - 4 Queued SPI Interface Channels (QSPI) with master and slave capability up to 50 Mbit/s
 - 1 MCMCAN Modules with 4 CAN nodes for high efficiency data handling via FIFO buffering
 - 6 Single Edge Nibble Transmission (SENT) channels for connection to sensors
 - One Capture / Compare 6 module (Two kernels CCU60 and CCU61)
 - One General Purpose 12 Timer Unit (GPT120)
 - 1 IEEE802.3 Ethernet MAC with RMII and MII interfaces (ETH)
- Versatile Successive Approximation ADC (VADC)
 - Cluster of 6 independent ADC kernels

Summary of Features

- Input voltage range from 0 V to 5.5V (ADC supply)
- Digital programmable I/O ports
- On-chip debug support for OCDS Level 1 (CPUs, DMA, On Chip Buses)
- Multi-core debugging, real time tracing, and calibration
- Four/five wire JTAG (IEEE 1149.1) or DAP (Device Access Port) interface
- Power Management System and on-chip regulators
- Clock Generation Unit with System PLL and Peripheral PLL
- Embedded Voltage Regulator
- Qualified for automotive application according to AEC-Q100 (only applicable after delivery release of the corresponding sales codes)
- ISO 26262 Safety Element out of Context for safety requirements up to ASIL D (only applicable for sales codes listed within a released Safety Package Release Note from IFX)

Ordering Information

The ordering code for Infineon microcontrollers provides an exact reference to the required product. This ordering code identifies:

- The derivative itself, i.e. its function set, the temperature range, and the supply voltage
- The package and the type of delivery.

Table 1-1 Platform Feature Overview

Feature		TC33xEXT
CPUs	Type	TC1.6.2
	Cores / Checker Cores	2 / 1
	Max. Freq.	300 MHz
Cache per CPU	Program	32 KB
	Data	16 KB
SRAM per CPU	PSPR	32 KB for CPU 0/ 64 KB for CPU 1
	DSPR	192 KB for CPU0/ 96 KB for CPU1
	DLMU	8 KB for CPU0/ 64 KB for CPU1
Extension Memory	TCM	1 MB
	XCM	-
	XTM	16 KB
Program Flash	Size	2 MB
	Banks	1 x 2 MB
Data Flash	Size (single-ended)	128 KB (DF0) + 128 KB (DF1)
DMA	Channels	64
CONVCTRL	Modules	1
EVADC	Primary Groups/Channels	6 / 40
	Secondary Groups/Channels	0 / 0
	Fast Compare Channels	-
EDSADC	Channels	-
GTM	Clusters	0
	TIM (8 ch)	0
	TOM (16 ch)	0
	ATOM (8 ch)	0
	MCS (8 ch)	0
	CMU / ICM	0 / 0
	PSM	0
	TBU channels	0
	SPE	0
	CMP / MON	0 / 0
	BRC / DPLL	0 / 0
	CDTM modules	0
	DTM modules	0

Summary of Features
Table 1-1 Platform Feature Overview (cont'd)

Feature		TC33xEXT
Timer	GPT12	1
	CCU6	1 module with 2 kernels
STM	Modules	2
FlexRay	Modules	0
	Channels	0
CAN	Modules	1
	Nodes	1 x 4
	of which support TT-CAN	0
QSPI	Modules	4
	HSIC Channels	2
ASCLIN	Modules	6
I2C	Interfaces	0
SENT	Channels	6
PSI5	Modules	0
PSI5-S	Modules	0
HSSL	Channels	0
MSC	Channels	0
EBU	External Bus	0
SDMMC	eMMC/SD Interface	1
Ethernet (10/100Mbit/1Gbit)	Modules	1
FCE	Modules	1
Safety Support	SMU	yes
	IOM	no
SPU	Modules	1
RIF	Modules	1
HSPDM	Modules	1
Security	HSM+	1
Debug	OCDS	yes
	MCDS	no
	miniMCDS	no
	miniMCDS TRAM	no
	MCDS light	yes
	AGBT ¹⁾	yes
Low Power Features	Standby RAM	1 - DLMU0
	SCR	yes
Packages	Type	LFBGA-292 / LFBGA-180
I/O	Type	5 V CMOS / 3.3 V CMOS / LVDS
T _{ambient}	Range	-40 ... +125°C

Summary of Features

1) The Aurora Gigabit Trace Module (AGBT) is a trace interface intended for development use only (not to be used in series production). It is only available on the specific emulation devices with feature package E, T, and on ADAS devices with feature package A, H of the TC39x, TC37xEXT, TC35x, TC33xEXT. AGBT I/O functions are only available for packages with 292 or more pins. For details on AGBT parameters see the "TC3xx Emulation Devices" Data Sheet.

2 TC33xEXT Pin Definition and Functions:

The following figures are showing the TC33xEXT Logic Symbols for the package variants:

- LFBGA-292 for feature package DA and DZ ([Figure 2-1](#))
- LFBGA-180 for feature package DA ([Figure 2-2](#))

TC33xEXT Pin Definition and Functions:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		
A	NC1	VEXT	NC	NC	NC	NC	NC	NC	NC	P11.11	P11.9	P11.2	P14.8	P14.5	P14.1	P15.6	P15.4	P15.1	VDDP3	VSS	A	
B	NC	VSS	VEXT	NC	NC	NC	NC	NC	NC	P11.12	P11.10	P11.3	P14.6	P14.3	P14.4	P14.0	P15.3	VDDP3	VSS	P15.0	B	
C	P50_1	P50_0																	P15.2	P20.14	C	
D	P50_3	P50_2	VSS	VFLEX	P11.15	P11.14	P11.5	P11.6	P11.4	P14.10	P14.9	P14.7	P15.8	P15.7	VDD	VSS			P20.12	P20.13	D	
E	P50_5	P50_4	P10.0	VSS	P11.13	P11.8	P11.7	P11.1	P11.0	P12.1	P12.0	P14.2	P15.5	VDD	VSS	P20.9			P20.10	P20.11	E	
F	P50_7	P50_6	P10.1	P10.2												ESR0	P20.6			P20.7	P20.8	F
G	P50_9	P50_8	P10.3	P10.4			VDD	VSS	DAPE2	DAPE1	VSS	VDD			ESR1	PORST			P20.1	P20.3	G	
H	P50_11	P50_10	P10.5	P10.6	VDD		VSS	VSS	VSS	VSS		VDD			P21.7 / TDO	P21.6 / TDI			P20.2	P20.0	H	
J	P02.0	P02.1	P10.8	P10.7	VSS	VSS		VSS	VSS		VSS	VSS			TCK	P21.1			P21.3	P21.5	J	
K	P02.2	P02.3	P02.4	P02.5	AGBTC LKP (VSS)	VSS	VSS	VSS	VSS	VSS	VSS	DAPE0			TMS	P21.0			P21.2	P21.4	K	
L	P02.6	P00.0	P02.7	P02.8	AGBTC LKN (VSS)	VSS	VSS	VSS	VSS	VSS	VSS	AGBTE RR (VSS)			NC	NC			TRST	VSS	L	
M	P00.2	P00.1	P00.3	P00.4	VSS	VSS		VSS	VSS		VSS	VSS			NC	NC			XTAL2	XTAL1	M	
N	P00.8	P00.7	P00.6	P00.5	VDD		VSS	VSS	VSS	VSS		VDD			NC	NC			VDD	VEXT	N	
P	NC	P00.9	NC	NC			VDD	VSS	AGBTT XN (VSS)	AGBTT XP (VSS)	VSS	VDD			P22.4	P22.5			P22.1	P22.0	P	
R	NC	P00.12	NC	AN23												NC	NC			P22.3	P22.2	R
T	NC	NC	NC	AN22	AN15	AN12	AN6	AN4	AN0	VEVRS B	P34.2	NC	NC	NC	VSS	P23.5			NC	NC	T	
U	NC	NC	NC1	AN17	AN14	AN9	AN7	AN3	AN1	P34.1	P34.3	NC	NC	NC	NC	VSS			P23.1	NC	U	
V	NC	NC																	VEXT	NC	V	
W	NC	NC	AN19	AN18	AN16	AN13	AN11	AN8	AN2	P33.0	P33.2	P33.4	P33.6	P33.8	P33.10	P33.12	P32.1 / VGATE 1P	P32.4	VSS	VEXT	W	
Y	NC1	AN21	AN20	VSSM	VDDM	VAREF 1	VAGND 1	AN10	AN5	P33.1	P33.3	P33.5	P33.7	P33.9	P33.11	P33.13	P32.0 / VGATE 1N	NC	NC	VSS	Y	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20		

Figure 2-1 TC33xEXT Logic Symbol for the package variant LFBGA-292 for feature package DA and DZ

TC33xEXT Pin Definition and Functions:

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	
A	NC	NC	P11.10	P11.9	P11.6	P11.2	P11.0	P12.1	P14.0	P15.3	P15.6	P15.1	P15.0	NC	A
B	NC	VSS	P11.8	P11.7	P11.5	P11.4	P11.1	P11.14	P14.5	P14.1	P15.4	P15.2	VSS	P20.14	B
C	P50_0	P50_1	VSS	P11.12	P11.3	P11.15	P11.13	P12.0	P14.6	P14.4	P14.3	VSS	P20.13	P20.10	C
D	P50_2	P50_3	NC	VSS	VFLEX	P11.11	NC	NC	P14.2	P15.5	VSS	P20.9	P20.12	P20.11	D
E	P50_4	P50_5	P10.1	P10.6	VSS					VSS	ESRT	P20.7	PORST	ESR0	E
F	P50_6	P50_7	P10.2	P10.5		VEXT	VDDP3	VEXT	VDD		P20.6	P20.8	P21.6 / TDI	P21.7 / TDO	F
G	P50_8	P50_9	P10.8	P10.3		VEXT	VSS	VSS	VDD		TMS	P20.3	P20.2	TCK	G
H	P50_10	P50_11	P10.7	P02.1		VDD	VSS	VSS	VEVRS B		NC	P21.0	P20.0	P21.5	H
J	P02.4	P02.5	P02.2	AN17		VSS	VDD	VDD	VEXT		NC	NC	P21.4	P21.3	J
K	P02.0	P02.7	AN20	AN16	VSS					VSS	NC	TRST	P21.2	VSS	K
L	P02.8	P02.6	NC	AN9	AN4	AN2	NC	P33.4	P33.8	P33.7	VSS	NC	XTAL2	XTAL1	L
M	P02.3	P00.0	AN21	AN12	AN5	AN3	NC	P33.1	P33.0	P33.12	P33.13	VSS	VDD	VEXT	M
N	AN15	AN14	AN13	AN11	AN8	AN0	NC	P33.3	P33.5	P33.11	P33.10	P32.1 / VGATE 1P	VSS	P23.1	N
P	NC	VSSM / VAGND 1	VDDM	VAREF 1	AN10	AN1	NC	P33.2	P33.6	P33.9	VEXT	P32.0 / VGATE 1N	P32.4	NC	P
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	

Figure 2-2 TC33xEXT Logic Symbol for the package variant LFBGA-180 for feature package DA

2.1 LFBGA-292 Package Variant Pin Configuration of TC33xEXT for feature package DA and DZ

Table 2-1 Port 00 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
L2	P00.0	I	FAST / PU1 / VEXT / ES	General-purpose input
	CCU61_CTRAPA			Trap input capture
	CCU60_T12HRE			External timer start 12
	GETH_MDIOA			MDIO Input
	P00.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASCLK	O2		Shift clock output
	ASCLIN3_ATX	O3		Transmit output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT63	O7		T13 PWM channel 63
	GETH_MDIO	O		MDIO Output
M2	P00.1	I	SLOW / PU1 / VEXT / ES	General-purpose input
	CCU60_CC60INB			T12 capture input 60
	ASCLIN3_ARXE			Receive input
	CCU61_CC60INA			T12 capture input 60
	SENT_SENT0B			Receive input channel 0
	P00.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	SENT_SPC0	O6		Transmit output
	CCU61_CC60	O7		T12 PWM channel 60

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-1 Port 00 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
M1	P00.2	I	SLOW / PU1 / VEXT / ES1	General-purpose input
	SENT_SENT1B			Receive input channel 1
	P00.2	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASCLK	O2		Shift clock output
	—	O3		Reserved
	—	O4		Reserved
	CAN03_TXD	O5		CAN transmit output node 3
	QSPI3_SLSO4	O6		Master slave select output
	CCU61_COUT60	O7		T12 PWM channel 60
M4	P00.3	I	SLOW / PU1 / VEXT / ES1	General-purpose input
	CCU60_CC61INB			T12 capture input 61
	CAN03_RXDA			CAN receive input node 3
	SENT_SENT2B			Receive input channel 2
	CCU61_CC61INA			T12 capture input 61
	P00.3	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASLSO	O2		Slave select signal output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	SENT_SPC2	O6		Transmit output
	CCU61_CC61	O7		T12 PWM channel 61
M5	P00.4	I	SLOW / PU1 / VEXT / ES1	General-purpose input
	SCU_E_REQ2_2			ERU Channel 2 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	SENT_SENT3B			Receive input channel 3
	P00.4	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	SENT_SPC3	O6		Transmit output
	CCU61_COUT61	O7		T12 PWM channel 61

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-1 Port 00 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
N5	P00.5	I	SLOW / PU1 / VEXT / ES1	General-purpose input
	CCU60_CC62INB			T12 capture input 62
	CCU61_CC62INA			T12 capture input 62
	SENT_SENT4B			Receive input channel 4
	P00.5	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI3_SLSO3	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	SENT_SPC4	O6		Transmit output
	CCU61_CC62	O7		T12 PWM channel 62
N4	P00.6	I	SLOW / PU1 / VEXT / ES1	General-purpose input
	SENT_SENT5B			Receive input channel 5
	ASCLIN5_ARXA			Receive input
	P00.6	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	EVADC_EMUX10	O5		Control of external analog multiplexer interface 1
	SENT_SPC5	O6		Transmit output
	CCU61_COUT62	O7		T12 PWM channel 62
N2	P00.7	I	SLOW / PU1 / VEXT / ES1	General-purpose input
	CCU61_CC60INC			T12 capture input 60
	GPT120_T2INA			Trigger/gate input of timer T2
	CCU61_CCPOS0A			Hall capture input 0
	CCU60_T12HRB			External timer start 12
	P00.7	O0		General-purpose output
	—	O1		Reserved
	ASCLIN5_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	EVADC_EMUX11	O5		Control of external analog multiplexer interface 1
	—	O6		Reserved
	CCU61_CC60	O7		T12 PWM channel 60

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-1 Port 00 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
N1	P00.8	I	SLOW / PU1 / VEXT / ES1	General-purpose input
	CCU61_CC61INC			T12 capture input 61
	GPT120_T2EUDA			Count direction control input of timer T2
	CCU61_CCPOS1A			Hall capture input 1
	CCU60_T13HRB			External timer start 13
	P00.8	O0		General-purpose output
	—	O1		Reserved
	QSPI3_SLSO6	O2		Master slave select output
	—	O3		Reserved
	—	O4		Reserved
	EVADC_EMUX12	O5		Control of external analog multiplexer interface 1
	—	O6		Reserved
	CCU61_CC61	O7		T12 PWM channel 61
P2	P00.9	I	SLOW / PU1 / VEXT / ES1	General-purpose input
	CCU61_CC62INC			T12 capture input 62
	CCU61_CCPOS2A			Hall capture input 2
	GPT120_T4EUDA			Count direction control input of timer T4
	CCU60_T13HRC			External timer start 13
	CCU60_T12HRC			External timer start 12
	P00.9	O0		General-purpose output
	—	O1		Reserved
	QSPI3_SLSO7	O2		Master slave select output
	ASCLIN3_ARTS	O3		Ready to send output
	—	O4		Reserved
	ASCLIN4_ATX	O5		Transmit output
	—	O6		Reserved
	CCU61_CC62	O7		T12 PWM channel 62
R2	P00.12	I	SLOW / PU1 / VEXT / ES1	General-purpose input
	ASCLIN3_ACTSA			Clear to send input
	ASCLIN4_ARXA			Receive input
	P00.12	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU61_COUT63	O7		T13 PWM channel 63

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-2 Port 02 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
J1	P02.0	I	FAST / PU1 / VEXT / ES	General-purpose input
	CCU61_CC60INB			T12 capture input 60
	ASCLIN2_ARXG			Receive input
	CCU60_CC60INA			T12 capture input 60
	SCU_E_REQ3_2			ERU Channel 3 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P02.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ATX	O2		Transmit output
	QSPI3_SLSO1	O3		Master slave select output
	—	O4		Reserved
	CAN00_TXD	O5		CAN transmit output node 0
	—	O6		Reserved
	CCU60_CC60	O7		T12 PWM channel 60
J2	P02.1	I	SLOW / PU1 / VEXT / ES	General-purpose input
	ASCLIN2_ARXB			Receive input
	CAN00_RXDA			CAN receive input node 0
	SCU_E_REQ2_1			ERU Channel 2 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P02.1	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI3_SLSO2	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT60	O7		T12 PWM channel 60

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-2 Port 02 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
K1	P02.2	I	FAST / PU1 / VEXT / ES	General-purpose input
	CCU61_CC61INB			T12 capture input 61
	CCU60_CC61INA			T12 capture input 61
	P02.2	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	QSPI3_SLSO3	O3		Master slave select output
	—	O4		Reserved
	CAN02_TXD	O5		CAN transmit output node 2
	—	O6		Reserved
	CCU60_CC61	O7		T12 PWM channel 61
K2	P02.3	I	SLOW / PU1 / VEXT / ES	General-purpose input
	CAN02_RXDB			CAN receive input node 2
	ASCLIN1_ARXG			Receive input
	P02.3	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ASLSO	O2		Slave select signal output
	QSPI3_SLSO4	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT61	O7		T12 PWM channel 61
K4	P02.4	I	FAST / PU1 / VEXT / ES	General-purpose input
	CCU61_CC62INB			T12 capture input 62
	QSPI3_SLSIA			Slave select input
	CCU60_CC62INA			T12 capture input 62
	P02.4	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ASCLK	O2		Shift clock output
	QSPI3_SLSO0	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_CC62	O7		T12 PWM channel 62

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-2 Port 02 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
K5	P02.5	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI3_MRSTA			Master SPI data input
	SENT_SENT3C			Receive input channel 3
	P02.5	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI3_MRST	O3		Slave SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT62	O7		T12 PWM channel 62
L1	P02.6	I	FAST / PU1 / VEXT / ES	General-purpose input
	CCU60_CC60INC			T12 capture input 60
	SENT_SENT2C			Receive input channel 2
	GPT120_T3INA			Trigger/gate input of core timer T3
	CCU60_CCPOS0A			Hall capture input 0
	CCU61_T12HRB			External timer start 12
	QSPI3_MTSRA			Slave SPI data input
	RIF0_RAMP1B			External RAMP B input
	P02.6	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI3_MTSR	O3		Master SPI data output
	—	O4		Reserved
	EVADC_EMUX00	O5		Control of external analog multiplexer interface 0
	—	O6		Reserved
	CCU60_CC60	O7		T12 PWM channel 60

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Table 2-2 Port 02 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
L4	P02.7	I	FAST / PU1 / VEXT / ES	General-purpose input
	CCU60_CC61INC			T12 capture input 61
	SENT_SENT1C			Receive input channel 1
	GPT120_T3EUDA			Count direction control input of core timer T3
	CCU60_CCPOS1A			Hall capture input 1
	QSPI3_SCLKA			Slave SPI clock inputs
	CCU61_T13HRB			External timer start 13
	P02.7	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI3_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	EVADC_EMUX01	O5		Control of external analog multiplexer interface 0
	SENT_SPC1	O6		Transmit output
	CCU60_CC61	O7		T12 PWM channel 61
L5	P02.8	I	SLOW / PU1 / VEXT / ES	General-purpose input
	CCU60_CC62INC			T12 capture input 62
	SENT_SENT0C			Receive input channel 0
	CCU60_CCPOS2A			Hall capture input 2
	GPT120_T4INA			Trigger/gate input of timer T4
	CCU61_T12HRC			External timer start 12
	CCU61_T13HRC			External timer start 13
	P02.8	O0		General-purpose output
	—	O1		Reserved
	QSPI3_SLSO5	O2		Master slave select output
	—	O3		Reserved
	—	O4		Reserved
	EVADC_EMUX02	O5		Control of external analog multiplexer interface 0
	GETH_MDC	O6		MDIO clock
	CCU60_CC62	O7		T12 PWM channel 62

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-3 Port 10 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
E4	P10.0	I	FAST / PU1 / VEXT / ES	General-purpose input
	GPT120_T6EUDB			Count direction control input of core timer T6
	GETH_RXERC			Receive Error MII
	P10.0	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI1_SLSO10	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
F4	P10.1	I	FAST / PU1 / VEXT / ES	General-purpose input
	GPT120_T5EUDB			Count direction control input of timer T5
	QSPI1_MRSTA			Master SPI data input
	P10.1	O0		General-purpose output
	—	O1		Reserved
	QSPI1_MTSR	O2		Master SPI data output
	QSPI1_MRST	O3		Slave SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
F5	P10.2	I	FAST / PU1 / VEXT / ES	General-purpose input
	CAN02_RXDE			CAN receive input node 2
	QSPI1_SCLKA			Slave SPI clock inputs
	GPT120_T6INB			Trigger/gate input of core timer T6
	SCU_E_REQ2_0			ERU Channel 2 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P10.2	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI1_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-3 Port 10 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
G4	P10.3	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI1_MTSRA			Slave SPI data input
	SCU_E_REQ3_0			ERU Channel 3 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	GPT120_T5INB			Trigger/gate input of timer T5
	P10.3	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI1_MTSR	O3		Master SPI data output
	—	O4		Reserved
	—	O5		Reserved
	CAN02_TXD	O6		CAN transmit output node 2
	—	O7		Reserved
G5	P10.4	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI1_MTSRC			Slave SPI data input
	CCU60_CCPOS0C			Hall capture input 0
	GPT120_T3INB			Trigger/gate input of core timer T3
	P10.4	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI1_SLSO8	O3		Master slave select output
	QSPI1_MTSR	O4		Master SPI data output
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
H4	P10.5	I	SLOW / PU2 / VEXT / ES	General-purpose input
	PMS_HWCFG4IN			HWCFG4 pin input
	P10.5	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ATX	O2		Transmit output
	QSPI3_SLSO8	O3		Master slave select output
	QSPI1_SLSO9	O4		Master slave select output
	GPT120_T6OUT	O5		External output for overflow/underflow detection of core timer T6
	ASCLIN2_ASLSO	O6		Slave select signal output
	—	O7		Reserved

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-3 Port 10 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
H5	P10.6	I	SLOW / PU2 / VEXT / ES	General-purpose input
	ASCLIN2_ARXD			Receive input
	QSPI3_MTSRB			Slave SPI data input
	PMS_HWCFG5IN			HWCFG5 pin input
	P10.6	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ASCLK	O2		Shift clock output
	QSPI3_MTSR	O3		Master SPI data output
	GPT120_T3OUT	O4		External output for overflow/underflow detection of core timer T3
	—	O5		Reserved
	QSPI1_MRST	O6		Slave SPI data output
	—	O7		Reserved
J5	P10.7	I	SLOW / PU1 / VEXT / ES	General-purpose input
	GPT120_T3EUDB			Count direction control input of core timer T3
	ASCLIN2_ACTSA			Clear to send input
	QSPI3_MRSTB			Master SPI data input
	SCU_E_REQ0_2			ERU Channel 0 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	CCU60_CCPOS1C			Hall capture input 1
	P10.7	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI3_MRST	O3		Slave SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-3 Port 10 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
J4	P10.8	I	SLOW / PU1 / VEXT / ES	General-purpose input
	GPT120_T4INB			Trigger/gate input of timer T4
	QSPI3_SCLKB			Slave SPI clock inputs
	SCU_E_REQ1_2			ERU Channel 1 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	CCU60_CCPOS2C			Hall capture input 2
	P10.8	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ARTS	O2		Ready to send output
	QSPI3_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

Table 2-4 Port 11 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
E10	P11.0	I	RFAST / PU1 / VFLEX / ES	General-purpose input
	ASCLIN3_ARXB			Receive input
	P11.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	GETH_TXD3	O6		Transmit Data
	—	O7		Reserved
E9	P11.1	I	RFAST / PU1 / VFLEX / ES	General-purpose input
	P11.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASCLK	O2		Shift clock output
	ASCLIN3_ATX	O3		Transmit output
	—	O4		Reserved
	—	O5		Reserved
	GETH_TXD2	O6		Transmit Data
	—	O7		Reserved

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Table 2-4 Port 11 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
A12	P11.2	I	RFAST / PU1 / VFLEX / ES	General-purpose input
	P11.2	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPIO_SLSO5	O3		Master slave select output
	QSPI1_SLSO5	O4		Master slave select output
	—	O5		Reserved
	GETH_TXD1	O6		Transmit Data
	CCU60_COUT63	O7		T13 PWM channel 63
B12	P11.3	I	RFAST / PU1 / VFLEX / ES	General-purpose input
	QSPI1_MRSTB			Master SPI data input
	P11.3	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI1_MRST	O3		Slave SPI data output
	—	O4		Reserved
	—	O5		Reserved
	GETH_TXD0	O6		Transmit Data
	CCU60_COUT62	O7		T12 PWM channel 62
D10	P11.4	I	RFAST / PU1 / VFLEX / ES	General-purpose input
	GETH_RXCLKB			Receive Clock MII
	P11.4	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASCLK	O2		Shift clock output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	GETH_TXER	O6		Transmit Error MII
	GETH_TXCLK	O7		Transmit Clock Output for RGMII

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Table 2-4 Port 11 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
D8	P11.5	I	SLOW / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	GETH_TXCLKA			Transmit Clock Input for MII
	GETH_GREFCLK			Gigabit Reference Clock input for RGMII (125 MHz high precision)
	P11.5	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
D9	P11.6	I	RFAST / PU1 / VFLEX / ES	General-purpose input
	QSPI1_SCLKB			Slave SPI clock inputs
	P11.6	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI1_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	—	O5		Reserved
	GETH_TXEN	O6		Transmit Enable MII and RMII
	GETH_TCTL			Transmit Control for RGMII
	CCU60_COUT61	O7		T12 PWM channel 61
E8	P11.7	I	SLOW / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	GETH_RXD3A			Receive Data 3 MII and RGMII (RGMII can use RXD3A only)
	P11.7	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-4 Port 11 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
E7	P11.8	I	SLOW / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	GETH_RXD2A			Receive Data 2 MII and RGMII (RGMII can use RXD2A only)
	P11.8	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
A11	P11.9	I	FAST / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	QSPI1_MTSRB			Slave SPI data input
	GETH_RXD1A			Receive Data 1 MII, RMII and RGMII (RGMII can use RXD1A only)
	P11.9	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI1_MTSR	O3		Master SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT60	O7		T12 PWM channel 60

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Table 2-4 Port 11 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
B11	P11.10	I	FAST / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	CAN03_RXDD			CAN receive input node 3
	ASCLIN1_ARXE			Receive input
	SCU_E_REQ6_3			ERU Channel 6 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	GETH_RXD0A			Receive Data 0 MII, RMII and RGMII (RGMII can use RXD0A only)
	QSPI1_SLSIA			Slave select input
	P11.10	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI0_SLSO3	O3		Master slave select output
	QSPI1_SLSO3	O4		Master slave select output
	—	O5		Reserved
	—	O6		Reserved
	CCU60_CC62	O7		T12 PWM channel 62
A10	P11.11	I	FAST / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	GETH_CRSDVA			Carrier Sense / Data Valid combi-signal for RMII
	GETH_RXDVA			Receive Data Valid MII
	GETH_CRSB			Carrier Sense MII
	GETH_RCTLA			Receive Control for RGMII
	P11.11	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI0_SLSO4	O3		Master slave select output
	QSPI1_SLSO4	O4		Master slave select output
	—	O5		Reserved
	—	O6		Reserved
	CCU60_CC61	O7		T12 PWM channel 61

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-4 Port 11 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
B10	P11.12	I	FAST / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	GETH_REFCLKA			Reference Clock input for RMII (50 MHz)
	GETH_TXCLKB			Transmit Clock Input for MII
	GETH_RXCLKA			Receive Clock MII
	P11.12	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	CAN03_TXD	O5		CAN transmit output node 3
	CCU_EXTCLK1	O6		External Clock 1
	CCU60_CC60	O7		T12 PWM channel 60
E6	P11.13	I	SLOW / PU1 / VFLEX / ES	General-purpose input
	GETH_RXERA			Receive Error MII
	P11.13	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
D7	P11.14	I	SLOW / PU1 / VFLEX / ES	General-purpose input
	GETH_CRSDVB			Carrier Sense / Data Valid combi-signal for RMII
	GETH_RXDVB			Receive Data Valid MII
	GETH_CRSA			Carrier Sense MII
	P11.14	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

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Table 2-4 Port 11 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
D6	P11.15	I	SLOW / PU1 / VFLEX / ES	General-purpose input
	GETH_COLA			Collision MII
	P11.15	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	—			Reserved

Table 2-5 Port 12 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
E12	P12.0	I	SLOW / PU1 / VFLEX / ES	General-purpose input
	CAN00_RXDC			CAN receive input node 0
	GETH_RXCLKC			Receive Clock MII
	P12.0	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	GETH_MDC	O6		MDIO clock
	—	O7		Reserved
E11	P12.1	I	SLOW / PU1 / VFLEX / ES	General-purpose input
	GETH_MDIOC			MDIO Input
	P12.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASLSO	O2		Slave select signal output
	—	O3		Reserved
	—	O4		Reserved
	CAN00_TXD	O5		CAN transmit output node 0
	—	O6		Reserved
	—	O7		Reserved
	GETH_MDIO	O		MDIO Output

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-6 Port 14 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
B16	P14.0	I	FAST / PU1 / VEXT / ES2	General-purpose input
	P14.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN0_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	CAN01_TXD	O5		CAN transmit output node 1
	ASCLIN0_ASCLK	O6		Shift clock output
	CCU60_COUT62	O7		T12 PWM channel 62
A15	P14.1	I	FAST / PU1 / VEXT / ES2	General-purpose input
	ASCLIN0_ARXA			Receive input
	CAN01_RXDB			CAN receive input node 1
	SCU_E_REQ3_1			ERU Channel 3 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	PMS_PINAWKP			PINA (P14.1) pin input
	P14.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN0_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT63	O7		T13 PWM channel 63
E13	P14.2	I	SLOW / PU2 / VEXT / ES	General-purpose input
	PMS_HWCFG2IN			HWCFG2 pin input
	P14.2	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ATX	O2		Transmit output
	QSPI2_SLSO1	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	ASCLIN2_ASCLK	O6		Shift clock output
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-6 Port 14 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
B14	P14.3	I	SLOW / PU2 / VEXT / ES	General-purpose input
	PMS_HWCFG3IN			HWCFG3 pin input
	ASCLIN2_ARXA			Receive input
	SCU_E_REQ1_0			ERU Channel 1 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P14.3	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ATX	O2		Transmit output
	QSPI2_SLSO3	O3		Master slave select output
	ASCLIN1_ASLSO	O4		Slave select signal output
	ASCLIN3_ASLSO	O5		Slave select signal output
	—	O6		Reserved
	—	O7		Reserved
B15	P14.4	I	SLOW / PU2 / VEXT / ES	General-purpose input
	PMS_HWCFG6IN			HWCFG6 pin input
	P14.4	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	GETH_PPS	O6		Pulse Per Second
	—	O7		Reserved
	—	—		—
A14	P14.5	I	FAST / PU2 / VEXT / ES	General-purpose input
	PMS_HWCFG1IN			HWCFG1 pin input
	P14.5	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	—	—		—

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-6 Port 14 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
B13	P14.6	I	FAST / PU1 / VEXT / ES	General-purpose input
	P14.6	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI2_SLSO2	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
D13	P14.7	I	SLOW / PU1 / VEXT / ES	General-purpose input
	CAN01_RXDE			CAN receive input node 1
	CAN02_RXDF			CAN receive input node 2
	P14.7	O0		General-purpose output
	—	O1		Reserved
	ASCLIN0_ARTS	O2		Ready to send output
	QSPI2_SLSO4	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
A13	P14.8	I	SLOW / PU1 / VEXT / ES	General-purpose input
	CAN02_RXDD			CAN receive input node 2
	ASCLIN1_ARXD			Receive input
	P14.8	O0		General-purpose output
	—	O1		Reserved
	ASCLIN5_ASLSO	O2		Slave select signal output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-6 Port 14 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
D12	P14.9	I	SLOW / PU1 / VEXT / ES	General-purpose input
	ASCLIN0_ACTSA			Clear to send input
	P14.9	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	CAN01_TXD	O4		CAN transmit output node 1
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
D11	P14.10	I	FAST / PU1 / VEXT / ES	General-purpose input
	P14.10	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	ASCLIN1_ATX	O4		Transmit output
	CAN02_TXD	O5		CAN transmit output node 2
	—	O6		Reserved
	—	O7		Reserved

Table 2-7 Port 15 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
B20	P15.0	I	FAST / PU1 / VEXT / ES	General-purpose input
	SDMMC0_DAT7_IN			read data in
	P15.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	QSPI0_SLSO13	O3		Master slave select output
	—	O4		Reserved
	CAN02_TXD	O5		CAN transmit output node 2
	ASCLIN1_ASCLK	O6		Shift clock output
	—	O7		Reserved
	SDMMC0_DAT7	O		write data out

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-7 Port 15 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
A18	P15.1	I	FAST / PU1 / VEXT / ES	General-purpose input
	CAN02_RXDA			CAN receive input node 2
	ASCLIN1_ARXA			Receive input
	QSPI2_SLSIB			Slave select input
	SCU_E_REQ7_2			ERU Channel 7 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P15.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	QSPI2_SLSO5	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	SDMMC0_CLK	O7		card clock
C19	P15.2	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI2_SLSIA			Slave select input
	QSPI2_MRSTE			Master SPI data input
	QSPI2_HSICINA			Highspeed capture channel
	P15.2	O0		General-purpose output
	—	O1		Reserved
	ASCLIN0_ATX	O2		Transmit output
	QSPI2_SLSO0	O3		Master slave select output
	—	O4		Reserved
	CAN01_TXD	O5		CAN transmit output node 1
	ASCLIN0_ASCLK	O6		Shift clock output
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-7 Port 15 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
B17	P15.3	I	FAST / PU1 / VEXT / ES	General-purpose input
	CAN01_RXDA			CAN receive input node 1
	ASCLIN0_ARXB			Receive input
	QSPI2_SCLKA			Slave SPI clock inputs
	QSPI2_HSICINB			Highspeed capture channel
	SDMMC0_CMD_IN			command in
	P15.3	O0		General-purpose output
	—	O1		Reserved
	ASCLIN0_ATX	O2		Transmit output
	QSPI2_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	SDMMC0_CMD	O		command out
A17	P15.4	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI2_MRSTA			Master SPI data input
	SCU_E_REQ0_0			ERU Channel 0 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P15.4	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	QSPI2_MRST	O3		Slave SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_CC62	O7		T12 PWM channel 62

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-7 Port 15 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
E14	P15.5	I	FAST / PU1 / VEXT / ES	General-purpose input
	ASCLIN1_ARXB			Receive input
	QSPI2_MTSRA			Slave SPI data input
	SCU_E_REQ4_3			ERU Channel 4 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P15.5	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	QSPI2_MTSR	O3		Master SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_CC61	O7		T12 PWM channel 61
A16	P15.6	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI2_MTSRB			Slave SPI data input
	P15.6	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ATX	O2		Transmit output
	QSPI2_MTSR	O3		Master SPI data output
	—	O4		Reserved
	QSPI2_SCLK	O5		Master SPI clock output
	ASCLIN3_ASCLK	O6		Shift clock output
	CCU60_CC60	O7		T12 PWM channel 60
D15	P15.7	I	FAST / PU1 / VEXT / ES	General-purpose input
	ASCLIN3_ARXA			Receive input
	QSPI2_MRSTB			Master SPI data input
	P15.7	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ATX	O2		Transmit output
	QSPI2_MRST	O3		Slave SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT60	O7		T12 PWM channel 60

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-7 Port 15 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
D14	P15.8	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI2_SCLKB			Slave SPI clock inputs
	SCU_E_REQ5_0			ERU Channel 5 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P15.8	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI2_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	—	O5		Reserved
	ASCLIN3_ASCLK	O6		Shift clock output
	CCU60_COUT61	O7		T12 PWM channel 61

Table 2-8 Port 20 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
H20	P20.0	I	FAST / PU1 / VEXT / ES	General-purpose input
	CAN03_RXDC			CAN receive input node 3
	CCU_PAD_SYSClk			Sysclk input
	CBS_TG10			Trigger input
	SCU_E_REQ6_0			ERU Channel 6 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	GPT120_T6EUDA			Count direction control input of core timer T6
	P20.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ATX	O2		Transmit output
	ASCLIN3_ASCLK	O3		Shift clock output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	CBS_TGO0	O		Trigger output

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-8 Port 20 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
G19	P20.1	I	SLOW / PU1 / VEXT / ES	General-purpose input
	CBS_TGI1			Trigger input
	P20.1	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	CBS_TGO1	O		Trigger output
H19	P20.2	I	S / PU / VEXT	General-purpose input This pin is latched at power on reset release to enter test mode.
	TESTMODE			Testmode Enable Input
G20	P20.3	I	SLOW / PU1 / VEXT / ES	General-purpose input
	ASCLIN3_ARXC			Receive input
	GPT120_T6INA			Trigger/gate input of core timer T6
	P20.3	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ATX	O2		Transmit output
	QSPIO_SLSO9	O3		Master slave select output
	QSPI2_SLSO9	O4		Master slave select output
	CAN03_TXD	O5		CAN transmit output node 3
	—	O6		Reserved
	—	O7		Reserved
F17	P20.6	I	SLOW / PU1 / VEXT / ES	General-purpose input
	P20.6	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ARTS	O2		Ready to send output
	QSPIO_SLSO8	O3		Master slave select output
	QSPI2_SLSO8	O4		Master slave select output
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-8 Port 20 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
F19	P20.7	I	FAST / PU1 / VEXT / ES	General-purpose input
	CAN00_RXDB			CAN receive input node 0
	ASCLIN1_ACTSA			Clear to send input
	SDMMC0_DAT0_IN			read data in
	P20.7	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU61_COUT63	O7		T13 PWM channel 63
	SDMMC0_DAT0	O		write data out
F20	P20.8	I	FAST / PU1 / VEXT / ES	General-purpose input
	SDMMC0_DAT1_IN			read data in
	P20.8	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ASLSO	O2		Slave select signal output
	QSPIO_SLSO0	O3		Master slave select output
	QSPIO_SLSO0	O4		Master slave select output
	CAN00_TXD	O5		CAN transmit output node 0
	—	O6		Reserved
	CCU61_CC60	O7		T12 PWM channel 60
	SDMMC0_DAT1	O		write data out
E17	P20.9	I	FAST / PU1 / VEXT / ES	General-purpose input
	CAN03_RXDE			CAN receive input node 3
	ASCLIN1_ARXC			Receive input
	QSPIO_SLSIB			Slave select input
	SCU_E_REQ7_0			ERU Channel 7 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P20.9	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPIO_SLSO1	O3		Master slave select output
	QSPIO_SLSO1	O4		Master slave select output
	—	O5		Reserved
	—	O6		Reserved
	CCU61_CC61	O7		T12 PWM channel 61

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-8 Port 20 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
E19	P20.10	I	FAST / PU1 / VEXT / ES	General-purpose input
	SDMMC0_DAT2_IN			read data in
	P20.10	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	QSPI0_SLSO6	O3		Master slave select output
	QSPI2_SLSO7	O4		Master slave select output
	CAN03_TXD	O5		CAN transmit output node 3
	ASCLIN1_ASCLK	O6		Shift clock output
	CCU61_CC62	O7		T12 PWM channel 62
	SDMMC0_DAT2	O		write data out
E20	P20.11	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI0_SCLKA			Slave SPI clock inputs
	SDMMC0_DAT3_IN			read data in
	P20.11	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI0_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU61_COUT60	O7		T12 PWM channel 60
	SDMMC0_DAT3	O		write data out
D19	P20.12	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI0_MRSTA			Master SPI data input
	SDMMC0_DAT4_IN			read data in
	P20.12	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI0_MRST	O3		Slave SPI data output
	QSPI0_MTSR	O4		Master SPI data output
	—	O5		Reserved
	—	O6		Reserved
	CCU61_COUT61	O7		T12 PWM channel 61
	SDMMC0_DAT4	O		write data out

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-8 Port 20 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
D20	P20.13	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPIO_SLSIA			Slave select input
	SDMMC0_DAT5_IN			read data in
	P20.13	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPIO_SLSO2	O3		Master slave select output
	QSPIO1_SLSO2	O4		Master slave select output
	QSPIO_SCLK	O5		Master SPI clock output
	—	O6		Reserved
	CCU61_COUT62	O7		T12 PWM channel 62
	SDMMC0_DAT5	O		write data out
C20	P20.14	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPIO_MTSRA			Slave SPI data input
	SDMMC0_DAT6_IN			read data in
	DMU_FDEST			Enter destructive debug mode
	P20.14	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPIO_MTSR	O3		Master SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	SDMMC0_DAT6	O		write data out

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Table 2-9 Port 21 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
K17	P21.0	I	FAST / PU1 / VEXT / ES	General-purpose input
	P21.0	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	HSM_HSM1	O		Pin Output Value
J17	P21.1	I	FAST / PU1 / VEXT / ES	General-purpose input
	P21.1	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	HSM_HSM2	O		Pin Output Value
K19	P21.2	I	FAST / PU1 / VEXT / ES	General-purpose input
	SCU_EMGSTOP_POR T_B			Emergency stop Port Pin B input request
	P21.2	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASLSO	O2		Slave select signal output
	—	O3		Reserved
	—	O4		Reserved
	GETH_MDC	O5		MDIO clock
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-9 Port 21 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
J19	P21.3	I	FAST / PU1 / VEXT / ES	General-purpose input
	GETH_MDIOD			MDIO Input
	P21.3	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	GETH_MDIO	O		MDIO Output
K20	P21.4	I	FAST / PU1 / VEXT / ES6	General-purpose input
	P21.4	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
J20	P21.5	I	FAST / PU1 / VEXT / ES6	General-purpose input
	P21.5	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASCLK	O2		Shift clock output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-9 Port 21 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
H17	P21.6/TDI	I	FAST / PD / PU2 / VEXT / ES3	General-purpose input PD during Reset and in DAP/DAPE or JTAG mode. After Reset release and when not in DAP/DAPE or JTAG mode: PU. In Standby mode: HighZ.
	GPT120_T5EUDA			Count direction control input of timer T5
	ASCLIN3_ARXF			Receive input
	CBS_TGI2			Trigger input
	TDI			JTAG Module Data Input
	P21.6	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASLSO	O2		Slave select signal output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	GPT120_T3OUT	O7		External output for overflow/underflow detection of core timer T3
	CBS_TGO2	O		Trigger output
	DAP3	I/O		DAP: DAP3 Data I/O
	DAPE1	I/O		DAPE: DAPE1 Data I/O
H16	P21.7/TDO	I	FAST / PU2 / VEXT / ES4	General-purpose input
	GPT120_T5INA			Trigger/gate input of timer T5
	CBS_TGI3			Trigger input
	GETH_RXERB			Receive Error MII
	P21.7	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ATX	O2		Transmit output
	ASCLIN3_ASCLK	O3		Shift clock output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	GPT120_T6OUT	O7		External output for overflow/underflow detection of core timer T6
	CBS_TGO3	O		Trigger output
	DAP2	I/O		DAP: DAP2 Data I/O
	DAPE2	I/O		DAPE: DAPE2 Data I/O
	TDO	O		JTAG Module Data Output

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-10 Port 22 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
P20	P22.0	I	FAST / PU1 / VEXT / ES6	General-purpose input
	P22.0	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
P19	P22.1	I	FAST / PU1 / VEXT / ES6	General-purpose input
	P22.1	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
R20	P22.2	I	FAST / PU1 / VEXT / ES6	General-purpose input
	P22.2	O0		General-purpose output
	—	O1		Reserved
	ASCLIN5_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

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Table 2-10 Port 22 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
R19	P22.3	I	FAST / PU1 / VEXT / ES6	General-purpose input
	ASCLIN5_ARXC			Receive input
	P22.3	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	HSPDM_MUTE	O7		Mute output from the micro controller which could be used to control an external Transmitter
P16	P22.4	I	FAST / PU1 / VEXT / ES	General-purpose input
	P22.4	O0		General-purpose output
	—	O1		Reserved
	ASCLIN4_ASLSO	O2		Slave select signal output
	—	O3		Reserved
	QSPIO_SLSO12	O4		Master slave select output
	—	O5		Reserved
	—	O6		Reserved
	HSPDM_BS0_OUT	O7		Bit stream 0 output to the pad
P17	P22.5	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPIO_MTSRC			Slave SPI data input
	P22.5	O0		General-purpose output
	—	O1		Reserved
	ASCLIN4_ATX	O2		Transmit output
	—	O3		Reserved
	QSPIO_MTSR	O4		Master SPI data output
	—	O5		Reserved
	—	O6		Reserved
	HSPDM_BS1_OUT	O7		Bit stream 1 output to the pad

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Table 2-11 Port 23 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
U19	P23.1	I	FAST / PU1 / VEXT / ES	General-purpose input
	P23.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ARTS	O2		Ready to send output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	CCU_EXTCLK0	O6		External Clock 0
	—	O7		Reserved
T17	P23.5	I	FAST / PU1 / VEXT / ES	General-purpose input
	P23.5	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

Table 2-12 Port 32 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
Y17	P32.0	I	SLOW / PU1 / VEXT / ES	General-purpose input P32.0 / SMPS mode: analog output. External Pass Device gate control for EVRC
	P32.0	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-12 Port 32 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
W17	P32.1	I	SLOW / PU1 / VEXT / ES	General-purpose input P32.1 / External Pass Device gate control for EVRC
	P32.1	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
W18	P32.4	I	FAST / PU1 / VEXT / ES	General-purpose input
	ASCLIN1_ACTSB			Clear to send input
	P32.4	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	CCU_EXTCLK1	O6		External Clock 1
	CCU60_COUT63	O7		T13 PWM channel 63
	PMS_DCDCSYNCO	O		DC-DC synchronization output

Table 2-13 Port 33 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
W10	P33.0	I	SLOW / PU1 / VEVRSB / ES5	General-purpose input
	P33.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN5_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-13 Port 33 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
Y10	P33.1	I	SLOW / PU1 / VEVR SB / ES5	General-purpose input
	P33.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASLSO	O2		Slave select signal output
	QSPI2_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	EVADC_EMUX02	O5		Control of external analog multiplexer interface 0
	—	O6		Reserved
	—	O7		Reserved
W11	P33.2	I	SLOW / PU1 / VEVR SB / ES5	General-purpose input
	P33.2	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASCLK	O2		Shift clock output
	QSPI2_SLSO10	O3		Master slave select output
	—	O4		Reserved
	EVADC_EMUX01	O5		Control of external analog multiplexer interface 0
	—	O6		Reserved
	—	O7		Reserved
Y11	P33.3	I	SLOW / PU1 / VEVR SB / ES5	General-purpose input
	P33.3	O0		General-purpose output
	—	O1		Reserved
	ASCLIN5_ASCLK	O2		Shift clock output
	—	O3		Reserved
	—	O4		Reserved
	EVADC_EMUX00	O5		Control of external analog multiplexer interface 0
	—	O6		Reserved
	—	O7		Reserved
W12	P33.4	I	SLOW / PU1 / VEVR SB / ES5	General-purpose input
	CCU61_CTRAPC			Trap input capture
	ASCLIN5_ARXB			Receive input
	P33.4	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ARTS	O2		Ready to send output
	QSPI2_SLSO12	O3		Master slave select output
	—	O4		Reserved
	EVADC_EMUX12	O5		Control of external analog multiplexer interface 1
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-13 Port 33 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
Y12	P33.5	I	SLOW / PU1 / VEVRSB / ES5	General-purpose input
	GPT120_T4EADB			Count direction control input of timer T4
	ASCLIN2_ACTSB			Clear to send input
	CCU61_CCPOS2C			Hall capture input 2
	SENT_SENT5C			Receive input channel 5
	P33.5	O0		General-purpose output
	—	O1		Reserved
	QSPI0_SLSO7	O2		Master slave select output
	QSPI1_SLSO7	O3		Master slave select output
	—	O4		Reserved
	EVADC_EMUX11	O5		Control of external analog multiplexer interface 1
	—	O6		Reserved
	ASCLIN5_ASLSO	O7		Slave select signal output
W13	P33.6	I	SLOW / PU1 / VEVRSB / ES5	General-purpose input
	GPT120_T2EADB			Count direction control input of timer T2
	SENT_SENT4C			Receive input channel 4
	CCU61_CCPOS1C			Hall capture input 1
	P33.6	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ASLSO	O2		Slave select signal output
	QSPI2_SLSO11	O3		Master slave select output
	—	O4		Reserved
	EVADC_EMUX10	O5		Control of external analog multiplexer interface 1
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-13 Port 33 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
Y13	P33.7	I	SLOW / PU1 / VEVR SB / ES5	General-purpose input
	CAN00_RXDE			CAN receive input node 0
	GPT120_T2INB			Trigger/gate input of timer T2
	CCU61_CCPOS0C			Hall capture input 0
	SCU_E_REQ4_0			ERU Channel 4 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P33.7	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ASCLK	O2		Shift clock output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
W14	P33.8	I	FAST / HighZ / VEVR SB	General-purpose input
	ASCLIN2_ARXE			Receive input
	SCU_EMGSTOP_POR T_A			Emergency stop Port Pin A input request
	P33.8	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	CAN00_TXD	O5		CAN transmit output node 0
	—	O6		Reserved
	CCU61_COUT62	O7		T12 PWM channel 62
	SMU_FSP0	O		FSP[1..0] Output Signals - Generated by SMU_core
Y14	P33.9	I	SLOW / PU1 / VEVR SB / ES5	General-purpose input
	QSPI3_HSI CINA			Highspeed capture channel
	P33.9	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ATX	O2		Transmit output
	—	O3		Reserved
	ASCLIN2_ASCLK	O4		Shift clock output
	CAN01_TXD	O5		CAN transmit output node 1
	ASCLIN0_ATX	O6		Transmit output
	CCU61_CC62	O7		T12 PWM channel 62

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-13 Port 33 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
W15	P33.10	I	FAST / PU1 / VEVR SB / ES5	General-purpose input
	QSPI3_HSICINB			Highspeed capture channel
	CAN01_RXDD			CAN receive input node 1
	ASCLIN0_ARXD			Receive input
	P33.10	O0		General-purpose output
	—	O1		Reserved
	QSPI1_SLSO6	O2		Master slave select output
	—	O3		Reserved
	ASCLIN1_ASLSO	O4		Slave select signal output
	—	O5		Reserved
	—	O6		Reserved
	CCU61_COUT61	O7		T12 PWM channel 61
	SMU_FSP1	O		FSP[1..0] Output Signals - Generated by SMU_core
Y15	P33.11	I	FAST / PU1 / VEVR SB / ES5	General-purpose input
	P33.11	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ASCLK	O2		Shift clock output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU61_CC61	O7		T12 PWM channel 61
W16	P33.12	I	FAST / PU1 / VEVR SB / ES5	General-purpose input
	CAN00_RXDD			CAN receive input node 0
	PMS_PINBWKP			PINB (P33.12) pin input
	P33.12	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	—	O3		Reserved
	ASCLIN1_ASCLK	O4		Shift clock output
	—	O5		Reserved
	—	O6		Reserved
	CCU61_COUT60	O7		T12 PWM channel 60

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-13 Port 33 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
Y16	P33.13	I	FAST / PU1 / VEVR SB / ES5	General-purpose input
	ASCLIN1_ARXF			Receive input
	P33.13	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	—	O3		Reserved
	QSPI2_SLSO6	O4		Master slave select output
	CAN00_TXD	O5		CAN transmit output node 0
	—	O6		Reserved
	CCU61_CC60	O7		T12 PWM channel 60

Table 2-14 Port 34 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
U11	P34.1	I	SLOW / PU1 / VEVR SB / ES5	General-purpose input
	P34.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN4_ATX	O2		Transmit output
	—	O3		Reserved
	CAN00_TXD	O4		CAN transmit output node 0
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT63	O7		T13 PWM channel 63
T12	P34.2	I	SLOW / PU1 / VEVR SB / ES	General-purpose input
	ASCLIN4_ARXB			Receive input
	CAN00_RXDG			CAN receive input node 0
	P34.2	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_CC60	O7		T12 PWM channel 60

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-14 Port 34 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
U12	P34.3	I	SLOW / PU1 / VEVRSB / ES	General-purpose input
	P34.3	O0		General-purpose output
	—	O1		Reserved
	ASCLIN4_ASCLK	O2		Shift clock output
	—	O3		Reserved
	QSPI2_SLSO10	O4		Master slave select output
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT60	O7		T12 PWM channel 60

Table 2-15 Port 50 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
C2	P50.0	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D1N			LVDS RX Input (inverted Data Bits of Channel #0)
C1	P50.1	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D1P			LVDS RX Input (Data Bits of Channel #0)
D2	P50.2	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D2N			LVDS RX Input (inverted Data Bits of Channel #1)
D1	P50.3	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D2P			LVDS RX Input (Data Bits of Channel #1)
E2	P50.4	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_CLKN			LVDS RX Input (inverted Serial Clock)
E1	P50.5	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_CLKP			LVDS RX Input (Serial Clock)
F2	P50.6	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_FRP			LVDS RX Input (FrameClock)

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Table 2-15 Port 50 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
F1	P50.7	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_FRN			LVDS RX Input (inverted FrameClock)
G2	P50.8	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D3P			LVDS RX Input (Data Bits of Channel #2)
G1	P50.9	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D3N			LVDS RX Input (inverted Data Bits of Channel #2)
H2	P50.10	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D4P			LVDS RX Input (Data Bits of Channel #3)
H1	P50.11	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D4N			LVDS RX Input (inverted Data Bits of Channel #3)

Table 2-16 Analog Inputs

Ball	Symbol	Ctrl.	Buffer Type	Function
T10	AN0	I	D / HighZ / VDDM	Analog Input 0
	EVADC_G0CH0			Analog input channel 0, group 0
U10	AN1	I	D / HighZ / VDDM	Analog Input 1
	EVADC_G0CH1			Analog input channel 1, group 0
W9	AN2	I	D / HighZ / VDDM	Analog Input 2
	EVADC_G0CH2			Analog input channel 2, group 0
U9	AN3	I	D / HighZ / VDDM	Analog Input 3
	EVADC_G0CH3			Analog input channel 3, group 0
T9	AN4	I	D / HighZ / VDDM	Analog Input 4
	EVADC_G0CH4			Analog input channel 4, group 0
	EVADC_G5CH0			Analog input channel 0, group 5
Y9	AN5	I	D / HighZ / VDDM	Analog Input 5
	EVADC_G0CH5			Analog input channel 5, group 0
	EVADC_G5CH1			Analog input channel 1, group 5
T8	AN6	I	D / HighZ / VDDM	Analog Input 6
	EVADC_G0CH6			Analog input channel 6, group 0
	EVADC_G5CH2			Analog input channel 2, group 5

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Table 2-16 Analog Inputs (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
U8	AN7	I	D / HighZ / VDDM	Analog Input 7
	EVADC_G0CH7			Analog input channel 7, group 0
	EVADC_G5CH3			Analog input channel 3, group 5
W8	AN8	I	D / HighZ / VDDM	Analog Input 8
	EVADC_G1CH0			Analog input channel 0, group 1
	EVADC_G5CH4			Analog input channel 4, group 5
U7	AN9	I	D / HighZ / VDDM	Analog Input 9
	EVADC_G1CH1			Analog input channel 1, group 1
	EVADC_G5CH5			Analog input channel 5, group 5
Y8	AN10	I	D / HighZ / VDDM	Analog Input 10
	EVADC_G1CH2			Analog input channel 2, group 1
	EVADC_G5CH6			Analog input channel 6, group 5
W7	AN11	I	D / HighZ / VDDM	Analog Input 11
	EVADC_G1CH3			Analog input channel 3, group 1
	EVADC_G5CH7			Analog input channel 7, group 5
T7	AN12	I	D / HighZ / VDDM	Analog Input 12
	EVADC_G1CH4			Analog input channel 4, group 1
	EVADC_G4CH0			Analog input channel 0, group 4
W6	AN13	I	D / HighZ / VDDM	Analog Input 13
	EVADC_G1CH5			Analog input channel 5, group 1
	EVADC_G4CH1			Analog input channel 1, group 4
U6	AN14	I	D / HighZ / VDDM	Analog Input 14
	EVADC_G1CH6			Analog input channel 6, group 1
	EVADC_G4CH2			Analog input channel 2, group 4
T6	AN15	I	D / HighZ / VDDM	Analog Input 15
	EVADC_G1CH7			Analog input channel 7, group 1
	EVADC_G4CH3			Analog input channel 3, group 4
W5	AN16	I	D / HighZ / VDDM	Analog Input 16
	EVADC_G2CH0			Analog input channel 0, group 2
	EVADC_G4CH4			Analog input channel 4, group 4
U5	AN17	I	D / HighZ / VDDM	Analog Input 17
	EVADC_G2CH1			Analog input channel 1, group 2
	EVADC_G4CH5			Analog input channel 5, group 4
W4	AN18	I	D / HighZ / VDDM	Analog Input 18
	EVADC_G2CH2			Analog input channel 2, group 2
	EVADC_G4CH6			Analog input channel 6, group 4

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Table 2-16 Analog Inputs (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
W3	AN19	I	D / HighZ / VDDM	Analog Input 19
	EVADC_G2CH3			Analog input channel 3, group 2
	EVADC_G4CH7			Analog input channel 7, group 4
Y3	AN20	I	D / HighZ / VDDM	Analog Input 20
	EVADC_G3CH0			Analog input channel 0, group 3
Y2	AN21	I	D / HighZ / VDDM	Analog Input 21
	EVADC_G3CH1			Analog input channel 1, group 3
T5	AN22	I	D / HighZ / VDDM	Analog Input 22
	EVADC_G3CH2			Analog input channel 2, group 3
R5	AN23	I	D / HighZ / VDDM	Analog Input 23
	EVADC_G3CH3			Analog input channel 3, group 3

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Note: Port Pins P32.0 and P32.1 are bidirectional pads and are having the following two functionalities implemented:

1. In case the pins are used as standard GPIOs the functions defined in the pin configuration tables of P32.0 and P32.1 are available.
2. In case the pins are used as pre-drivers for external MOSFETs (internal DCDC usecase) P32.0 and P32.1 act as analog IOs named VGATE1N and VGATE1P.

Table 2-17 System I/O

Ball	Symbol	Ctrl.	Buffer Type	Function
L7	AGBTCLKN (VSS)	I	AGBT_C LK / VEXT	Input PAD (negative pole) for the external 100 MHz differential clock. AGBT Input
K7	AGBTCLKP (VSS)	I	AGBT_C LK / VEXT	Input PAD (positive pole) for the external 100 MHz differential clock. AGBT Input
P10	AGBTTXN (VSS)	O	AGBT_T X / VEXT	Off-chip driver output PAD of the 2.5Gbps transmitter, negative pole AGBT Output
P11	AGBTTXP (VSS)	O	AGBT_T X / VEXT	Off-chip driver output PAD of the 2.5Gbps transmitter, positive pole AGBT Output
L14	AGBTERR (VSS)	I	FAST / PD / VEXT	Input PAD for CRC error from FPGA. AGBT Input
Y17	VGATE1N	O	—	DCDC N ch. MOSFET gate driver output P32.0 / SMPS mode: analog output. External Pass Device gate control for EVRC
W17	VGATE1P	O	—	DCDC P ch. MOSFET gate driver output P32.1 / External Pass Device gate control for EVRC
M20	XTAL1	I	XTAL / VEXT	XTAL pad1 XTAL1. Main Oscillator/PLL/Clock Generator Input.
M19	XTAL2	O	XTAL / VEXT	XTAL pad2 XTAL2. Main Oscillator/PLL/Clock Generator OUTPUT
K14	DAPE0	I	FAST / PD2 / VEXT	DAPE: DAPE0 Clock Input DAPE: DAPE0 clock input
L19	TRST	I	FAST /	JTAG Module Reset/Enable Input
	DAPE0	I	PU2 / VEXT	DAPE: DAPE0 Clock Input
K16	TMS	I	FAST /	JTAG Module State Machine Control Input
	DAP1	I/O	PD2 / VEXT	DAP: DAP1 Data I/O

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Table 2-17 System I/O (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
J16	TCK	I	FAST /	JTAG Module Clock Input
	DAP0	I	PD2 / VEXT	DAP: DAP0 Clock Input
G10	DAPE2	I/O	FAST / PD2 / VEXT	DAPE: DAPE2 Data I/O DAPE: DAPE2 Data I/O
G11	DAPE1	I/O	FAST / PD2 / VEXT	DAPE: DAPE1 Data I/O DAPE: DAPE1 Data I/O
G17	PORST	I/O	PORST / PD / VEXT	PORST pin Power On Reset Input. Additional strong PD in case of power fail.
F16	ESR0	I	FAST / OD / VEXT	ESR0 Port Pin input - can be used to trigger a reset or an NMI ESR0: External System Request Reset 0. Default configuration during and after reset is open-drain driver. The driver drives low during power-on reset. This is valid additionally after deactivation of PORST_N until the internal reset phase has finished. See also SCU chapter for details. Default after power-on can be different. See also SCU chapter 'Reset Control Unit' and SCU_IOCRR register description. PMS_EVRWUP: EVR Wakeup Pin
	PMS_ESR0WKP	I		ESR0 pin input
G16	ESR1	I	FAST / PU1 / VEXT	ESR1 Port Pin input - can be used to trigger a reset or an NMI ESR1: External System Request Reset 1. Default NMI function. See also SCU chapter for details. Default after power-on can be different. See also SCU chapter 'Reset Control Unit' and SCU_IOCRR register description. PMS_EVRWUP: EVR Wakeup Pin
	PMS_ESR1WKP	I		ESR1 pin input

TC33xEXT Pin Definition and Functions:LFBGA-292 Package Variant Pin

Table 2-18 Supply

Ball	Symbol	Ctrl.	Buffer Type	Function
Y5	VDDM	I	—	ADC Analog Power Supply (5V / 3.3V)
P9, P12, N9, N10, N11, N12, M7, M8, M10, M11, M13, M14, L8, L9, L10, L11, L12, L13, K8, K9, K10, K11, K12, K13, J7, J8, J10, J11, J13, J14, H9, H10, H11, H12, G9, G12	VSS	I	—	Digital Ground
P8, P13, N7, N14, E15, H14, D16, G13, G8, H7	VDD	I	—	Digital Core Power Supply (1.25V)
A2, B3, V19, W20	VEXT	I	—	External Power Supply (5V / 3.3V)
D5	VFLEX	I	—	Digital Power Supply for Flex Port Pads (5V / 3.3V)
B18, A19	VDDP3	I	—	Flash Power Supply (3.3V)
B2, D4, E5, T16, U17, W19, Y20, E16, D17, B19, A20	VSS	I	—	Digital Ground
Y4	VSSM	I	—	Analog Ground for VDDM
L20	VSS	I	—	Oscillator Ground, VSS(OSC)
Y6	VAREF1	I	—	Positive Analog Reference Voltage 1
Y7	VAGND1	I	—	Negative Analog Reference Voltage 1

TC33xEXT Pin Definition and Functions: LFBGA-292 Package Variant Pin

Table 2-18 Supply (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
A3, A4, A5, A6, A7, A8, A9, B1, B4, B5, B6, B7, B8, B9, L16, L17, M16, M17, N16, N17, P1, P4, P5, R1, R4, R16, R17, T1, T2, T4, T13, T14, T15, T19, T20, U1, U2, U13, U14, U15, U16, U20, V1, V2, V20, W1, W2, Y18, Y19	NC	I	—	Not connected. These pins are reserved for future extensions and shall not be connected externally
A1, Y1, U4	NC1	I	—	Not connected. These pins are not connected on package level and will not be used for future extensions
T11	VEVRSB	I	—	Standby Power Supply (5V / 3.3V) for the Standby SRAM
N19	VDD	I	—	Digital Power Supply for Oscillator (1.25V), VDD(OSC)
N20	VEXT	I	—	Digital Power Supply for Oscillator (shall be supplied with same level as used for VEXT), VEXT(OSC)

2.2 LFBGA-180 Package Variant Pin Configuration of TC33xEXT for feature package DA

Table 2-19 Port 00 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
M2	P00.0	I	FAST / PU1 / VEXT / ES	General-purpose input
	CCU61_CTRAPA			Trap input capture
	CCU60_T12HRE			External timer start 12
	GETH_MDIOA			MDIO Input
	P00.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASCLK	O2		Shift clock output
	ASCLIN3_ATX	O3		Transmit output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT63	O7		T13 PWM channel 63
	GETH_MDIO	O		MDIO Output

Table 2-20 Port 02 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
K1	P02.0	I	FAST / PU1 / VEXT / ES	General-purpose input
	CCU61_CC60INB			T12 capture input 60
	ASCLIN2_ARXG			Receive input
	CCU60_CC60INA			T12 capture input 60
	SCU_E_REQ3_2			ERU Channel 3 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P02.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ATX	O2		Transmit output
	QSPI3_SLSO1	O3		Master slave select output
	—	O4		Reserved
	CAN00_TXD	O5		CAN transmit output node 0
	—	O6		Reserved
	CCU60_CC60	O7		T12 PWM channel 60

TC33xEXT Pin Definition and Functions: LFBGA-180 Package Variant Pin

Table 2-20 Port 02 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
H4	P02.1	I	SLOW / PU1 / VEXT / ES	General-purpose input
	ASCLIN2_ARXB			Receive input
	CAN00_RXDA			CAN receive input node 0
	SCU_E_REQ2_1			ERU Channel 2 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P02.1	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI3_SLSO2	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT60	O7		T12 PWM channel 60
J3	P02.2	I	FAST / PU1 / VEXT / ES	General-purpose input
	CCU61_CC61INB			T12 capture input 61
	CCU60_CC61INA			T12 capture input 61
	P02.2			General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	QSPI3_SLSO3	O3		Master slave select output
	—	O4		Reserved
	CAN02_TXD	O5		CAN transmit output node 2
	—	O6		Reserved
	CCU60_CC61	O7		T12 PWM channel 61
M1	P02.3	I	SLOW / PU1 / VEXT / ES	General-purpose input
	CAN02_RXDB			CAN receive input node 2
	ASCLIN1_ARXG			Receive input
	P02.3			General-purpose output
	—	O1		Reserved
	ASCLIN2_ASLSO	O2		Slave select signal output
	QSPI3_SLSO4	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT61	O7		T12 PWM channel 61

TC33xEXT Pin Definition and Functions: LFBGA-180 Package Variant Pin

Table 2-20 Port 02 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
J1	P02.4	I	FAST / PU1 / VEXT / ES	General-purpose input
	CCU61_CC62INB			T12 capture input 62
	QSPI3_SLSIA			Slave select input
	CCU60_CC62INA			T12 capture input 62
	P02.4	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ASCLK	O2		Shift clock output
	QSPI3_SLSO0	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_CC62	O7		T12 PWM channel 62
J2	P02.5	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI3_MRSTA			Master SPI data input
	SENT_SENT3C			Receive input channel 3
	P02.5	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI3_MRST	O3		Slave SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT62	O7		T12 PWM channel 62

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-20 Port 02 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
L2	P02.6	I	FAST / PU1 / VEXT / ES	General-purpose input
	CCU60_CC60INC			T12 capture input 60
	SENT_SENT2C			Receive input channel 2
	GPT120_T3INA			Trigger/gate input of core timer T3
	CCU60_CCPOS0A			Hall capture input 0
	CCU61_T12HRB			External timer start 12
	QSPI3_MTSRA			Slave SPI data input
	RIFO_RAMP1B			External RAMP B input
	P02.6	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI3_MTSR	O3		Master SPI data output
	—	O4		Reserved
	EVADC_EMUX00	O5		Control of external analog multiplexer interface 0
	—	O6		Reserved
	CCU60_CC60	O7		T12 PWM channel 60
K2	P02.7	I	FAST / PU1 / VEXT / ES	General-purpose input
	CCU60_CC61INC			T12 capture input 61
	SENT_SENT1C			Receive input channel 1
	GPT120_T3EUDA			Count direction control input of core timer T3
	CCU60_CCPOS1A			Hall capture input 1
	QSPI3_SCLKA			Slave SPI clock inputs
	CCU61_T13HRB			External timer start 13
	P02.7	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI3_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	EVADC_EMUX01	O5		Control of external analog multiplexer interface 0
	SENT_SPC1	O6		Transmit output
	CCU60_CC61	O7		T12 PWM channel 61

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-20 Port 02 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
L1	P02.8	I	SLOW / PU1 / VEXT / ES	General-purpose input
	CCU60_CC62INC			T12 capture input 62
	SENT_SENT0C			Receive input channel 0
	CCU60_CCPOS2A			Hall capture input 2
	GPT120_T4INA			Trigger/gate input of timer T4
	CCU61_T12HRC			External timer start 12
	CCU61_T13HRC			External timer start 13
	P02.8	O0		General-purpose output
	—	O1		Reserved
	QSPI3_SLSO5	O2		Master slave select output
	—	O3		Reserved
	—	O4		Reserved
	EVADC_EMUX02	O5		Control of external analog multiplexer interface 0
	GETH_MDC	O6		MDIO clock
	CCU60_CC62	O7		T12 PWM channel 62

Table 2-21 Port 10 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
E3	P10.1	I	FAST / PU1 / VEXT / ES	General-purpose input
	GPT120_T5EUDB			Count direction control input of timer T5
	QSPI1_MRSTA			Master SPI data input
	P10.1	O0		General-purpose output
	—	O1		Reserved
	QSPI1_MTSR	O2		Master SPI data output
	QSPI1_MRST	O3		Slave SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-21 Port 10 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
F3	P10.2	I	FAST / PU1 / VEXT / ES	General-purpose input
	CAN02_RXDE			CAN receive input node 2
	QSPI1_SCLKA			Slave SPI clock inputs
	GPT120_T6INB			Trigger/gate input of core timer T6
	SCU_E_REQ2_0			ERU Channel 2 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P10.2	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI1_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
G4	P10.3	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI1_MTSRA			Slave SPI data input
	SCU_E_REQ3_0			ERU Channel 3 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	GPT120_T5INB			Trigger/gate input of timer T5
	P10.3	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI1_MTSR	O3		Master SPI data output
	—	O4		Reserved
	—	O5		Reserved
	CAN02_TXD	O6		CAN transmit output node 2
	—	O7		Reserved
F4	P10.5	I	SLOW / PU2 / VEXT / ES	General-purpose input
	PMS_HWC4IN			HWC4 pin input
	P10.5	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ATX	O2		Transmit output
	QSPI3_SLSO8	O3		Master slave select output
	QSPI1_SLSO9	O4		Master slave select output
	GPT120_T6OUT	O5		External output for overflow/underflow detection of core timer T6
	ASCLIN2_ASLSO	O6		Slave select signal output
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-21 Port 10 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
E4	P10.6	I	SLOW / PU2 / VEXT / ES	General-purpose input
	ASCLIN2_ARXD			Receive input
	QSPI3_MTSRB			Slave SPI data input
	PMS_HWCFG5IN			HWCFG5 pin input
	P10.6	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ASCLK	O2		Shift clock output
	QSPI3_MTSR	O3		Master SPI data output
	GPT120_T3OUT	O4		External output for overflow/underflow detection of core timer T3
	—	O5		Reserved
	QSPI1_MRST	O6		Slave SPI data output
	—	O7		Reserved
H3	P10.7	I	SLOW / PU1 / VEXT / ES	General-purpose input
	GPT120_T3EUDB			Count direction control input of core timer T3
	ASCLIN2_ACTSA			Clear to send input
	QSPI3_MRSTB			Master SPI data input
	SCU_E_REQ0_2			ERU Channel 0 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	CCU60_CCPOS1C			Hall capture input 1
	P10.7	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI3_MRST	O3		Slave SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions: LFBGA-180 Package Variant Pin

Table 2-21 Port 10 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
G3	P10.8	I	SLOW / PU1 / VEXT / ES	General-purpose input
	GPT120_T4INB			Trigger/gate input of timer T4
	QSPI3_SCLKB			Slave SPI clock inputs
	SCU_E_REQ1_2			ERU Channel 1 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	CCU60_CCPOS2C			Hall capture input 2
	P10.8	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ARTS	O2		Ready to send output
	QSPI3_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

Table 2-22 Port 11 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
A7	P11.0	I	RFAST / PU1 / VFLEX / ES	General-purpose input
	ASCLIN3_ARXB			Receive input
	P11.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	GETH_TXD3	O6		Transmit Data
	—	O7		Reserved
B7	P11.1	I	RFAST / PU1 / VFLEX / ES	General-purpose input
	P11.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASCLK	O2		Shift clock output
	ASCLIN3_ATX	O3		Transmit output
	—	O4		Reserved
	—	O5		Reserved
	GETH_TXD2	O6		Transmit Data
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-22 Port 11 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
A6	P11.2	I	RFAST / PU1 / VFLEX / ES	General-purpose input
	P11.2	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPIO_SLSO5	O3		Master slave select output
	QSPI1_SLSO5	O4		Master slave select output
	—	O5		Reserved
	GETH_TXD1	O6		Transmit Data
	CCU60_COUT63	O7		T13 PWM channel 63
C5	P11.3	I	RFAST / PU1 / VFLEX / ES	General-purpose input
	QSPI1_MRSTB			Master SPI data input
	P11.3	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI1_MRST	O3		Slave SPI data output
	—	O4		Reserved
	—	O5		Reserved
	GETH_TXD0	O6		Transmit Data
	CCU60_COUT62	O7		T12 PWM channel 62
B6	P11.4	I	RFAST / PU1 / VFLEX / ES	General-purpose input
	GETH_RXCLKB			Receive Clock MII
	P11.4	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASCLK	O2		Shift clock output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	GETH_TXER	O6		Transmit Error MII
	GETH_TXCLK	O7		Transmit Clock Output for RGMII

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-22 Port 11 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
B5	P11.5	I	SLOW / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	GETH_TXCLKA			Transmit Clock Input for MII
	GETH_GREFCLK			Gigabit Reference Clock input for RGMII (125 MHz high precision)
	P11.5	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
A5	P11.6	I	RFAST / PU1 / VFLEX / ES	General-purpose input
	QSPI1_SCLKB			Slave SPI clock inputs
	P11.6	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI1_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	—	O5		Reserved
	GETH_TXEN	O6		Transmit Enable MII and RMII
	GETH_TCTL			Transmit Control for RGMII
	CCU60_COUT61	O7		T12 PWM channel 61
B4	P11.7	I	SLOW / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	GETH_RXD3A			Receive Data 3 MII and RGMII (RGMII can use RXD3A only)
	P11.7	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-22 Port 11 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
B3	P11.8	I	SLOW / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	GETH_RXD2A			Receive Data 2 MII and RGMII (RGMII can use RXD2A only)
	P11.8	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
A4	P11.9	I	FAST / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	QSPI1_MTSRB			Slave SPI data input
	GETH_RXD1A			Receive Data 1 MII, RMII and RGMII (RGMII can use RXD1A only)
	P11.9	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI1_MTSR	O3		Master SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT60	O7		T12 PWM channel 60

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-22 Port 11 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
A3	P11.10	I	FAST / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	CAN03_RXDD			CAN receive input node 3
	ASCLIN1_ARXE			Receive input
	SCU_E_REQ6_3			ERU Channel 6 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	GETH_RXD0A			Receive Data 0 MII, RMII and RGMII (RGMII can use RXD0A only)
	QSPI1_SLSIA			Slave select input
	P11.10	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI0_SLSO3	O3		Master slave select output
	QSPI1_SLSO3	O4		Master slave select output
	—	O5		Reserved
	—	O6		Reserved
	CCU60_CC62	O7		T12 PWM channel 62
D6	P11.11	I	FAST / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	GETH_CRSDVA			Carrier Sense / Data Valid combi-signal for RMII
	GETH_RXDVA			Receive Data Valid MII
	GETH_CRSB			Carrier Sense MII
	GETH_RCTLA			Receive Control for RGMII
	P11.11	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI0_SLSO4	O3		Master slave select output
	QSPI1_SLSO4	O4		Master slave select output
	—	O5		Reserved
	—	O6		Reserved
	CCU60_CC61	O7		T12 PWM channel 61

TC33xEXT Pin Definition and Functions: LFBGA-180 Package Variant Pin

Table 2-22 Port 11 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
C4	P11.12	I	FAST / RGMII_In put / PU1 / VFLEX / ES	General-purpose input
	GETH_REFCLKA			Reference Clock input for RMII (50 MHz)
	GETH_TXCLKB			Transmit Clock Input for MII
	GETH_RXCLKA			Receive Clock MII
	P11.12	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	CAN03_TXD	O5		CAN transmit output node 3
	CCU_EXTCLK1	O6		External Clock 1
	CCU60_CC60	O7		T12 PWM channel 60
C7	P11.13	I	SLOW / PU1 / VFLEX / ES	General-purpose input
	GETH_RXERA			Receive Error MII
	P11.13	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
B8	P11.14	I	SLOW / PU1 / VFLEX / ES	General-purpose input
	GETH_CRSDVB			Carrier Sense / Data Valid combi-signal for RMII
	GETH_RXDVB			Receive Data Valid MII
	GETH_CRSA			Carrier Sense MII
	P11.14	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions: LFBGA-180 Package Variant Pin

Table 2-22 Port 11 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
C6	P11.15	I	SLOW / PU1 / VFLEX / ES	General-purpose input
	GETH_COLA			Collision MII
	P11.15	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	—			Reserved

Table 2-23 Port 12 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
C8	P12.0	I	SLOW / PU1 / VFLEX / ES	General-purpose input
	CAN00_RXDC			CAN receive input node 0
	GETH_RXCLKC			Receive Clock MII
	P12.0	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	GETH_MDC	O6		MDIO clock
	—	O7		Reserved
A8	P12.1	I	SLOW / PU1 / VFLEX / ES	General-purpose input
	GETH_MDIOC			MDIO Input
	P12.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASLSO	O2		Slave select signal output
	—	O3		Reserved
	—	O4		Reserved
	CAN00_TXD	O5		CAN transmit output node 0
	—	O6		Reserved
	—	O7		Reserved
	GETH_MDIO	O		MDIO Output

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-24 Port 14 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
A9	P14.0	I	FAST / PU1 / VEXT / ES2	General-purpose input
	P14.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN0_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	CAN01_TXD	O5		CAN transmit output node 1
	ASCLIN0_ASCLK	O6		Shift clock output
	CCU60_COUT62	O7		T12 PWM channel 62
B10	P14.1	I	FAST / PU1 / VEXT / ES2	General-purpose input
	ASCLIN0_ARXA			Receive input
	CAN01_RXDB			CAN receive input node 1
	SCU_E_REQ3_1			ERU Channel 3 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	PMS_PINAWKP			PINA (P14.1) pin input
	P14.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN0_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_COUT63	O7		T13 PWM channel 63
D9	P14.2	I	SLOW / PU2 / VEXT / ES	General-purpose input
	PMS_HWCFG2IN			HWCFG2 pin input
	P14.2	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ATX	O2		Transmit output
	QSPI2_SLSO1	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	ASCLIN2_ASCLK	O6		Shift clock output
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-24 Port 14 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
C11	P14.3	I	SLOW / PU2 / VEXT / ES	General-purpose input
	PMS_HWCFG3IN			HWCFG3 pin input
	ASCLIN2_ARXA			Receive input
	SCU_E_REQ1_0			ERU Channel 1 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P14.3	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ATX	O2		Transmit output
	QSPI2_SLSO3	O3		Master slave select output
	ASCLIN1_ASLSO	O4		Slave select signal output
	ASCLIN3_ASLSO	O5		Slave select signal output
	—	O6		Reserved
	—	O7		Reserved
C10	P14.4	I	SLOW / PU2 / VEXT / ES	General-purpose input
	PMS_HWCFG6IN			HWCFG6 pin input
	P14.4	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	GETH_PPS	O6		Pulse Per Second
	—	O7		Reserved
	—	—		—
B9	P14.5	I	FAST / PU2 / VEXT / ES	General-purpose input
	PMS_HWCFG1IN			HWCFG1 pin input
	P14.5	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	—	—		—

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-24 Port 14 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
C9	P14.6	I	FAST / PU1 / VEXT / ES	General-purpose input
	P14.6	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPI2_SLSO2	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved

Table 2-25 Port 15 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
A13	P15.0	I	FAST / PU1 / VEXT / ES	General-purpose input
	SDMMC0_DAT7_IN			read data in
	P15.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	QSPI0_SLSO13	O3		Master slave select output
	—	O4		Reserved
	CAN02_TXD	O5		CAN transmit output node 2
	ASCLIN1_ASCLK	O6		Shift clock output
	—	O7		Reserved
	SDMMC0_DAT7	O		write data out
A12	P15.1	I	FAST / PU1 / VEXT / ES	General-purpose input
	CAN02_RXDA			CAN receive input node 2
	ASCLIN1_ARXA			Receive input
	QSPI2_SLSIB			Slave select input
	SCU_E_REQ7_2			ERU Channel 7 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P15.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	QSPI2_SLSO5	O3		Master slave select output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	SDMMC0_CLK	O7		card clock

TC33xEXT Pin Definition and Functions: LFBGA-180 Package Variant Pin

Table 2-25 Port 15 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
B12	P15.2	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI2_SLSIA			Slave select input
	QSPI2_MRSTE			Master SPI data input
	QSPI2_HSICINA			Highspeed capture channel
	P15.2	O0		General-purpose output
	—	O1		Reserved
	ASCLIN0_ATX	O2		Transmit output
	QSPI2_SLSO0	O3		Master slave select output
	—	O4		Reserved
	CAN01_TXD	O5		CAN transmit output node 1
	ASCLIN0_ASCLK	O6		Shift clock output
	—	O7		Reserved
A10	P15.3	I	FAST / PU1 / VEXT / ES	General-purpose input
	CAN01_RXDA			CAN receive input node 1
	ASCLIN0_ARXB			Receive input
	QSPI2_SCLKA			Slave SPI clock inputs
	QSPI2_HSICINB			Highspeed capture channel
	SDMMC0_CMD_IN			command in
	P15.3	O0		General-purpose output
	—	O1		Reserved
	ASCLIN0_ATX	O2		Transmit output
	QSPI2_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	SDMMC0_CMD	O		command out

TC33xEXT Pin Definition and Functions: LFBGA-180 Package Variant Pin

Table 2-25 Port 15 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
B11	P15.4	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI2_MRSTA			Master SPI data input
	SCU_E_REQ0_0			ERU Channel 0 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P15.4	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	QSPI2_MRST	O3		Slave SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_CC62	O7		T12 PWM channel 62
D10	P15.5	I	FAST / PU1 / VEXT / ES	General-purpose input
	ASCLIN1_ARXB			Receive input
	QSPI2_MTSRA			Slave SPI data input
	SCU_E_REQ4_3			ERU Channel 4 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P15.5	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	QSPI2_MTSR	O3		Master SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU60_CC61	O7		T12 PWM channel 61
A11	P15.6	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPI2_MTSRB			Slave SPI data input
	P15.6	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ATX	O2		Transmit output
	QSPI2_MTSR	O3		Master SPI data output
	—	O4		Reserved
	QSPI2_SCLK	O5		Master SPI clock output
	ASCLIN3_ASCLK	O6		Shift clock output
	CCU60_CC60	O7		T12 PWM channel 60

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Table 2-26 Port 20 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
H13	P20.0	I	FAST / PU1 / VEXT / ES	General-purpose input
	CAN03_RXDC			CAN receive input node 3
	CCU_PAD_SYSCLK			Sysclk input
	CBS_TG10			Trigger input
	SCU_E_REQ6_0			ERU Channel 6 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	GPT120_T6EUDA			Count direction control input of core timer T6
	P20.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ATX	O2		Transmit output
	ASCLIN3_ASCLK	O3		Shift clock output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	CBS_TGO0	O		Trigger output
G13	P20.2	I	S / PU / VEXT	General-purpose input This pin is latched at power on reset release to enter test mode.
	$\overline{\text{TESTMODE}}$			Testmode Enable Input
G12	P20.3	I	SLOW / PU1 / VEXT / ES	General-purpose input
	ASCLIN3_ARXC			Receive input
	GPT120_T6INA			Trigger/gate input of core timer T6
	P20.3	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ATX	O2		Transmit output
	QSPIO_SLSO9	O3		Master slave select output
	QSPI2_SLSO9	O4		Master slave select output
	CAN03_TXD	O5		CAN transmit output node 3
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-26 Port 20 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
F11	P20.6	I	SLOW / PU1 / VEXT / ES	General-purpose input
	P20.6	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ARTS	O2		Ready to send output
	QSPI0_SLSO8	O3		Master slave select output
	QSPI2_SLSO8	O4		Master slave select output
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
E12	P20.7	I	FAST / PU1 / VEXT / ES	General-purpose input
	CAN00_RXDB			CAN receive input node 0
	ASCLIN1_ACTSA			Clear to send input
	SDMMC0_DAT0_IN			read data in
	P20.7	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU61_COUT63	O7		T13 PWM channel 63
	SDMMC0_DAT0	O		write data out
F12	P20.8	I	FAST / PU1 / VEXT / ES	General-purpose input
	SDMMC0_DAT1_IN			read data in
	P20.8	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ASLSO	O2		Slave select signal output
	QSPI0_SLSO0	O3		Master slave select output
	QSPI1_SLSO0	O4		Master slave select output
	CAN00_TXD	O5		CAN transmit output node 0
	—	O6		Reserved
	CCU61_CC60	O7		T12 PWM channel 60
	SDMMC0_DAT1	O		write data out

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-26 Port 20 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
D12	P20.9	I	FAST / PU1 / VEXT / ES	General-purpose input
	CAN03_RXDE			CAN receive input node 3
	ASCLIN1_ARXC			Receive input
	QSPIO_SLSIB			Slave select input
	SCU_E_REQ7_0			ERU Channel 7 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P20.9	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPIO_SLSO1	O3		Master slave select output
	QSPI1_SLSO1	O4		Master slave select output
	—	O5		Reserved
	—	O6		Reserved
	CCU61_CC61	O7		T12 PWM channel 61
C14	P20.10	I	FAST / PU1 / VEXT / ES	General-purpose input
	SDMMC0_DAT2_IN			read data in
	P20.10	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	QSPIO_SLSO6	O3		Master slave select output
	QSPI2_SLSO7	O4		Master slave select output
	CAN03_TXD	O5		CAN transmit output node 3
	ASCLIN1_ASCLK	O6		Shift clock output
	CCU61_CC62	O7		T12 PWM channel 62
	SDMMC0_DAT2	O		write data out
D14	P20.11	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPIO_SCLKA			Slave SPI clock inputs
	SDMMC0_DAT3_IN			read data in
	P20.11	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPIO_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU61_COUT60	O7		T12 PWM channel 60
	SDMMC0_DAT3	O		write data out

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-26 Port 20 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
D13	P20.12	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPIO_MRSTA			Master SPI data input
	SDMMC0_DAT4_IN			read data in
	P20.12	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPIO_MRST	O3		Slave SPI data output
	QSPIO_MTSR	O4		Master SPI data output
	—	O5		Reserved
	—	O6		Reserved
	CCU61_COUT61	O7		T12 PWM channel 61
	SDMMC0_DAT4	O		write data out
C13	P20.13	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPIO_SLSIA			Slave select input
	SDMMC0_DAT5_IN			read data in
	P20.13	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPIO_SLSO2	O3		Master slave select output
	QSPIO1_SLSO2	O4		Master slave select output
	QSPIO_SCLK	O5		Master SPI clock output
	—	O6		Reserved
	CCU61_COUT62	O7		T12 PWM channel 62
	SDMMC0_DAT5	O		write data out
B14	P20.14	I	FAST / PU1 / VEXT / ES	General-purpose input
	QSPIO_MTSRA			Slave SPI data input
	SDMMC0_DAT6_IN			read data in
	DMU_FDEST			Enter destructive debug mode
	P20.14	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	QSPIO_MTSR	O3		Master SPI data output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	SDMMC0_DAT6	O		write data out

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Table 2-27 Port 21 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
H12	P21.0	I	FAST / PU1 / VEXT / ES	General-purpose input
	P21.0	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	HSM_HSM1	O		Pin Output Value
K13	P21.2	I	FAST / PU1 / VEXT / ES	General-purpose input
	SCU_EMGSTOP_POR T_B			Emergency stop Port Pin B input request
	P21.2	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASLSO	O2		Slave select signal output
	—	O3		Reserved
	—	O4		Reserved
	GETH_MDC	O5		MDIO clock
	—	O6		Reserved
	—	O7		Reserved
J14	P21.3	I	FAST / PU1 / VEXT / ES	General-purpose input
	GETH_MDIOD			MDIO Input
	P21.3	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
	GETH_MDIO	O		MDIO Output

TC33xEXT Pin Definition and Functions: LFBGA-180 Package Variant Pin

Table 2-27 Port 21 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
J13	P21.4	I	FAST / PU1 / VEXT / ES6	General-purpose input
	P21.4	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
H14	P21.5	I	FAST / PU1 / VEXT / ES6	General-purpose input
	P21.5	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASCLK	O2		Shift clock output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
F13	P21.6/TDI	I	FAST / PD / PU2 / VEXT / ES3	General-purpose input PD during Reset and in DAP/DAPE or JTAG mode. After Reset release and when not in DAP/DAPE or JTAG mode: PU. In Standby mode: HighZ.
	GPT120_T5EUDA			Count direction control input of timer T5
	ASCLIN3_ARXF			Receive input
	CBS_TGI2			Trigger input
	TDI			JTAG Module Data Input
	P21.6	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASLSO	O2		Slave select signal output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	GPT120_T3OUT	O7		External output for overflow/underflow detection of core timer T3
	CBS_TGO2	O		Trigger output
	DAP3	I/O		DAP: DAP3 Data I/O
	DAPE1	I/O		DAPE: DAPE1 Data I/O

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Table 2-27 Port 21 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
F14	P21.7/TDO	I	FAST / PU2 / VEXT / ES4	General-purpose input
	GPT120_T5INA			Trigger/gate input of timer T5
	CBS_TGI3			Trigger input
	GETH_RXERB			Receive Error MII
	P21.7	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ATX	O2		Transmit output
	ASCLIN3_ASCLK	O3		Shift clock output
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	GPT120_T6OUT	O7		External output for overflow/underflow detection of core timer T6
	CBS_TGO3	O		Trigger output
	DAP2	I/O		DAP: DAP2 Data I/O
	DAPE2	I/O		DAPE: DAPE2 Data I/O
	TDO	O		JTAG Module Data Output

Table 2-28 Port 23 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
N14	P23.1	I	FAST / PU1 / VEXT / ES	General-purpose input
	P23.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ARTS	O2		Ready to send output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	CCU_EXTCLK0	O6		External Clock 0
	—	O7		Reserved

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Table 2-29 Port 32 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
P12	P32.0	I	SLOW / PU1 / VEXT / ES	General-purpose input P32.0 / SMPS mode: analog output. External Pass Device gate control for EVRC
	P32.0	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
N12	P32.1	I	SLOW / PU1 / VEXT / ES	General-purpose input P32.1 / External Pass Device gate control for EVRC
	P32.1	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
P13	P32.4	I	FAST / PU1 / VEXT / ES	General-purpose input
	ASCLIN1_ACTSB			Clear to send input
	P32.4	O0		General-purpose output
	—	O1		Reserved
	—	O2		Reserved
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	CCU_EXTCLK1	O6		External Clock 1
	CCU60_COUT63	O7		T13 PWM channel 63
	PMS_DCDCSYNCO	O		DC-DC synchronization output

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Table 2-30 Port 33 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
M9	P33.0	I	SLOW / PU1 / VEVRSB / ES5	General-purpose input
	P33.0	O0		General-purpose output
	—	O1		Reserved
	ASCLIN5_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
M8	P33.1	I	SLOW / PU1 / VEVRSB / ES5	General-purpose input
	P33.1	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASLSO	O2		Slave select signal output
	QSPI2_SCLK	O3		Master SPI clock output
	—	O4		Reserved
	EVADC_EMUX02	O5		Control of external analog multiplexer interface 0
	—	O6		Reserved
	—	O7		Reserved
P8	P33.2	I	SLOW / PU1 / VEVRSB / ES5	General-purpose input
	P33.2	O0		General-purpose output
	—	O1		Reserved
	ASCLIN3_ASCLK	O2		Shift clock output
	QSPI2_SLSO10	O3		Master slave select output
	—	O4		Reserved
	EVADC_EMUX01	O5		Control of external analog multiplexer interface 0
	—	O6		Reserved
	—	O7		Reserved
N8	P33.3	I	SLOW / PU1 / VEVRSB / ES5	General-purpose input
	P33.3	O0		General-purpose output
	—	O1		Reserved
	ASCLIN5_ASCLK	O2		Shift clock output
	—	O3		Reserved
	—	O4		Reserved
	EVADC_EMUX00	O5		Control of external analog multiplexer interface 0
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-30 Port 33 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
L8	P33.4	I	SLOW / PU1 / VEVRSB / ES5	General-purpose input
	CCU61_CTRAPC			Trap input capture
	ASCLIN5_ARXB			Receive input
	P33.4	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ARTS	O2		Ready to send output
	QSPI2_SLSO12	O3		Master slave select output
	—	O4		Reserved
	EVADC_EMUX12	O5		Control of external analog multiplexer interface 1
	—	O6		Reserved
	—	O7		Reserved
N9	P33.5	I	SLOW / PU1 / VEVRSB / ES5	General-purpose input
	GPT120_T4EUDB			Count direction control input of timer T4
	ASCLIN2_ACTSB			Clear to send input
	CCU61_CCPOS2C			Hall capture input 2
	SENT_SENT5C			Receive input channel 5
	P33.5	O0		General-purpose output
	—	O1		Reserved
	QSPI0_SLSO7	O2		Master slave select output
	QSPI1_SLSO7	O3		Master slave select output
	—	O4		Reserved
	EVADC_EMUX11	O5		Control of external analog multiplexer interface 1
	—	O6		Reserved
	ASCLIN5_ASLSO	O7		Slave select signal output
P9	P33.6	I	SLOW / PU1 / VEVRSB / ES5	General-purpose input
	GPT120_T2EUDB			Count direction control input of timer T2
	SENT_SENT4C			Receive input channel 4
	CCU61_CCPOS1C			Hall capture input 1
	P33.6	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ASLSO	O2		Slave select signal output
	QSPI2_SLSO11	O3		Master slave select output
	—	O4		Reserved
	EVADC_EMUX10	O5		Control of external analog multiplexer interface 1
	—	O6		Reserved
	—	O7		Reserved

TC33xEXT Pin Definition and Functions: LFBGA-180 Package Variant Pin

Table 2-30 Port 33 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
L10	P33.7	I	SLOW / PU1 / VEVR SB / ES5	General-purpose input
	CAN00_RXDE			CAN receive input node 0
	GPT120_T2INB			Trigger/gate input of timer T2
	CCU61_CCPOS0C			Hall capture input 0
	SCU_E_REQ4_0			ERU Channel 4 inputs 0 to 5 (0 is the LSB and 5 is the MSB)
	P33.7	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ASCLK	O2		Shift clock output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	—	O7		Reserved
L9	P33.8	I	FAST / HighZ / VEVR SB	General-purpose input
	ASCLIN2_ARXE			Receive input
	SCU_EMGSTOP_PORT_A			Emergency stop Port Pin A input request
	P33.8	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ATX	O2		Transmit output
	—	O3		Reserved
	—	O4		Reserved
	CAN00_TXD	O5		CAN transmit output node 0
	—	O6		Reserved
	CCU61_COUT62	O7		T12 PWM channel 62
	SMU_FSP0	O		FSP[1..0] Output Signals - Generated by SMU_core
P10	P33.9	I	SLOW / PU1 / VEVR SB / ES5	General-purpose input
	QSPI3_HSIICINA			Highspeed capture channel
	P33.9	O0		General-purpose output
	—	O1		Reserved
	ASCLIN2_ATX	O2		Transmit output
	—	O3		Reserved
	ASCLIN2_ASCLK	O4		Shift clock output
	CAN01_TXD	O5		CAN transmit output node 1
	ASCLIN0_ATX	O6		Transmit output
	CCU61_CC62	O7		T12 PWM channel 62

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-30 Port 33 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
N11	P33.10	I	FAST / PU1 / VEVR SB / ES5	General-purpose input
	QSPI3_HSICINB			Highspeed capture channel
	CAN01_RXDD			CAN receive input node 1
	ASCLIN0_ARXD			Receive input
	P33.10	O0		General-purpose output
	—	O1		Reserved
	QSPI1_SLSO6	O2		Master slave select output
	—	O3		Reserved
	ASCLIN1_ASLSO	O4		Slave select signal output
	—	O5		Reserved
	—	O6		Reserved
	CCU61_COUT61	O7		T12 PWM channel 61
	SMU_FSP1	O		FSP[1..0] Output Signals - Generated by SMU_core
N10	P33.11	I	FAST / PU1 / VEVR SB / ES5	General-purpose input
	P33.11	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ASCLK	O2		Shift clock output
	—	O3		Reserved
	—	O4		Reserved
	—	O5		Reserved
	—	O6		Reserved
	CCU61_CC61	O7		T12 PWM channel 61
M10	P33.12	I	FAST / PU1 / VEVR SB / ES5	General-purpose input
	CAN00_RXDD			CAN receive input node 0
	PMS_PINBWKP			PINB (P33.12) pin input
	P33.12	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	—	O3		Reserved
	ASCLIN1_ASCLK	O4		Shift clock output
	—	O5		Reserved
	—	O6		Reserved
	CCU61_COUT60	O7		T12 PWM channel 60

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-30 Port 33 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
M11	P33.13	I	FAST / PU1 / VEVR SB / ES5	General-purpose input
	ASCLIN1_ARXF			Receive input
	P33.13	O0		General-purpose output
	—	O1		Reserved
	ASCLIN1_ATX	O2		Transmit output
	—	O3		Reserved
	QSPI2_SLSO6	O4		Master slave select output
	CAN00_TXD	O5		CAN transmit output node 0
	—	O6		Reserved
	CCU61_CC60	O7		T12 PWM channel 60

Table 2-31 Port 50 Functions

Ball	Symbol	Ctrl.	Buffer Type	Function
C1	P50.0	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D1N			LVDS RX Input (inverted Data Bits of Channel #0)
C2	P50.1	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D1P			LVDS RX Input (Data Bits of Channel #0)
D1	P50.2	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D2N			LVDS RX Input (inverted Data Bits of Channel #1)
D2	P50.3	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D2P			LVDS RX Input (Data Bits of Channel #1)
E1	P50.4	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_CLKN			LVDS RX Input (inverted Serial Clock)
E2	P50.5	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_CLKP			LVDS RX Input (Serial Clock)
F1	P50.6	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_FRP			LVDS RX Input (FrameClock)

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-31 Port 50 Functions (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
F2	P50.7	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_FRN			LVDS RX Input (inverted FrameClock)
G1	P50.8	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D3P			LVDS RX Input (Data Bits of Channel #2)
G2	P50.9	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D3N			LVDS RX Input (inverted Data Bits of Channel #2)
H1	P50.10	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D4P			LVDS RX Input (Data Bits of Channel #3)
H2	P50.11	I	LVDS_R X / HighZ / VEXT / ES	—
	RIF0_D4N			LVDS RX Input (inverted Data Bits of Channel #3)

Table 2-32 Analog Inputs

Ball	Symbol	Ctrl.	Buffer Type	Function
N6	AN0	I	D / HighZ / VDDM	Analog Input 0
	EVADC_G0CH0			Analog input channel 0, group 0
P6	AN1	I	D / HighZ / VDDM	Analog Input 1
	EVADC_G0CH1			Analog input channel 1, group 0
L6	AN2	I	D / HighZ / VDDM	Analog Input 2
	EVADC_G0CH2			Analog input channel 2, group 0
M6	AN3	I	D / HighZ / VDDM	Analog Input 3
	EVADC_G0CH3			Analog input channel 3, group 0
L5	AN4	I	D / HighZ / VDDM	Analog Input 4
	EVADC_G0CH4			Analog input channel 4, group 0
	EVADC_G5CH0			Analog input channel 0, group 5
M5	AN5	I	D / HighZ / VDDM	Analog Input 5
	EVADC_G0CH5			Analog input channel 5, group 0
	EVADC_G5CH1			Analog input channel 1, group 5
N5	AN8	I	D / HighZ / VDDM	Analog Input 8
	EVADC_G1CH0			Analog input channel 0, group 1
	EVADC_G5CH4			Analog input channel 4, group 5

TC33xEXT Pin Definition and Functions:LFBGA-180 Package Variant Pin

Table 2-32 Analog Inputs (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
L4	AN9	I	D / HighZ / VDDM	Analog Input 9
	EVADC_G1CH1			Analog input channel 1, group 1
	EVADC_G5CH5			Analog input channel 5, group 5
P5	AN10	I	D / HighZ / VDDM	Analog Input 10
	EVADC_G1CH2			Analog input channel 2, group 1
	EVADC_G5CH6			Analog input channel 6, group 5
N4	AN11	I	D / HighZ / VDDM	Analog Input 11
	EVADC_G1CH3			Analog input channel 3, group 1
	EVADC_G5CH7			Analog input channel 7, group 5
M4	AN12	I	D / HighZ / VDDM	Analog Input 12
	EVADC_G1CH4			Analog input channel 4, group 1
	EVADC_G4CH0			Analog input channel 0, group 4
N3	AN13	I	D / HighZ / VDDM	Analog Input 13
	EVADC_G1CH5			Analog input channel 5, group 1
	EVADC_G4CH1			Analog input channel 1, group 4
N2	AN14	I	D / HighZ / VDDM	Analog Input 14
	EVADC_G1CH6			Analog input channel 6, group 1
	EVADC_G4CH2			Analog input channel 2, group 4
N1	AN15	I	D / HighZ / VDDM	Analog Input 15
	EVADC_G1CH7			Analog input channel 7, group 1
	EVADC_G4CH3			Analog input channel 3, group 4
K4	AN16	I	D / HighZ / VDDM	Analog Input 16
	EVADC_G2CH0			Analog input channel 0, group 2
	EVADC_G4CH4			Analog input channel 4, group 4
J4	AN17	I	D / HighZ / VDDM	Analog Input 17
	EVADC_G2CH1			Analog input channel 1, group 2
	EVADC_G4CH5			Analog input channel 5, group 4
K3	AN20	I	D / HighZ / VDDM	Analog Input 20
	EVADC_G3CH0			Analog input channel 0, group 3
M3	AN21	I	D / HighZ / VDDM	Analog Input 21
	EVADC_G3CH1			Analog input channel 1, group 3

TC33xEXT Pin Definition and Functions: LFBGA-180 Package Variant Pin

Note: Port Pins P32.0 and P32.1 are bidirectional pads and are having the following two functionalities implemented:

3. In case the pins are used as standard GPIOs the functions defined in the pin configuration tables of P32.0 and P32.1 are available.
4. In case the pins are used as pre-drivers for external MOSFETs (internal DCDC usecase) P32.0 and P32.1 act as analog IOs named VGATE1N and VGATE1P.

Table 2-33 System I/O

Ball	Symbol	Ctrl.	Buffer Type	Function
P12	VGATE1N	O	—	DCDC N ch. MOSFET gate driver output P32.0 / SMPS mode: analog output. External Pass Device gate control for EVRC
N12	VGATE1P	O	—	DCDC P ch. MOSFET gate driver output P32.1 / External Pass Device gate control for EVRC
L14	XTAL1	I	XTAL / VEXT	XTAL pad1 XTAL1. Main Oscillator/PLL/Clock Generator Input.
L13	XTAL2	O	XTAL / VEXT	XTAL pad2 XTAL2. Main Oscillator/PLL/Clock Generator OUTPUT
K12	TRST	I	FAST /	JTAG Module Reset/Enable Input
	DAPE0	I	PU2 / VEXT	DAPE: DAPE0 Clock Input
G11	TMS	I	FAST /	JTAG Module State Machine Control Input
	DAP1	I/O	PD2 / VEXT	DAP: DAP1 Data I/O
G14	TCK	I	FAST /	JTAG Module Clock Input
	DAP0	I	PD2 / VEXT	DAP: DAP0 Clock Input
E13	PORST	I/O	PORST / PD / VEXT	PORST pin Power On Reset Input. Additional strong PD in case of power fail.
E14	ESR0	I	FAST / OD / VEXT	ESR0 Port Pin input - can be used to trigger a reset or an NMI ESR0: External System Request Reset 0. Default configuration during and after reset is open-drain driver. The driver drives low during power-on reset. This is valid additionally after deactivation of PORST_N until the internal reset phase has finished. See also SCU chapter for details. Default after power-on can be different. See also SCU chapter 'Reset Control Unit' and SCU_IOCRR register description. PMS_EVRWUP: EVR Wakeup Pin
	PMS_ESR0WKP	I		ESR0 pin input

TC33xEXT Pin Definition and Functions: LFBGA-180 Package Variant Pin

Table 2-33 System I/O (cont'd)

Ball	Symbol	Ctrl.	Buffer Type	Function
E11	ESR1	I	FAST / PU1 / VEXT	ESR1 Port Pin input - can be used to trigger a reset or an NMI ESR1: External System Request Reset 1. Default NMI function. See also SCU chapter for details. Default after power-on can be different. See also SCU chapter 'Reset Control Unit' and SCU_IOCRR register description. PMS_EVRWUP: EVR Wakeup Pin
	PMS_ESR1WKP	I		ESR1 pin input

Table 2-34 Supply

Ball	Symbol	Ctrl.	Buffer Type	Function
P3	VDDM	I	—	ADC Analog Power Supply (5V / 3.3V)
B13, C12, D11, E10, G7, G8, H7, H8, J6, K5	VSS	I	—	Digital Ground
F9, G9, H6, J7, J8	VDD	I	—	Digital Core Power Supply (1.25V)
F6, F8, G6, J9, P11	VEXT	I	—	External Power Supply (5V / 3.3V)
D5	VFLEX	I	—	Digital Power Supply for Flex Port Pads (5V / 3.3V)
F7	VDDP3	I	—	Flash Power Supply (3.3V)
B2, C3, D4, E5, K10, L11, M12, N13	VSS	I	—	Digital Ground
K14	VSS	I	—	Oscillator Ground, VSS(OSC)
P4	VAREF1	I	—	Positive Analog Reference Voltage 1
A1, A2, A14, B1, D3, D7, D8, H11, J11, J12, K11, L3, L7, L12, M7, N7, P1, P7, P14	NC	I	—	Not connected. These pins are reserved for future extensions and shall not be connected externally
H9	VEVRSB	I	—	Standby Power Supply (5V / 3.3V) for the Standby SRAM
M13	VDD	I	—	Digital Power Supply for Oscillator (1.25V), VDD(OSC)
M14	VEXT	I	—	Digital Power Supply for Oscillator (shall be supplied with same level as used for VEXT), VEXT(OSC)
P2	VSSM/VAGND1	I	—	Analog Ground for VDDM / Negative Analog Reference Voltage 1

2.3 Sequence of Pads in Pad Frame for feature package DA and DZ

Table 2-35 Pad List

Number	Pad Name	Pad Type	X	Y	Comment
1	P15.0	FAST / PU1 / VEXT / ES	255699	175644	General-purpose I/O
2	P15.1	FAST / PU1 / VEXT / ES	356697	175644	General-purpose I/O
3	P15.2	FAST / PU1 / VEXT / ES	425997	322614	General-purpose I/O
4	P15.3	FAST / PU1 / VEXT / ES	476397	175644	General-purpose I/O
5	P15.4	FAST / PU1 / VEXT / ES	526797	322614	General-purpose I/O
6	P15.5	FAST / PU1 / VEXT / ES	577197	175644	General-purpose I/O
7	P15.6	FAST / PU1 / VEXT / ES	627597	322614	General-purpose I/O
8	P15.7	FAST / PU1 / VEXT / ES	677997	175644	General-purpose I/O
9	P15.8	FAST / PU1 / VEXT / ES	724995	322614	General-purpose I/O
10	VSS	Vx	816993	175644	Supply Voltage
11	VDD	Vx	917991	322614	Supply Voltage
12	VSS	Vx	1009989	175644	Supply Voltage
13	VEXT	Vx	1056987	322614	Supply Voltage
14	P14.0	FAST / PU1 / VEXT / ES2	1103985	175644	General-purpose I/O
15	P14.1	FAST / PU1 / VEXT / ES2	1150983	322614	General-purpose I/O
16	P14.2	SLOW / PU2 / VEXT / ES	1197981	175644	General-purpose I/O
17	P14.3	SLOW / PU2 / VEXT / ES	1244979	322614	General-purpose I/O
18	P14.4	SLOW / PU2 / VEXT / ES	1291977	175644	General-purpose I/O
19	P14.5	FAST / PU2 / VEXT / ES	1338975	322614	General-purpose I/O
20	P14.6	FAST / PU1 / VEXT / ES	1385973	175644	General-purpose I/O
21	P14.7	SLOW / PU1 / VEXT / ES	1432971	322614	General-purpose I/O
22	P14.8	SLOW / PU1 / VEXT / ES	1479969	175644	General-purpose I/O

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for
Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
23	P14.9	SLOW / PU1 / VEXT / ES	1526967	322614	General-purpose I/O
24	VSS	Vx	1609965	175644	Supply Voltage
25	VDDP3	—	1674963	322614	Supply Voltage
26	VSS	Vx	1748961	175644	Supply Voltage
27	P14.10	FAST / PU1 / VEXT / ES	1827459	322614	General-purpose I/O
28	VSS	Vx	1874457	175644	Supply Voltage
29	VDD	Vx	1921455	322614	Supply Voltage
30	VDD	Vx	1997955	175644	Supply Voltage
31	VEXT	Vx	2179953	322614	Supply Voltage
32	VEXT	Vx	2253951	175644	Supply Voltage
33	VDD_EXT_IO	Vx	2327949	322614	Supply Voltage
34	VDDP3	Vx	2401947	175644	Supply Voltage
35	VDDP3	Vx	2475945	322614	Supply Voltage
36	VDDP3	Vx	2567943	175644	Supply Voltage
37	VDD	Vx	2641941	322614	Supply Voltage
38	VDD	Vx	2697939	175644	Supply Voltage
39	VDD	Vx	2762937	322614	Supply Voltage
40	VSS	Vx	2809935	175644	Supply Voltage
41	VSS	Vx	2928933	175644	Supply Voltage
42	P12.0	SLOW / PU1 / VFLEX / ES	3030147	322614	General-purpose I/O
43	P12.1	SLOW / PU1 / VFLEX / ES	3077145	175644	General-purpose I/O
44	P11.0	RFAST / PU1 / VFLEX / ES	3181149	322614	General-purpose I/O
45	VFLEX	Vx	3228147	175644	Supply Voltage
46	P11.1	RFAST / PU1 / VFLEX / ES	3303351	322614	General-purpose I/O
47	VSS	Vx	3350349	175644	Supply Voltage
48	P11.2	RFAST / PU1 / VFLEX / ES	3425553	322614	General-purpose I/O
49	VDD	Vx	3472551	175644	Supply Voltage
50	P11.4	RFAST / PU1 / VFLEX / ES	3588255	322614	General-purpose I/O
51	VSS	Vx	3675753	175644	Supply Voltage
52	P11.3	RFAST / PU1 / VFLEX / ES	3750957	322614	General-purpose I/O
53	VFLEX	Vx	3797955	175644	Supply Voltage

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for
Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
54	P11.6	RFAST / PU1 / VFLEX / ES	3873159	322614	General-purpose I/O
55	VSS	Vx	3920157	175644	Supply Voltage
56	P11.5	SLOW / RGMII_Input / PU1 / VFLEX / ES	3967155	322614	General-purpose I/O
57	P11.7	SLOW / RGMII_Input / PU1 / VFLEX / ES	4022253	175644	General-purpose I/O
58	P11.9	FAST / RGMII_Input / PU1 / VFLEX / ES	4072851	322614	General-purpose I/O
59	VFLEX	Vx	4123449	175644	Supply Voltage
60	P11.8	SLOW / RGMII_Input / PU1 / VFLEX / ES	4170447	322614	General-purpose I/O
61	P11.10	FAST / RGMII_Input / PU1 / VFLEX / ES	4221045	175644	General-purpose I/O
62	P11.11	FAST / RGMII_Input / PU1 / VFLEX / ES	4271643	322614	General-purpose I/O
63	VSS	Vx	4322241	175644	Supply Voltage
64	P11.12	FAST / RGMII_Input / PU1 / VFLEX / ES	4369239	322614	General-purpose I/O
65	VDD	Vx	4502637	322614	Supply Voltage
66	VSS	Vx	4603635	175644	Supply Voltage
67	VSS	—	4704633	175644	Supply Voltage
68	VDD	—	4796631	322614	Supply Voltage
69	P11.14	SLOW / PU1 / VFLEX / ES	4888629	175644	General-purpose I/O
70	P11.15	SLOW / PU1 / VFLEX / ES	4935627	322614	General-purpose I/O
71	P11.13	SLOW / PU1 / VFLEX / ES	4982625	175644	General-purpose I/O
72	VDD	Vx	5128839	322614	Supply Voltage

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for
Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
73	VSS	—	5247837	175644	Supply Voltage
74	VSS	Vx	5498496	255699	Supply Voltage
75	VEXT	Vx	5498496	356697	Supply Voltage
76	VSS	Vx	5498496	457695	Supply Voltage
77	VDD	Vx	5351526	522693	Supply Voltage
78	P50.0	LVDS_RX / HighZ / VEXT / ES	5498496	604692	General-purpose I/O
79	P50.1	LVDS_RX / HighZ / VEXT / ES	5498496	685692	General-purpose I/O
80	P50.2	LVDS_RX / HighZ / VEXT / ES	5498496	766692	General-purpose I/O
81	P50.3	LVDS_RX / HighZ / VEXT / ES	5498496	847692	General-purpose I/O
82	P50.4	LVDS_RX / HighZ / VEXT / ES	5498496	928692	General-purpose I/O
83	P50.5	LVDS_RX / HighZ / VEXT / ES	5498496	1009692	General-purpose I/O
84	P50.6	LVDS_RX / HighZ / VEXT / ES	5498496	1126692	General-purpose I/O
85	P50.7	LVDS_RX / HighZ / VEXT / ES	5498496	1207692	General-purpose I/O
86	P50.8	LVDS_RX / HighZ / VEXT / ES	5498496	1288692	General-purpose I/O
87	P50.9	LVDS_RX / HighZ / VEXT / ES	5498496	1369692	General-purpose I/O
88	P50.10	LVDS_RX / HighZ / VEXT / ES	5498496	1450692	General-purpose I/O
89	P50.11	LVDS_RX / HighZ / VEXT / ES	5498496	1531692	General-purpose I/O
90	VEXT	Vx	5351526	1595691	Supply Voltage
91	VSS	Vx	5498496	1642689	Supply Voltage

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for
Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
92	VDD	Vx	5351526	1752687	Supply Voltage
93	VSS	Vx	5498496	1853685	Supply Voltage
94	VDD	Vx	5351526	1954683	Supply Voltage
95	VSS	Vx	5498496	2055681	Supply Voltage
96	P10.0	FAST / PU1 / VEXT / ES	5351526	2156679	General-purpose I/O
97	P10.1	FAST / PU1 / VEXT / ES	5498496	2203677	General-purpose I/O
98	P10.6	SLOW / PU2 / VEXT / ES	5351526	2250675	General-purpose I/O
99	P10.4	FAST / PU1 / VEXT / ES	5498496	2297673	General-purpose I/O
100	P10.5	SLOW / PU2 / VEXT / ES	5351526	2344671	General-purpose I/O
101	P10.3	FAST / PU1 / VEXT / ES	5498496	2391669	General-purpose I/O
102	P10.8	SLOW / PU1 / VEXT / ES	5351526	2438667	General-purpose I/O
103	P10.2	FAST / PU1 / VEXT / ES	5498496	2485665	General-purpose I/O
104	P10.7	SLOW / PU1 / VEXT / ES	5351526	2532663	General-purpose I/O
105	VEXT	Vx	5498496	2579661	Supply Voltage
106	P02.0	FAST / PU1 / VEXT / ES	5351526	2626659	General-purpose I/O
107	VSS	Vx	5498496	2673657	Supply Voltage
108	VDD	Vx	5351526	2758455	Supply Voltage
109	VSS	Vx	5498496	2848653	Supply Voltage
110	P02.4	FAST / PU1 / VEXT / ES	5351526	2904651	General-purpose I/O
111	P02.1	SLOW / PU1 / VEXT / ES	5498496	2951649	General-purpose I/O
112	P02.5	FAST / PU1 / VEXT / ES	5351526	2998647	General-purpose I/O
113	P02.2	FAST / PU1 / VEXT / ES	5498496	3045645	General-purpose I/O
114	P02.3	SLOW / PU1 / VEXT / ES	5351526	3092643	General-purpose I/O
115	P02.7	FAST / PU1 / VEXT / ES	5498496	3139641	General-purpose I/O
116	P02.6	FAST / PU1 / VEXT / ES	5351526	3186639	General-purpose I/O

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for
Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
117	P02.8	SLOW / PU1 / VEXT / ES	5498496	3233637	General-purpose I/O
118	P00.0	FAST / PU1 / VEXT / ES	5351526	3280635	General-purpose I/O
119	VEXT	Vx	5498496	3327633	Supply Voltage
120	P00.1	SLOW / PU1 / VEXT / ES	5351526	3374631	General-purpose I/O
121	VSS	Vx	5498496	3421629	Supply Voltage
122	VDD	Vx	5351526	3486627	Supply Voltage
123	RESERVED	Vx	5498496	3533625	OTPMust be bonded to VSS
124	P00.2	SLOW / PU1 / VEXT / ES1	5351526	3580623	General-purpose I/O
125	VSS	Vx	5498496	3645621	Supply Voltage
126	P00.7	SLOW / PU1 / VEXT / ES1	5351526	3710619	General-purpose I/O
127	P00.3	SLOW / PU1 / VEXT / ES1	5498496	3757617	General-purpose I/O
128	P00.8	SLOW / PU1 / VEXT / ES1	5351526	3804615	General-purpose I/O
129	VSS	—	5498496	3851613	Supply Voltage
130	VEXT	—	5351526	3898611	Supply Voltage
131	P00.4	SLOW / PU1 / VEXT / ES1	5498496	3945609	General-purpose I/O
132	P00.9	SLOW / PU1 / VEXT / ES1	5351526	3992607	General-purpose I/O
133	P00.5	SLOW / PU1 / VEXT / ES1	5498496	4039605	General-purpose I/O
134	P00.12	SLOW / PU1 / VEXT / ES1	5351526	4086603	General-purpose I/O
135	P00.6	SLOW / PU1 / VEXT / ES1	5498496	4133601	General-purpose I/O
136	VDD	—	5351526	4180599	Supply Voltage
137	VSS	—	5498496	4245597	Supply Voltage
138	AN23	D / HighZ / VDDM	5351526	4383022	Analog Input 23
139	AN22	D / HighZ / VDDM	5498496	4430020	Analog Input 22
140	AN21	D / HighZ / VDDM	5351526	4477018	Analog Input 21
141	AN20	D / HighZ / VDDM	5498496	4527018	Analog Input 20

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for
Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
142	AN19	D / HighZ / VDDM	5351526	4577018	Analog Input 19
143	AN18	D / HighZ / VDDM	5498496	4627017	Analog Input 18
144	AN17	D / HighZ / VDDM	5351526	4677016	Analog Input 17
145	AN16	D / HighZ / VDDM	5498496	4727016	Analog Input 16
146	AN15	D / HighZ / VDDM	5351526	4777016	Analog Input 15
147	AN14	D / HighZ / VDDM	5498496	4827015	Analog Input 14
148	AN13	D / HighZ / VDDM	5498496	4927014	Analog Input 13
149	AN12	D / HighZ / VDDM	5426514	4998996	Analog Input 12
150	AN11	D / HighZ / VDDM	5326515	4998996	Analog Input 11
151	AN10	D / HighZ / VDDM	5226516	4998996	Analog Input 10
152	AN9	D / HighZ / VDDM	5126517	4998996	Analog Input 9
153	AN8	D / HighZ / VDDM	5076518	4852026	Analog Input 8
154	AN7	D / HighZ / VDDM	5026518	4998996	Analog Input 7
155	AN6	D / HighZ / VDDM	4976518	4852026	Analog Input 6
156	VAGND0	Vx	4926519	4998996	Supply Voltage
157	VAREF0	Vx	4876520	4852026	Supply Voltage
158	VSSM	Vx	4829522	4998996	Supply Voltage
159	VDDM	Vx	4782524	4852026	Supply Voltage
160	VSSM	Vx	4735526	4998996	Supply Voltage
161	VDDM	Vx	4688528	4852026	Supply Voltage
162	AN5	D / HighZ / VDDM	4641530	4998996	Analog Input 5
163	AN4	D / HighZ / VDDM	4594532	4852026	Analog Input 4
164	AN3	D / HighZ / VDDM	4547534	4998996	Analog Input 3
165	AN2	D / HighZ / VDDM	4500536	4852026	Analog Input 2

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for
Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
166	AN1	D / HighZ / VDDM	4453538	4998996	Analog Input 1
167	AN0	D / HighZ / VDDM	4406540	4852026	Analog Input 0
168	VSS	Vx	4278141	4998996	Supply Voltage
169	VDD	Vx	4177143	4852026	Supply Voltage
170	AGBTCLKN (VSS)	AGBT_CLK / VEXT	4023963	4999041	Input PAD (negative pole) for the external 100 MHz differential clock. AGBT Input
171	AGBTCLKP (VSS)	AGBT_CLK / VEXT	3942873	4999041	Input PAD (positive pole) for the external 100 MHz differential clock. AGBT Input
172	VEXT	Vx	3861963	4989536	Supply Voltage
173	VSS	Vx	3772503	4989537	Supply Voltage
174	AGBTTXN (VSS)	AGBT_TX / VEXT	3673031	4999041	Off-chip driver output PAD of the 2.5Gbps transmitter, negative pole AGBT Output
175	AGBTTXP (VSS)	AGBT_TX / VEXT	3591941	4999041	Off-chip driver output PAD of the 2.5Gbps transmitter, positive pole AGBT Output
176	AGBTERR (VSS)	FAST / PD / VEXT	3401743	4998902	Input PAD for CRC error from FPGA. AGBT Input
177	VSS	Vx	3249022	4998996	Supply Voltage
178	VDD	Vx	3202024	4852026	Supply Voltage
179	VSS	Vx	3083026	4998996	Supply Voltage
180	VDD	Vx	3036028	4852026	Supply Voltage
181	VEVRSB	Vx	2962030	4998996	Supply Voltage
182	P33.1	SLOW / PU1 / VEVRSB / ES5	2915032	4852026	General-purpose I/O
183	P33.0	SLOW / PU1 / VEVRSB / ES5	2868034	4998996	General-purpose I/O
184	P33.2	SLOW / PU1 / VEVRSB / ES5	2821036	4852026	General-purpose I/O
185	VSS	Vx	2774038	4998996	Supply Voltage

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for
Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
186	P33.3	SLOW / PU1 / VEVRSB / ES5	2727040	4852026	General-purpose I/O
187	VSS	Vx	2626042	4998996	Supply Voltage
188	VDD	Vx	2579044	4852026	Supply Voltage
189	P33.5	SLOW / PU1 / VEVRSB / ES5	2473546	4852026	General-purpose I/O
190	P34.1	SLOW / PU1 / VEVRSB / ES5	2426548	4998996	General-purpose I/O
191	P33.4	SLOW / PU1 / VEVRSB / ES5	2379550	4852026	General-purpose I/O
192	P34.2	SLOW / PU1 / VEVRSB / ES	2332552	4998996	General-purpose I/O
193	P33.6	SLOW / PU1 / VEVRSB / ES5	2285554	4852026	General-purpose I/O
194	P34.3	SLOW / PU1 / VEVRSB / ES	2238556	4998996	General-purpose I/O
195	P33.7	SLOW / PU1 / VEVRSB / ES5	2191558	4852026	General-purpose I/O
196	P33.8	FAST / HighZ / VEVRSB	2144560	4998996	General-purpose I/O
197	P33.9	SLOW / PU1 / VEVRSB / ES5	2097562	4852026	General-purpose I/O
198	VSS	Vx	1987564	4998996	Supply Voltage
199	VDD	Vx	1940566	4852026	Supply Voltage
200	VEVRSB	Vx	1792570	4852026	Supply Voltage
201	VEVRSB	Vx	1745572	4998996	Supply Voltage
202	VSS	Vx	1653574	4998996	Supply Voltage
203	P33.10	FAST / PU1 / VEVRSB / ES5	1561576	4998996	General-purpose I/O
204	P33.11	FAST / PU1 / VEVRSB / ES5	1469578	4998996	General-purpose I/O
205	P33.12	FAST / PU1 / VEVRSB / ES5	1377580	4998996	General-purpose I/O

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for
Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
206	P33.13	FAST / PU1 / VEVRSB / ES5	1285582	4998996	General-purpose I/O
207	VDD	Vx	1150641	4998996	Supply Voltage
208	VSS	Vx	1015641	4998996	Supply Voltage
209	VSS	Vx	894141	4998996	Supply Voltage
210	VDD	Vx	799641	4998996	Supply Voltage
211	VEXT	Vx	728397	4852026	Supply Voltage
212	P32.0	SLOW / PU1 / VEXT / ES	677997	4998996	General-purpose I/O
213	VGATE1N	Vx	627597	4852026	DCDC N ch. MOSFET gate driver output
214	P32.1	SLOW / PU1 / VEXT / ES	577197	4998996	General-purpose I/O
215	VGATE1P	Vx	526797	4852026	DCDC P ch. MOSFET gate driver output
216	VSS	Vx	476397	4998996	Supply Voltage
217	P32.4	FAST / PU1 / VEXT / ES	425997	4852026	General-purpose I/O
218	VSS	Vx	356697	4998996	Supply Voltage
219	VDD	Vx	255699	4998996	Supply Voltage
220	P23.5	FAST / PU1 / VEXT / ES	175644	4918941	General-purpose I/O
221	P23.1	FAST / PU1 / VEXT / ES	175644	4817943	General-purpose I/O
222	P22.0	FAST / PU1 / VEXT / ES6	175644	4716945	General-purpose I/O
223	VSS	Vx	175644	4615947	Supply Voltage
224	VEXT	Vx	322614	4565547	Supply Voltage
225	P22.1	FAST / PU1 / VEXT / ES6	175644	4515147	General-purpose I/O
226	VDD	Vx	322614	4441149	Supply Voltage
227	VSS	Vx	175644	4362651	Supply Voltage
228	P22.2	FAST / PU1 / VEXT / ES6	322614	4288653	General-purpose I/O
229	P22.3	FAST / PU1 / VEXT / ES6	175644	4241655	General-purpose I/O
230	P22.4	FAST / PU1 / VEXT / ES	322614	4194657	General-purpose I/O
231	P22.5	FAST / PU1 / VEXT / ES	175644	4147659	General-purpose I/O

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for
Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
232	VDD	Vx	175644	4037661	Supply Voltage
233	VDD	Vx	322614	3927663	Supply Voltage
234	VSS	Vx	175644	3817665	Supply Voltage
235	VSS	Vx	175644	3707667	Supply Voltage
236	VDD	Vx	362646	3569337	Supply Voltage
237	VSS	Vx	185148	3519927	Supply Voltage
238	XTAL1	XTAL / VEXT	185148	3362778	XTAL pad1 XTAL1. Main Oscillator/PLL/Clock Generator Input.
239	XTAL2	XTAL / VEXT	185148	3263778	XTAL pad2 XTAL2. Main Oscillator/PLL/Clock Generator OUTPUT
240	VSS	Vx	185148	3106629	Supply Voltage
241	VEXT	Vx	362646	3057219	Supply Voltage
242	DAPE0	FAST / PD2 / VEXT	175644	2962917	DAPE: DAPE0 Clock Input DAPE: DAPE0 clock input
243	$\overline{\text{TRST}}$	FAST / PU2 / VEXT	322614	2915919	JTAG Module Reset/Enable Input
244	TMS	FAST / PD2 / VEXT	175644	2868921	JTAG Module State Machine Control Input
245	P21.2	FAST / PU1 / VEXT / ES	322614	2821923	General-purpose I/O
246	P21.0	FAST / PU1 / VEXT / ES	175644	2774925	General-purpose I/O
247	P21.4	FAST / PU1 / VEXT / ES6	322614	2727927	General-purpose I/O
248	P21.1	FAST / PU1 / VEXT / ES	175644	2680929	General-purpose I/O
249	VEXT	Vx	322614	2633931	Supply Voltage
250	VSS	Vx	175644	2586933	Supply Voltage
251	VDD	Vx	322614	2485935	Supply Voltage
252	VDD	Vx	175644	2384937	Supply Voltage
253	VSS	Vx	175644	2274939	Supply Voltage
254	VSS	Vx	175644	2173941	Supply Voltage
255	P21.3	FAST / PU1 / VEXT / ES	322614	2072943	General-purpose I/O
256	TCK	FAST / PD2 / VEXT	175644	2025945	JTAG Module Clock Input
257	P21.5	FAST / PU1 / VEXT / ES6	322614	1978947	General-purpose I/O

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for
Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
258	P20.2	S / PU / VEXT	175644	1931949	General-purpose I/O This pin is latched at power on reset release to enter test mode.
259	P20.0	FAST / PU1 / VEXT / ES	322614	1884951	General-purpose I/O
260	P21.6/TDI	FAST / PD / PU2 / VEXT / ES3	175644	1837953	General-purpose I/O PD during Reset and in DAP/DAPE or JTAG mode. After Reset release and when not in DAP/DAPE or JTAG mode: PU. In Standby mode: HighZ.
261	P21.7/TDO	FAST / PU2 / VEXT / ES4	322614	1790955	General-purpose I/O
262	DAPE2	FAST / PD2 / VEXT	175644	1743957	DAPE: DAPE2 Data I/O DAPE: DAPE2 Data I/O
263	VDD	Vx	322614	1642959	Supply Voltage
264	VSS	Vx	175644	1532961	Supply Voltage
265	VEXT	Vx	322614	1431963	Supply Voltage
266	DAPE1	FAST / PD2 / VEXT	175644	1384965	DAPE: DAPE1 Data I/O DAPE: DAPE1 Data I/O
267	P20.1	SLOW / PU1 / VEXT / ES	322614	1337967	General-purpose I/O
268	VSS	Vx	175644	1290969	Supply Voltage
269	P20.3	SLOW / PU1 / VEXT / ES	322614	1243971	General-purpose I/O
270	P20.6	SLOW / PU1 / VEXT / ES	175644	1196973	General-purpose I/O
271	P20.8	FAST / PU1 / VEXT / ES	322614	1149975	General-purpose I/O
272	P20.7	FAST / PU1 / VEXT / ES	175644	1102977	General-purpose I/O
273	PORST	PORST / PD / VEXT	322614	1055979	PORST pin Power On Reset Input. Additional strong PD in case of power fail.

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for
Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
274	<u>ESR0</u>	FAST / OD / VEXT	175644	1008981	ESR0 Port Pin input - can be used to trigger a reset or an NMI ESR0: External System Request Reset 0. Default configuration during and after reset is open-drain driver. The driver drives low during power-on reset. This is valid additionally after deactivation of PORST_N until the internal reset phase has finished. See also SCU chapter for details. Default after power-on can be different. See also SCU chapter 'Reset Control Unit' and SCU_IOCRR register description. PMS_EVRWUP: EVR Wakeup Pin
275	VDD	Vx	322614	925983	Supply Voltage
276	VSS	Vx	175644	833985	Supply Voltage
277	P20.11	FAST / PU1 / VEXT / ES	322614	728505	General-purpose I/O
278	<u>ESR1</u>	FAST / PU1 / VEXT	175644	678087	ESR1 Port Pin input - can be used to trigger a reset or an NMI ESR1: External System Request Reset 1. Default NMI function. See also SCU chapter for details. Default after power-on can be different. See also SCU chapter 'Reset Control Unit' and SCU_IOCRR register description. PMS_EVRWUP: EVR Wakeup Pin
279	P20.10	FAST / PU1 / VEXT / ES	322614	627669	General-purpose I/O
280	P20.9	FAST / PU1 / VEXT / ES	175644	577251	General-purpose I/O
281	P20.12	FAST / PU1 / VEXT / ES	322614	526833	General-purpose I/O
282	VSS	Vx	175644	476415	Supply Voltage
283	VEXT	Vx	322614	425997	Supply Voltage

TC33xEXT Pin Definition and Functions: Sequence of Pads in Pad Frame for

Table 2-35 Pad List (cont'd)

Number	Pad Name	Pad Type	X	Y	Comment
284	P20.13	FAST / PU1 / VEXT / ES	175644	356697	General-purpose I/O
285	P20.14	FAST / PU1 / VEXT / ES	175644	255699	General-purpose I/O

Whenever in table of section 3 'Electrical Specification' the term 'neighbor pads' is used, the detailed definition is provided by [Figure 2-35](#). This statement is also valid for next/nearest neighbor pads.

In order to find out who is affecting operation on a target pad (interfering) a number of active close-neighbor pads (ACNP) has to be defined.

Finding close-neighbor pads.

The Pad Ring has four edges: bottom, left, top, right. Each edge is limited, i.e. it has two ends.

Each pad has two direct (first) neighbors unless it is located at the end of the edge. In that case it only has one neighbor. Similarly, each pad has two indirect (second) neighbors unless it or its first neighbor is located at the end of the edge. These first and second neighbors we will collectively call Close-Neighbor pads. Therefore each pad has 2 to 4 close-neighbor pads.

Finding close-neighbors can be done with the following sequence:

- 1.) Choose a target pad and lookup its "X" and "Y" coordinates in tables [Figure 2-35](#).
- 2.) Find first and second neighbors by calculating "X" and "Y" distance from the selected pad. [Figure 2-35](#) is sorted by "Y" coordinate, which might help locate the 4 close-neighbor candidates (if the pad is near the edge, it might end up with less than 4 close-neighbors).

Defining active pads:

Pad is active if it is currently in use and if it doesn't have "Vxx" in the name.

Figuring out number of active close-neighbor pads follow next rules:

- If the first neighbor is active, then we count it and also check if second neighbor (on the same side of selected pad) is active.
- If the first neighbor is not active, then we do not check the second on the same side.

2.4 Legend

The data in this chapter 2 match with the file TC33xed_IO_Spirit_v1.0.0.1.16.xml.

Column “Ctrl.”:

I = Input (for GPIO port lines with IOCR bit field Selection PCx = 0XXX_B)

O = Output (for GPIO port lines the ‘O’ represents in most cases the port HWOUT function)

O0 = Output with IOCR bit field selection PCx = 1X000_B

O1 = Output with IOCR bit field selection PCx = 1X001_B (ALT1)

O2 = Output with IOCR bit field selection PCx = 1X010_B (ALT2)

O3 = Output with IOCR bit field selection PCx = 1X011_B (ALT3)

O4 = Output with IOCR bit field selection PCx = 1X100_B (ALT4)

O5 = Output with IOCR bit field selection PCx = 1X101_B (ALT5)

O6 = Output with IOCR bit field selection PCx = 1X110_B (ALT6)

O7 = Output with IOCR bit field selection PCx = 1X111_B (ALT7)

Column “Buffer Type”:

FAST = Pad class FAST (5V/3.3V)

SLOW = Pad class SLOW (5V/3.3V)

LVDS_TX = Pad class LVDS Transmit

LVDS_RX = Pad class LVDS Receive

S = Pad class S (Analog Input overlayed with General Purpose Input)

D = Pad class D (Analog Input)

Porst = Porst input Pad

XTAL1 = XTAL1 input Pad

XTAL2 = XTAL2 input Pad

PU = with pull-up device connected during reset ($\overline{\text{PORST}} = 0$)

PU1 = with pull-up device connected during reset ($\overline{\text{PORST}} = 0$)¹⁾

PU2 = with pull-up device connected during startup and reset, HighZ in Standby mode

PD = with pull-down device connected during reset ($\overline{\text{PORST}} = 0$)

PD1 = with pull-down device connected during reset ($\overline{\text{PORST}} = 0$)¹⁾

PD2 = with pull-down device connected during startup and reset, HighZ in Standby mode

OD = open drain during reset ($\overline{\text{PORST}} = 0$)

ES = Supports Emergency Stop

ES1 = ES. ES can be overruled by VADC, control via P00_PCSR

ES2 = ES. ES can be overruled by DXCPL - DAP over CAN physical layer, No overruling for DXCM - Debug over CAN message

ES3 = ES. ES can be overruled by JTAG mode if this pin is used as TDI

ES4 = ES. ES can be overruled by JTAG or Three Pin DAP mode

ES5 = ES. ES can be overruled by the Standby Controller - SCR - if implemented. Overruling can be disabled via the control register P33_PCSR and P34_PCSR

¹⁾ The default state of GPIOs (Px.y) during and after PORST active is controlled via HWCFG6 (P14.4). Pls. see also chapter PMS, HWCFG[6].

TC33xEXT Pin Definition and Functions:Legend

ES6 = ES. On LVDS TX pads the ES affects the pads only in CMOS mode, not in LVDS mode. Thus, only when LPCRx.TX_EN selects the CMOS Mode, the output is switched off in the ES event.

3 Electrical Specification

3.1 Parameter Interpretation

The parameters listed in this section partly represent the characteristics of the TC33xEXT and partly its requirements on the system. To aid interpreting the parameters easily when evaluating them for a design, they are marked with an two-letter abbreviation in column "Symbol":

- **CC**
Such parameters indicate **C**ontroller **C**haracteristics which are a distinctive feature of the TC33xEXT and must be regarded for a system design.
- **SR**
Such parameters indicate **S**ystem **R**equirements which must be provided by the microcontroller system in which the TC33xEXT designed in.

Electrical Specification Absolute Maximum Ratings

3.2 Absolute Maximum Ratings

Stresses above the values listed under “Absolute Maximum Ratings” may cause permanent damage to the device. This is a stress rating only and functional operation of the device at these or any other conditions above those indicated in the Operational Conditions of this specification is not implied. Exposure to absolute maximum rating conditions may affect device reliability.

Table 3-1 Absolute Maximum Ratings

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Storage Temperature	T_{ST} SR	-65	-	150	°C	upto 65h @ $T_J = 150^{\circ}\text{C}$
Voltage at V_{DD} power supply pins with respect to V_{SS} ^{1) 2)}	V_{DD} SR	-	-	1.65	V	upto 2.8h
		-	-	1.45	V	upto 72h
Voltage at V_{DDP3} power supply pins with respect to V_{SS}	V_{DDP3} SR	-	-	4.43	V	
Voltage at V_{DDM} , V_{EXT} , V_{FLEX} and V_{EVRSB} power supply pins with respect to V_{SS}	V_{DDM} SR	-	-	6.75	V	upto 2.8h
		-	-	5.6	V	upto 72h
Voltage on all analog and class S input pins with respect to V_{SS} ³⁾	V_{IN} SR	-0.7	-	6.75	V	
Voltage on all other input pins with respect to V_{SS} ³⁾	V_{IN} SR	-0.7	-	6.75	V	
Input current on any pin during overload condition ^{4) 5)}	I_{IN} SR	-10	-	10	mA	
Absolute maximum sum of all input circuit currents during overload condition. ⁴⁾	ΣI_{IN} SR	-100	-	100	mA	

- 1) Valid for cumulated for up to 2.8h and pulse forms followed a power supply switch on phase, where the rise and fall times are related to the system capacities and coils.
- 2) Due to EVRC output voltage oscillation during switch off phase V_{DD} can drop down to -0.72V. For V_{DD} an input level down to -0.72V during switch off phase will not cause any damage or reliability problem.
- 3) Voltages below V_{INmin} have no Impact to the device reliability as Long as the times and currents defined in section Pin Reliability in Overload for the affected pad(s) are not violated.
- 4) This parameter is an Absolute Maximum Rating. Exposure to Absolute Maximum Ratings for extended periods of time may damage the device.
- 5) The specified min. and max. values represent the current limits, which have to be maintained, in case of a short circuit condition on the output of any Fast, RFast, Slow and Class S pad, not being used during operation.
This covers also output currents due to switching in operation for $C_L=200\text{pF}$.

3.3 Pin Reliability in Overload

When receiving signals from higher voltage devices, low-voltage devices experience overload currents and voltages that go beyond their own IO power supplies specification.

The following table defines overload conditions that will not cause any negative reliability impact if all the following conditions are met:

- allowed time interval (defined in Note column) for overload condition is not exceeded. If no time limit is defined the allowed time includes both 'Operation Lifetime hours' and 'Inactive Lifetime hours'. The number of hours in [Table 3-54](#) and [Table 3-55](#) are examples only and the applicable numbers are defined by the customer profiles accepted by Infineon.
- Operating Conditions** are met for
 - pad supply levels
 - temperature

If a pin current is out of the **Operating Conditions** but within the overload parameters, then the parameters functionality of this pin as stated in the Operating Conditions can no longer be guaranteed. Operation is still possible in most cases but with relaxed parameters.

Table 3-2 Overload Parameters

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Input current on any digital pin during overload condition	I_{IN}	-5	-	5	mA	except LVDS pins
		-15 ¹⁾	-	15 ¹⁾	mA	except LVDS pins; limited to max. 20 pulses with 1ms pulse length
Input current on LVDS pin during overload condition	I_{INLVDS}	-3	-	3	mA	
Input current on analog input pin during overload condition	I_{INANA}	-3	-	3	mA	
		-5	-	5	mA	limited to 60h over lifetime
Absolute sum of all analog input currents for analog inputs during overload condition	I_{INSA}	-20	-	20	mA	
Absolute maximum sum of all input circuit currents during overload condition (digital and analog combined)	ΣI_{INS}	-100	-	100	mA	
Signal voltage over/undershoot at GPIOs	V_{OUS}	$V_{SS} - 2$	-	$V_{EXT/FLEX} + 2$	V	limited to 60h over lifetime; Valid for non LVDS and analog pads
Sum of all inactive device pin currents	I_{IDS}	-100	-	100	mA	
Static pin output current	$I_{OUT CC}$	-	-	2.5	mA	100% duty cycle; output driver = medium
		-	-	5	mA	100% duty cycle; output driver = strong

Electrical Specification Pin Reliability in Overload

Table 3-2 Overload Parameters (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Overload coupling factor for digital inputs, negative	$K_{\text{OVDN CC}}$	-	-	$3 \cdot 10^{-4}$		Overload injected on GPIO non LVDS pad and affecting neighbor fast pads; $-5\text{mA} < I_{\text{IN}} < 0\text{mA}$
		-	-	$2 \cdot 10^{-3}$		Overload injected on GPIO non LVDS pad and affecting neighbor slow pads VGASTE1N and VGATE1P; $-5\text{mA} < I_{\text{IN}} < 0\text{mA}$
		-	-	$1 \cdot 10^{-4}$		Overload injected on GPIO non LVDS pad and affecting neighbor slow pads; $-5\text{mA} < I_{\text{IN}} < 0\text{mA}$
		-	-	0.8		Overload injected on LVDS RX pad and affecting neighbor LVDS pads
Overload coupling factor for digital inputs, positive	$K_{\text{OVDP CC}}$	-	-	$1.5 \cdot 10^{-3}$		Overload injected on GPIO non LVDS pad and affecting neighbor GPIO non LVDS pads
		-	-	1		Overload injected on LVDS RX pad and affecting neighbor LVDS pads
Overload coupling factor for analog inputs, negative ²⁾	$K_{\text{OVAN CC}}$	-	-	$1 \cdot 10^{-4}$		Analog inputs overlaid with slow pads or pull down diagnostics; $-5\text{mA} < I_{\text{IN}} < 0\text{mA}$
		-	-	$1 \cdot 10^{-5}$		else; $-5\text{mA} < I_{\text{IN}} < 0\text{mA}$
Overload coupling factor for analog inputs, positive ²⁾	$K_{\text{OVAP CC}}$	-	-	$2 \cdot 10^{-4}$		Analog inputs overlaid with slow pads or pull down diagnostics; $0\text{mA} < I_{\text{IN}} < 5\text{mA}$
		-	-	$2 \cdot 10^{-5}$		else; $0\text{mA} < I_{\text{IN}} < 5\text{mA}$

1) Reduced VADC result accuracy and / or GPIO input levels (V_{IL} and V_{IH}) can differ from specified parameters.

2) Overload coupling on analog inputs is caused by parasitic effects between pads, input multiplexers and surrounding structures. The given parameters have been verified for all permutations of channels. Also watch multiple connections of a pin to several channels.

3.4 Operating Conditions

The following operating conditions must not be exceeded in order to ensure correct operation and reliability of the TC33xEXT. All parameters specified in the following tables refer to these operating conditions, unless otherwise noticed.

Digital supply voltages applied to the TC33xEXT must be static regulated voltages.

All parameters specified in the following tables refer to these operating conditions (see table below), unless otherwise noticed in the Note / Test Condition column.

Table 3-3 Operating Conditions

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
SRI frequency	f_{SRI} SR	-	-	300	MHz	
CPU Frequency (All CPUs)	f_{CPUx} SR	-	-	300	MHz	
PLL0 output frequency	f_{PLL0} SR	20	-	300	MHz	
SPB frequency	f_{SPB} SR	-	-	100	MHz	
FSI2 frequency	f_{FSI2} SR	-	-	300	MHz	
FSI frequency	f_{FSI} SR	20	-	100	MHz	
GTM frequency	f_{GTM} SR	-	-	200	MHz	
STM frequency	f_{STM} SR	-	-	100	MHz	
BBB frequency	f_{BBB} SR	-	-	150	MHz	
VADC frequency	f_{ADC} SR	-	-	160	MHz	
ASCLIN Operating Frequency	f_{ASCLINx} SR	-	-	200	MHz	
CAN frequency	f_{CAN} SR	-	-	80	MHz	
PLL1 output frequency from PER PLL	f_{PLL1} SR	20	-	320	MHz	
PLL2 output frequency from PER PLL	f_{PLL2} SR	20	-	200	MHz	
QSPI Frequency	f_{QSPI} SR	-	-	200	MHz	
ADAS clock frequency	f_{ADAS} CC	200	-	300	MHz	
MCANH frequency	f_{MCANH} CC	-	-	100	MHz	
GETH frequency	f_{GETH} CC	100	-	150	MHz	
Ambient Temperature	T_{A} SR	-40	-	125	°C	valid for all SAK products
Junction Temperature	T_{J} SR	-40	-	150	°C	valid for all SAK products
Core Supply Voltage	V_{DD} SR	1.125 ¹⁾	1.25	1.375 ²⁾	V	
ADC analog supply voltage	V_{DDM} SR	2.97	5.0	5.5 ³⁾	V	

Electrical Specification Operating Conditions
Table 3-3 Operating Conditions (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Digital external supply voltage for pads and EVR	V_{EXT} SR	4.5	5.0	5.5 ³⁾	V	Nominal 5V Pad / Port Pin supply range. 5V pad parameters are valid.
		2.97	3.3	3.63	V	Nominal 3.3V Pad / Port Pin supply range with VDDP3 supplied externally and EVR33 inactive. 3.3V pad parameters are valid.
		3.6	-	4.5	V	Flash configured in cranking mode; Flash read operation with reduced performance. EVR33 active in low voltage mode. 3.3V pad parameters are valid.
		2.97	-	3.6	V	In case EVR33 is active, Flash configured in sleep mode and execution switched to RAM. 3.3V pad parameters are valid.
Digital supply voltage for Flex port	V_{FLEX} SR	2.97	-	4.0	V	3.3V pad parameters are valid
		4.5	5.0	5.5 ³⁾	V	5V pad parameters are valid
Digital supply voltage for Flash	V_{DDP3} SR	2.97	3.3	3.63 ⁴⁾	V	
		2.6	-	3.63	V	Flash configured in cranking mode; Flash read operation with reduced performance.
Digital ground voltage	V_{SS} SR	0	-	-	V	
Analog ground voltage for V_{DDM}	V_{SSM} CC	-0.1	0	0.1	V	
Digital external supply voltage for EVR and during Standby mode	V_{EVRSB} SR	2.97 ⁵⁾	-	5.5	V	V_{EVRSB} is bonded together with V_{EXT} supply pin in smaller LQFP packages.
Voltage to ensure defined pad states	V_{DDPPA} CC	1.3 ⁶⁾	-	-	V	

1) For V_{DD} $1.08V \leq V_{DD} < 1.125V$ operation is still possible but with relaxed parameters.

2) Voltage overshoot to 1.69V is permissible, provided the duration is less than 2h cumulated. Reduced ADC accuracy and leakage is increased.

Electrical Specification Operating Conditions

- 3) Voltage overshoot to 6.5V is permissible, provided the duration is less than 2h cumulated. Reduced ADC accuracy and leakage is increased.
- 4) Voltage overshoot to 4.29V is permissible, provided the duration is less than 2h cumulated. Reduced ADC accuracy and leakage is increased.
- 5) $V_{\text{EVR SB}}$ supply voltage can drop down upto 2.6V during Standby mode. It is required to have a capacitor of 100nF on $V_{\text{EVR SB}}$ supply pin.
- 6) HWCFG[6] pin is latched and pull-up or tristate is activated at Port pins when VEXT has reached this level.

Limitation of Supply Voltage over Time

The maximum operation voltage for $V_{\text{EXT/FLEX/DDM}}$ supply rails is limited over the complete lifetime.

The following voltage profile is an example. Application specific voltage profiles need to be aligned and approved by Infineon Technologies for the fulfillment of quality and reliability targets.

Table 3-4 Example Voltage Profile

$V_{\text{EXT/FLEX/DDM}}$	Duration [h]
$5.4 \text{ V} < V_{\text{EXT/FLEX/DDM}} \leq 5.5 \text{ V}$	$\leq 5\%$ of lifetime
$5.15 \text{ V} < V_{\text{EXT/FLEX/DDM}} \leq 5.4 \text{ V}$	$\leq 15\%$ of lifetime
$4.85 \text{ V} < V_{\text{EXT/FLEX/DDM}} \leq 5.15 \text{ V}$	$\leq 60\%$ of lifetime
$4.6 \text{ V} < V_{\text{EXT/FLEX/DDM}} \leq 4.85 \text{ V}$	$\leq 15\%$ of lifetime
$4.5 \text{ V} < V_{\text{EXT/FLEX/DDM}} \leq 4.6 \text{ V}$	$\leq 5\%$ of lifetime

The maximum operation voltage for V_{DD} supply rails is limited over the complete lifetime.

The following voltage profile is an example. Application specific voltage profiles need to be aligned and approved by Infineon Technologies for the fulfillment of quality and reliability targets.

Table 3-5 Example Voltage Profile

V_{DD}	Duration [h]
$1.325 \text{ V} < V_{\text{DD}} \leq 1.375 \text{ V}$	$\leq 5\%$ of lifetime
$1.275 \text{ V} < V_{\text{DD}} \leq 1.325 \text{ V}$	$\leq 15\%$ of lifetime
$1.225 \text{ V} < V_{\text{DD}} \leq 1.275 \text{ V}$	$\leq 60\%$ of lifetime
$1.175 \text{ V} < V_{\text{DD}} \leq 1.225 \text{ V}$	$\leq 15\%$ of lifetime
$1.125 \text{ V} < V_{\text{DD}} \leq 1.175 \text{ V}$	$\leq 5\%$ of lifetime

3.5 5 V / 3.3 V switchable Pads

Pad classes slow GPIO and fast GPIO support both Automotive Level (AL) or TTL level (TTL) operation. Parameters are defined for AL operation and degrade in TTL operation.

Table 3-6 $\overline{\text{PORST}}$ Pad

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
$\overline{\text{PORST}}$ pad Output current	I_{PORST} CC	13	-	-	mA	$V_{\text{EXT}} = 2.97\text{V}$; $V_{\text{PORST}} = 0.9\text{V}$
Spike filter always blocked pulse duration	t_{SF1} CC	-	-	80	ns	
Spike filter pass-through blocked pulse duration	t_{SF2} CC	260	-	-	ns	without additional $\overline{\text{PORST}}$ Digital Filter active ($\text{PORSTDF} = 0$).
Input hysteresis ¹⁾	HYS CC	$0.055 \cdot V_{\text{EXT}}$	-	-	V	non of the neighbor pads are used as output; TTL (degraded, used for CIF)
Pull-down current ²⁾	I_{PDL} CC	-	-	130	μA	V_{IH} ; TTL (degraded, used for CIF)
		15	-	-	μA	V_{IL} ; TTL (degraded, used for CIF)
Input leakage current	I_{OZ} CC	-450	-	450	nA	$T_J \leq 150^\circ\text{C}$; $(0.1 \cdot V_{\text{EXT}}) < V_{\text{IN}} < (0.9 \cdot V_{\text{EXT}})$
		-500	-	500	nA	$T_J \leq 150^\circ\text{C}$; else
		-900	-	900	nA	$T_J \leq 170^\circ\text{C}$; $(0.1 \cdot V_{\text{EXT}}) < V_{\text{IN}} < (0.9 \cdot V_{\text{EXT}})$
		-950	-	950	nA	$T_J \leq 170^\circ\text{C}$; else
Input high voltage level	V_{IH} SR	1.4	-	-	V	TTL (degraded, used for CIF); $V_{\text{EXT}} = 2.97\text{V}$
		2.0	-	-	V	TTL; $V_{\text{EXT}} = 4.5\text{V}$
Input low voltage level	V_{IL} SR	-	-	0.5	V	TTL (degraded, used for CIF); $V_{\text{EXT}} = 2.97\text{V}$
		-	-	0.8	V	TTL; $V_{\text{EXT}} = 4.5\text{V}$
Pin capacitance	C_{IO} CC	-	2	3	pF	in addition 2.5pF from package to be added

1) Hysteresis is implemented to avoid metastable states and switching due to internal ground bounce. It can't be guaranteed that it suppresses switching due to external system noise.

2) Values for Pull-down resistor is defined via parameter R_{MDD} in table VADC 5V.

Electrical Specification 5 V / 3.3 V switchable Pads
Table 3-7 Fast 5V GPIO

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
On-Resistance of pad output	$R_{\text{DS(on) CC}}$	125	225	320	Ohm	medium driver; $I_{\text{OH/OL}} = 2\text{mA}$
		31	55	80	Ohm	strong driver; $I_{\text{OH/OL}} = 8\text{mA}$
Rise / Fall time ^{1) 2)}	$t_{\text{RF CC}}$	1.6	-	3.2	ns	$C_L = 25\text{pF}$; driver = strong sharp edge; from 0.2 * $V_{\text{EXT/FLEX/EVRSB}}$ to 0.8 * $V_{\text{EXT/FLEX/EVRSB}}$
		$4+0.55 \cdot C_L$	$4+0.75 \cdot C_L$	$12+1.0 \cdot C_L$	ns	driver = medium; $C_L \leq 200\text{pF}$
		$1.0+0.18 \cdot C_L$	$2.5+0.27 \cdot C_L$	$5.0+0.35 \cdot C_L$	ns	driver = strong edge = medium; $C_L \leq 200\text{pF}$
		$0.5+0.08 \cdot C_L$	$0.5+0.11 \cdot C_L$	$1.0+0.17 \cdot C_L$	ns	driver = strong edge = sharp ; $C_L \leq 200\text{pF}$
Asymmetry of sending	$t_{\text{TX_ASYM CC}}$	-1	-	1	ns	valid for all data rates excluding clock tolerance
Input frequency	$f_{\text{IN CC}}$	-	-	160	MHz	
Input hysteresis ³⁾	$HYS \text{ CC}$	$0.09 \cdot V_{\text{EXT/FLEX/EVRSB}}$	-	-	V	none of the neighbor pads are used as output; AL
		$0.075 \cdot V_{\text{EXT/FLEX/EVRSB}}$	-	-	V	none of the neighbor pads are used as output; TTL
		75	-	-	mV	two of the neighbor pads are used as output with driver=strong and edge=sharp; AL
Pull-up current ⁴⁾	$I_{\text{PUH CC}}$	30	-	-	μA	V_{IH} ; AL or TTL
		-	-	130	μA	V_{IL} ; AL or TTL
Pull-down current ⁵⁾	$I_{\text{PDL CC}}$	-	-	130	μA	V_{IH} ; AL or TTL
		30	-	-	μA	V_{IL} ; AL
		28	-	-	μA	V_{IL} ; TTL

Electrical Specification 5 V / 3.3 V switchable Pads
Table 3-7 Fast 5V GPIO (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Input leakage current	I_{OZ} CC	-1100	-	1100	nA	$T_J \leq 150^\circ\text{C}$; $(0.1 * V_{EXT/FLEX/EVRSB}) < V_{IN} < (0.9 * V_{EXT/FLEX/EVRSB})$
		-6000	-	6000	nA	$T_J \leq 150^\circ\text{C}$; LVDS_RX / Fast pad type ; else
		-3200	-	3200	nA	$T_J \leq 150^\circ\text{C}$; LVDS_TX / Fast pad type ; else
		-1500	-	1500	nA	$T_J \leq 150^\circ\text{C}$; else
		-2000	-	2000	nA	$T_J \leq 170^\circ\text{C}$; $(0.1 * V_{EXT/FLEX/EVRSB}) < V_{IN} < (0.9 * V_{EXT/FLEX/EVRSB})$
		-13500	-	13500	nA	$T_J \leq 170^\circ\text{C}$; LVDS_RX / Fast pad type ; else
		-2500	-	2500	nA	$T_J \leq 170^\circ\text{C}$; else
Input high voltage level	V_{IH} SR	$0.7 * V_{EXT/FLEX/EVRSB}$	-	-	V	AL
		2.0	-	-	V	TTL
Input low voltage level	V_{IL} SR	-	-	$0.44 * V_{EXT/FLEX/EVRSB}$	V	AL
		-	-	0.8	V	TTL
Input low threshold variation	V_{ILD} SR	-50	-	50	mV	max. variation of 1ms; $V_{EXT/FLEX/EVRSB} = \text{constant}$; AL
Pin capacitance	C_{IO} CC	-	2	3	pF	in addition 2.5pF from package to be added
Pad set-up time to get an software update of the configuration active	t_{SET} CC	-	-	100	ns	

- 1) In the formulas the value of C_L needs to be entered in pF to obtain results in ns.
- 2) Rise / fall times are defined 10% - 90% of pad supply voltage.
- 3) Hysteresis is implemented to avoid metastable states and switching due to internal ground bounce. It can't be guaranteed that it suppresses switching due to external system noise.
- 4) Values for Pull-up resistor is defined via parameter R_{MDU} in table VADC 5V.
- 5) Values for Pull-down resistor is defined via parameter R_{MDD} in table VADC 5V.

Electrical Specification 5 V / 3.3 V switchable Pads

Table 3-8 Fast 3.3V GPIO

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
On-Resistance of pad output	$R_{\text{DS(on) CC}}$	125	225	320	Ohm	medium driver; $I_{\text{OH/OL}} = 2\text{mA}$
		31	55	80	Ohm	strong driver; $I_{\text{OH/OL}} = 8\text{mA}$
Rise / Fall time ^{1) 2)}	$t_{\text{RF CC}}$	1.6	-	4.5	ns	$C_L = 25\text{pF}$; driver = strong sharp edge; from 0.2 * $V_{\text{EXT/FLEX/EVRSB}}$ to 0.8 * $V_{\text{EXT/FLEX/EVRSB}}$
		-	-	5	ns	$C_L = 25\text{pF}$; driver = strong sharp edge; from 0.8V to 2.0V (RMII)
		$2+0.57 \cdot C_L$	$5.5+0.75 \cdot C_L$	$10+1.25 \cdot C_L$	ns	driver = medium; $C_L \leq 200\text{pF}$
		$1.5+0.18 \cdot C_L$	$1.5+0.28 \cdot C_L$	$8+0.4 \cdot C_L$	ns	driver = strong edge = medium; $C_L \leq 200\text{pF}$
		$0.75+0.08 \cdot C_L$	$0.75+0.11 \cdot C_L$	$2.5+0.21 \cdot C_L$	ns	driver = strong edge = sharp; $C_L \leq 200\text{pF}$
Asymmetry of sending	$t_{\text{TX_ASYM CC}}$	-1	-	1	ns	valid for all data rates excluding clock tolerance
Input frequency	$f_{\text{IN CC}}$	-	-	160	MHz	
Input hysteresis ³⁾	$HYS \text{ CC}$	$0.055 \cdot V_{\text{EXT/FLEX/EVRSB}}$	-	-	V	none of the neighbor pads are used as output; AL
		$0.09 \cdot V_{\text{EXT/FLEX/EVRSB}}$	-	-	V	none of the neighbor pads are used as output; TTL
		$0.055 \cdot V_{\text{EXT/FLEX/EVRSB}}$	-	-	V	none of the neighbor pads are used as output; TTL (degraded, used for CIF)
		125	-	-	mV	two of the neighbor pads are used as output with driver=strong and edge=sharp; AL

Electrical Specification 5 V / 3.3 V switchable Pads
Table 3-8 Fast 3.3V GPIO (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Pull-up current ⁴⁾	I_{PUH} CC	17	-	-	μA	V_{IH} ; AL and TTL (degraded, used for CIF)
		11	-	-	μA	V_{IH} ; TTL
		-	-	80	μA	V_{IL} ; AL and TTL and TTL (degraded, used for CIF)
Pull-down current ⁵⁾	I_{PDL} CC	-	-	105	μA	V_{IH} ; AL and TTL (degraded, used for CIF)
		-	-	115	μA	V_{IH} ; TTL
		19	-	-	μA	V_{IL} ; AL and TTL
		15	-	-	μA	V_{IL} ; TTL (degraded, used for CIF)
Input leakage current	I_{OZ} CC	-1100	-	1100	nA	$T_J \leq 150^\circ\text{C}$; $(0.1 * V_{EXT/FLEX/EVRSB}) < V_{IN} < (0.9 * V_{EXT/FLEX/EVRSB})$
		-6000	-	6000	nA	$T_J \leq 150^\circ\text{C}$; LVDS_RX / Fast pad type ; else
		-1500	-	1500	nA	$T_J \leq 150^\circ\text{C}$; else
		-2000	-	2000	nA	$T_J \leq 170^\circ\text{C}$; $(0.1 * V_{EXT/FLEX/EVRSB}) < V_{IN} < (0.9 * V_{EXT/FLEX/EVRSB})$
		-13500	-	13500	nA	$T_J \leq 170^\circ\text{C}$; LVDS_RX / Fast pad type ; else
		-2500	-	2500	nA	$T_J \leq 170^\circ\text{C}$; else
Input high voltage level	V_{IH} SR	0.7 * $V_{EXT/FLEX/EVRSB}$	-	-	V	AL
		2.0	-	-	V	TTL
		1.4	-	-	V	TTL (degraded, used for CIF)
Input low voltage level	V_{IL} SR	-	-	0.42 * $V_{EXT/FLEX/EVRSB}$	V	AL
		-	-	0.8	V	TTL
		-	-	0.5	V	TTL (degraded, used for CIF)
Input low/high voltage level	V_{ILH} SR	1.0	-	1.9	V	RGMII; no hysteresis available

Electrical Specification 5 V / 3.3 V switchable Pads

Table 3-8 Fast 3.3V GPIO (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Input low threshold variation	V_{ILD} SR	-33	-	33	mV	max. variation of 1ms; $V_{EXT/FLEX/EVRSB} =$ constant; AL
Pin capacitance	C_{IO} CC	-	2	3	pF	in addition 2.5pF from package to be added
Pad set-up time to get an software update of the configuration active	t_{SET} CC	-	-	100	ns	

- 1) In the formulas the value of C_L needs to be entered in pF to obtain results in ns.
- 2) Rise / fall times are defined 10% - 90% of pad supply voltage.
- 3) Hysteresis is implemented to avoid metastable states and switching due to internal ground bounce. It can't be guaranteed that it suppresses switching due to external system noise.
- 4) Values for Pull-up resistor is defined via parameter R_{MDU} in table VADC 5V.
- 5) Values for Pull-down resistor is defined via parameter R_{MDD} in table VADC 5V.

Table 3-9 Slow 5V GPIO

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
On-Resistance of pad output	R_{DSON} CC	125	225	320	Ohm	medium driver; $I_{OH/OL} =$ 2mA
Rise / Fall time ^{1) 2)}	t_{RF} CC	$4+0.55 \cdot C_L$	$4+0.75 \cdot C_L$	$12+1 \cdot C_L$	ns	driver = medium edge = medium ; $C_L \leq 200$ pF
		$1.5+0.25 \cdot C_L$	$2.5+0.40 \cdot C_L$	$7+0.55 \cdot C_L$	ns	driver = medium edge = sharp ; $C_L \leq 200$ pF
Asymmetry of sending	t_{TX_ASYM} CC	-1	-	1	ns	valid for all data rates excluding clock tolerance
Input frequency	f_{IN} CC	-	-	160	MHz	
Input hysteresis ³⁾	HYS CC	$0.09 \cdot V_{EXT/FLEX/EVRSB}$	-	-	V	none of the neighbor pads are used as output; AL
		$0.075 \cdot V_{EXT/FLEX/EVRSB}$	-	-	V	none of the neighbor pads are used as output; TTL
		75	-	-	mV	two of the neighbor pads are used as output with driver=strong and edge=sharp; AL

Electrical Specification 5 V / 3.3 V switchable Pads
Table 3-9 Slow 5V GPIO (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Pull-up current ⁴⁾	I_{PUH} CC	30	-	-	μA	V_{IH} ; AL or TTL; except VGATE1P; except VGATE1N and $T_J > 150^\circ\text{C}$
		-	-	130	μA	V_{IL} ; AL or TTL; except VGATE1P; except VGATE1N and $T_J > 150^\circ\text{C}$
Pull-down current ⁵⁾	I_{PDL} CC	-	-	130	μA	V_{IH} ; AL or TTL
		30	-	-	μA	V_{IL} ; AL
		28	-	-	μA	V_{IL} ; TTL
Input leakage current	I_{OZ} CC	-300	-	300	nA	$T_J \leq 150^\circ\text{C}$; $(0.1 * V_{EXT/FLEX/EVRSB}) < V_{IN} < (0.9 * V_{EXT/FLEX/EVRSB})$
		-400	-	400	nA	$T_J \leq 150^\circ\text{C}$; else
		-600	-	600	nA	$T_J \leq 170^\circ\text{C}$; $(0.1 * V_{EXT/FLEX/EVRSB}) < V_{IN} < (0.9 * V_{EXT/FLEX/EVRSB})$
		-750	-	750	nA	$T_J \leq 170^\circ\text{C}$; else
		-18000	-	18000	nA	P32.0 and P32.1; $T_J \leq 150^\circ\text{C}$
		-38000	-	38000	nA	P32.0 and P32.1; $T_J \leq 170^\circ\text{C}$
Input high voltage level	V_{IH} SR	$0.7 * V_{EXT/FLEX/EVRSB}$	-	-	V	AL
		2.0	-	-	V	TTL
Input low voltage level	V_{IL} SR	-	-	$0.44 * V_{EXT/FLEX/EVRSB}$	V	AL
		-	-	0.8	V	TTL
Input low threshold variation	V_{ILD} SR	-50	-	50	mV	max. variation of 1ms; $V_{EXT/FLEX/EVRSB} = \text{constant}$; AL
Pin capacitance	C_{IO} CC	-	2	3	pF	in addition 2.5pF from package to be added
Pad set-up time to get an software update of the configuration active	t_{SET} CC	-	-	100	ns	

1) In the formulas the value of C_L needs to be entered in pF to obtain results in ns.

2) Rise / fall times are defined 10% - 90% of pad supply voltage.

3) Hysteresis is implemented to avoid metastable states and switching due to internal ground bounce. It can't be guaranteed that it suppresses switching due to external system noise.

Electrical Specification 5 V / 3.3 V switchable Pads

- 4) Values for Pull-up resistor is defined via parameter R_{MDU} in table VADC 5V.
 5) Values for Pull-down resistor is defined via parameter R_{MDD} in table VADC 5V.

Table 3-10 Slow 3.3V GPIO

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
On-Resistance of pad output	R_{DSON} CC	125	225	320	Ohm	medium driver; $I_{OH/OL} = 2mA$
Rise / Fall time ^{1) 2)}	t_{RF} CC	$2+0.57 \cdot C_L$	$5.5+0.75 \cdot C_L$	$10+1.25 \cdot C_L$	ns	driver = medium edge = medium ; $C_L \leq 200pF$
		$2+0.30 \cdot C_L$	$3.5+0.50 \cdot C_L$	$5+0.70 \cdot C_L$	ns	driver = medium edge = sharp ; $C_L \leq 200pF$
Asymmetry of sending	t_{TX_ASYM} CC	-1	-	1	ns	valid for all data rates excluding clock tolerance
Input frequency	f_{IN} CC	-	-	160	MHz	
Input hysteresis ³⁾	HYS CC	$0.055 \cdot V_{EXT/FLEX/EVRSB}$	-	-	V	none of the neighbor pads are used as output; AL
		$0.09 \cdot V_{EXT/FLEX/EVRSB}$	-	-	V	none of the neighbor pads are used as output; TTL
		$0.055 \cdot V_{EXT/FLEX/EVRSB}$	-	-	V	none of the neighbor pads are used as output;TTL (degraded, used for CIF)
		125	-	-	mV	two of the neighbor pads are used as output with driver=strong and edge=sharp; AL
Pull-up current ⁴⁾	I_{PUH} CC	17	-	-	μA	V_{IH} ; AL and TTL (degraded, used for CIF); except VGATE1P; except VGATE1N and $T_J > 150^\circ C$
		11	-	-	μA	V_{IH} ; TTL; except VGATE1P; except VGATE1N and $T_J > 150^\circ C$
		-	-	80	μA	V_{IL} ; AL and TTL and TTL (degraded, used for CIF); except VGATE1P; except VGATE1N and $T_J > 150^\circ C$

Electrical Specification 5 V / 3.3 V switchable Pads
Table 3-10 Slow 3.3V GPIO (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Pull-down current ⁵⁾	I_{PDL} CC	-	-	105	μA	V_{IH} ; AL and TTL (degraded, used for CIF)
		-	-	115	μA	V_{IH} ; TTL
		19	-	-	μA	V_{IL} ; AL and TTL
		15	-	-	μA	V_{IL} ; TTL (degraded, used for CIF)
Input leakage current	I_{OZ} CC	-300	-	300	nA	$T_J \leq 150^\circ\text{C}$; $(0.1 * V_{EXT/FLEX/EVRSB}) < V_{IN} < (0.9 * V_{EXT/FLEX/EVRSB})$
		-400	-	400	nA	$T_J \leq 150^\circ\text{C}$; else
		-600	-	600	nA	$T_J \leq 170^\circ\text{C}$; $(0.1 * V_{EXT/FLEX/EVRSB}) < V_{IN} < (0.9 * V_{EXT/FLEX/EVRSB})$
		-750	-	750	nA	$T_J \leq 170^\circ\text{C}$; else
		-18000	-	18000	nA	P32.0 and P32.1; $T_J \leq 150^\circ\text{C}$
		-38000	-	38000	nA	P32.0 and P32.1; $T_J \leq 170^\circ\text{C}$
Input high voltage level	V_{IH} SR	$0.7 * V_{EXT/FLEX/EVRSB}$	-	-	V	AL
		2.0	-	-	V	TTL
		1.4	-	-	V	TTL (degraded, used for CIF)
Input low voltage level	V_{IL} SR	-	-	$0.42 * V_{EXT/FLEX/EVRSB}$	V	AL
		-	-	0.8	V	TTL
		-	-	0.5	V	TTL (degraded, used for CIF)
Input low/high voltage level	V_{ILH} SR	1.0	-	1.9	V	RGMII; no hysteresis available
Input low threshold variation	V_{ILD} SR	-33	-	33	mV	max. variation of 1ms; $V_{EXT/FLEX/EVRSB} = \text{constant}$; AL
Pin capacitance	C_{IO} CC	-	2	3	pF	in addition 2.5pF from package to be added
Pad set-up time to get an software update of the configuration active	t_{SET} CC	-	-	100	ns	

1) In the formulas the value of C_L needs to be entered in pF to obtain results in ns.

Electrical Specification 5 V / 3.3 V switchable Pads

- 2) Rise / fall times are defined 10% - 90% of pad supply voltage.
- 3) Hysteresis is implemented to avoid metastable states and switching due to internal ground bounce. It can't be guaranteed that it suppresses switching due to external system noise.
- 4) Values for Pull-up resistor is defined via parameter R_{MDU} in table VADC 5V.
- 5) Values for Pull-down resistor is defined via parameter R_{MDD} in table VADC 5V.

Table 3-11 RFast 5V GPIO

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
On-Resistance of pad output	$R_{DS(on)}$ CC	125	225	320	Ohm	medium driver; $I_{OH/OL} = 2mA$
		31	55	80	Ohm	strong driver; $I_{OH/OL} = 8mA$
Rise / Fall time ^{1) 2)}	t_{RF} CC	1.6	-	3.2	ns	$C_L = 25pF$; driver = strong sharp edge; from $0.2 * V_{FLEX}$ to $0.8 * V_{FLEX}$
		$4+0.55*C_L$	$4+0.75*C_L$	$12+1.0*C_L$	ns	driver = medium; $C_L \leq 200pF$
		$1.0+0.18*C_L$	$2.5+0.27*C_L$	$5.0+0.35*C_L$	ns	driver = strong edge = medium; $C_L \leq 200pF$
		$0.5+0.08*C_L$	$0.5+0.11*C_L$	$1.0+0.17*C_L$	ns	driver = strong edge = sharp ; $C_L \leq 200pF$
Asymmetry of sending	t_{TX_ASYM} CC	-0.5	-	0.5	ns	valid for all data rates excluding clock tolerance
Input frequency	f_{IN} CC	-	-	160	MHz	
Input hysteresis ³⁾	HYS CC	$0.09 * V_{FLEX}$	-	-	V	none of the neighbor pads are used as output; AL
		$0.075 * V_{FLEX}$	-	-	V	none of the neighbor pads are used as output; TTL
		75	-	-	mV	two of the neighbor pads are used as output with driver=strong and edge=sharp; AL
Pull-up current ⁴⁾	I_{PUH} CC	30	-	-	μA	V_{IH} ; AL or TTL
		-	-	130	μA	V_{IL} ; AL or TTL
Pull-down current ⁵⁾	I_{PDL} CC	-	-	130	μA	V_{IH} ; AL or TTL
		30	-	-	μA	V_{IL} ; AL
		28	-	-	μA	V_{IL} ; TTL

Electrical Specification 5 V / 3.3 V switchable Pads

Table 3-11 RFast 5V GPIO (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Input leakage current	I_{OZ} CC	-1700	-	1700	nA	$T_J \leq 150^\circ\text{C}$; (0.1 * V_{FLEX}) < V_{IN} < (0.9 * V_{FLEX})
		-2100	-	2100	nA	$T_J \leq 150^\circ\text{C}$; else
		-3000	-	3000	nA	$T_J \leq 170^\circ\text{C}$; (0.1 * V_{FLEX}) < V_{IN} < (0.9 * V_{FLEX})
		-4000	-	4000	nA	$T_J \leq 170^\circ\text{C}$; else
Input high voltage level	V_{IH} SR	0.7 * V_{FLEX}	-	-	V	AL
		2.0	-	-	V	TTL
Input low voltage level	V_{IL} SR	-	-	0.44 * V_{FLEX}	V	AL
		-	-	0.8	V	TTL
Input low threshold variation	V_{ILD} SR	-50	-	50	mV	max. variation of 1ms; V_{FLEX} = constant; AL
Pin capacitance	C_{IO} CC	-	2	3.5	pF	in addition 2.5pF from package to be added
Pad set-up time to get an software update of the configuration active	t_{SET} CC	-	-	100	ns	

- 1) In the formulas the value of C_L needs to be entered in pF to obtain results in ns.
- 2) Rise / fall times are defined 10% - 90% of pad supply voltage.
- 3) Hysteresis is implemented to avoid metastable states and switching due to internal ground bounce. It can't be guaranteed that it suppresses switching due to external system noise.
- 4) Values for Pull-up resistor is defined via parameter R_{MDU} in table VADC 5V.
- 5) Values for Pull-down resistor is defined via parameter R_{MDD} in table VADC 5V.

Table 3-12 RFast 3.3V pad

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
On-Resistance of pad output	$R_{DS(on)}$ CC	8	20	30	Ohm	Driver = RGMII; $I_{OH/OL}$ = 8mA
		125	225	320	Ohm	medium driver; $I_{OH/OL}$ = 2mA
		31	55	80	Ohm	strong driver; $I_{OH/OL}$ = 8mA
Input Duty Cycle	f_D SR	47.5	50	52.5		

Electrical Specification 5 V / 3.3 V switchable Pads
Table 3-12 RFast 3.3V pad (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Rise / Fall time ^{1) 2)}	t_{RF} CC	1.6	-	4.5	ns	$C_L = 25\text{pF}$; driver = strong sharp edge; from $0.2 * V_{FLEX}$ to $0.8 * V_{FLEX}$
		-	-	5	ns	$C_L = 25\text{pF}$; driver = strong sharp edge; from 0.8V to 2.0V (RMII)
		-	-	1	ns	Driver = RGMII; from 20%V to 80%V; $C_L = 15\text{pF}$
		$2+0.57*C_L$	$5.5+0.75*C_L$	$10+1.25*C_L$	ns	driver = medium; $C_L \leq 200\text{pF}$
		$1.5+0.18*C_L$	$1.5+0.28*C_L$	$8+0.4*C_L$	ns	driver = strong edge = medium; $C_L \leq 200\text{pF}$
		$0.75+0.08*C_L$	$0.75+0.11*C_L$	$2.5+0.21*C_L$	ns	driver = strong edge = sharp ; $C_L \leq 200\text{pF}$
Asymmetry of sending	t_{TX_ASYM} CC	-0.4	-	0.4	ns	valid for all data rates excluding clock tolerance
Input frequency	f_{IN} CC	-	-	160	MHz	
Input hysteresis ³⁾	HYS CC	$0.055 * V_{FLEX}$	-	-	V	none of the neighbor pads are used as output; AL
		$0.09 * V_{FLEX}$	-	-	V	none of the neighbor pads are used as output; TTL
		$0.055 * V_{FLEX}$	-	-	V	none of the neighbor pads are used as output;TTL (degraded, used for CIF)
		125	-	-	mV	two of the neighbor pads are used as output with driver=strong and edge=sharp; AL
Pull-up current ⁴⁾	I_{PUH} CC	17	-	-	μA	V_{IH} ; AL and TTL (degraded, used for CIF)
		11	-	-	μA	V_{IH} ; TTL
		-	-	80	μA	V_{IL} ; AL and TTL and TTL (degraded, used for CIF)

Electrical Specification 5 V / 3.3 V switchable Pads

Table 3-12 RFast 3.3V pad (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Pull-down current ⁵⁾	I_{PDL} CC	-	-	105	μA	V_{IH} ; AL and TTL (degraded, used for CIF)
		-	-	115	μA	V_{IH} ; TTL
		19	-	-	μA	V_{IL} ; AL and TTL
		15	-	-	μA	V_{IL} ; TTL (degraded, used for CIF)
Input leakage current	I_{OZ} CC	-1700	-	1700	nA	$T_J \leq 150^\circ\text{C}$; $(0.1 * V_{FLEX}) < V_{IN} < (0.9 * V_{FLEX})$
		-2100	-	2100	nA	$T_J \leq 150^\circ\text{C}$; else
		-3000	-	3000	nA	$T_J \leq 170^\circ\text{C}$; $(0.1 * V_{FLEX}) < V_{IN} < (0.9 * V_{FLEX})$
		-4000	-	4000	nA	$T_J \leq 170^\circ\text{C}$; else
Input high voltage level	V_{IH} SR	$0.7 * V_{FLEX}$	-	-	V	AL
		2.0	-	-	V	TTL
		1.4	-	-	V	TTL (degraded, used for CIF)
Input low voltage level	V_{IL} SR	-	-	$0.42 * V_{FLEX}$	V	AL
		-	-	0.8	V	TTL
		-	-	0.5	V	TTL (degraded, used for CIF)
Input low threshold variation	V_{ILD} SR	-33	-	33	mV	max. variation of 1ms; V_{FLEX} = constant; AL
Pin capacitance	C_{IO} CC	-	2	3.5	pF	in addition 2.5pF from package to be added
Pad set-up time to get an software update of the configuration active	t_{SET} CC	-	-	100	ns	

1) In the formulas the value of C_L needs to be entered in pF to obtain results in ns.

2) Rise / fall times are defined 10% - 90% of pad supply voltage.

3) Hysteresis is implemented to avoid metastable states and switching due to internal ground bounce. It can't be guaranteed that it suppresses switching due to external system noise.

4) Values for Pull-up resistor is defined via parameter R_{MDU} in table VADC 5V.

5) Values for Pull-down resistor is defined via parameter R_{MDD} in table VADC 5V.

Electrical Specification 5 V / 3.3 V switchable Pads

Table 3-13 Class D

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Input leakage current	$I_{OZ\ CC}$	-150	-	150	nA	$T_J \leq 150^\circ\text{C}$; else
		-300 ¹⁾	-	300 ¹⁾	nA	$T_J \leq 150^\circ\text{C}$; PDD option available, or AltRef option available
		-300	-	300	nA	$T_J \leq 170^\circ\text{C}$; else
		-600 ²⁾	-	600 ²⁾	nA	$T_J \leq 170^\circ\text{C}$; PDD option available, or AltRef option available
Pin capacitance	$C_{IO\ CC}$	-	2	3	pF	in addition 2.5pF from package to be added

1) For AN11, 100 nA need to be added.

2) For AN11, 200 nA need to be added .

Table 3-14 ADC Reference Pads

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Input leakage current for V_{AREF}	$I_{OZ2\ CC}$	-1	-	1	μA	$T_J \leq 150^\circ\text{C}$; $V_{AREF} < V_{DDM}$; used for EVADC
		-2	-	2	μA	$T_J \leq 170^\circ\text{C}$; $V_{AREF} < V_{DDM}$; used for EVADC
		-3.5	-	3.5	μA	$T_J \leq 150^\circ\text{C}$; $V_{AREF} \leq V_{DDM} + 50\text{mV}$; used for EVADC
		-7	-	7	μA	$T_J \leq 170^\circ\text{C}$; $V_{AREF} \leq V_{DDM} + 50\text{mV}$; used for EVADC

Table 3-15 Driver Mode Selection for Slow Pads

PDx.2	PDx.1	PDx.0	Port Functionality	Driver Setting
X	X	0	Speed grade 1	medium sharp edge (sm)
X	X	1	Speed grade 2	medium medium edge (m)

Table 3-16 Driver Mode Selection for Fast Pads

PDx.2	PDx.1	PDx.0	Port Functionality	Driver Setting
X	0	0	Speed grade 1	Strong sharp edge (ss)
X	0	1	Speed grade 2	Strong medium edge (sm)
X	1	0	Speed grade 3	medium (m)
X	1	1	Speed grade 4	Reserved, do not use this combination

Table 3-17 Driver Mode Selection for RFast Pads

PDx.2	PDx.1	PDx.0	Port Functionality	Driver Setting
X	0	0	Speed grade 1	Strong sharp edge (ss)
X	0	1	Speed grade 2	Strong medium edge (sm)
X	1	0	Speed grade 3	medium (m)
X	1	1	Speed grade 4	RGMII function is active

3.6 High performance LVDS Pads

This LVDS pad type is used for the high speed chip to chip communication interface of the new TC33xEXT. It consists of a LVDS pad and a fast pad.

$C_L = 2.5 \text{ pF}$ for all LVDS parameters.

Table 3-18 LVDS - IEEE standard LVDS general purpose link (GPL)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Input voltage range	V_I SR	0	-	1600	mV	Driver ground potential difference < 925 mV; $R_T = 100 \text{ Ohm} \pm 10\%$
		0	-	2400	mV	Driver ground potential difference < 925 mV; $R_T = 100 \text{ Ohm} \pm 20\%$
Input differential threshold	V_{idth} SR	-100	-	100	mV	Driver ground potential difference < 900 mV; VDIFFADJ=10 and 11
		-100	-	100	mV	Driver ground potential difference < 925 mV; VDIFFADJ=00 and 01
Receiver differential input impedance	R_{in} CC	80	-	120	Ohm	$V_I \leq 2400 \text{ mV}$
Pad set-up time	t_{SET_LVDS} CC	-	10	13	μs	
Duty cycle	t_{duty} CC	45	-	55	%	

Note: Driver ground potential difference is defined as driver-receiver potential difference, that can result in a voltage shift when comparing driver output voltage level and receiver input voltage level of a transmitted signal.

Note: R_T in table 'LVDS - IEEE standard LVDS general purpose Link (GPL)' is as termination resistor of the receiver according to figure 3-5 in IEEE Std 1596.3-1996 and is represented in [Figure 3-1](#) either by R_{in} or by $R_T=100\text{Ohm}$ but not both. If R_T is mentioned in column Note / Test Condition always the internal resistor R_{in} in [Figure 3-1](#) is the selected one.

default after start-up = CMOS function

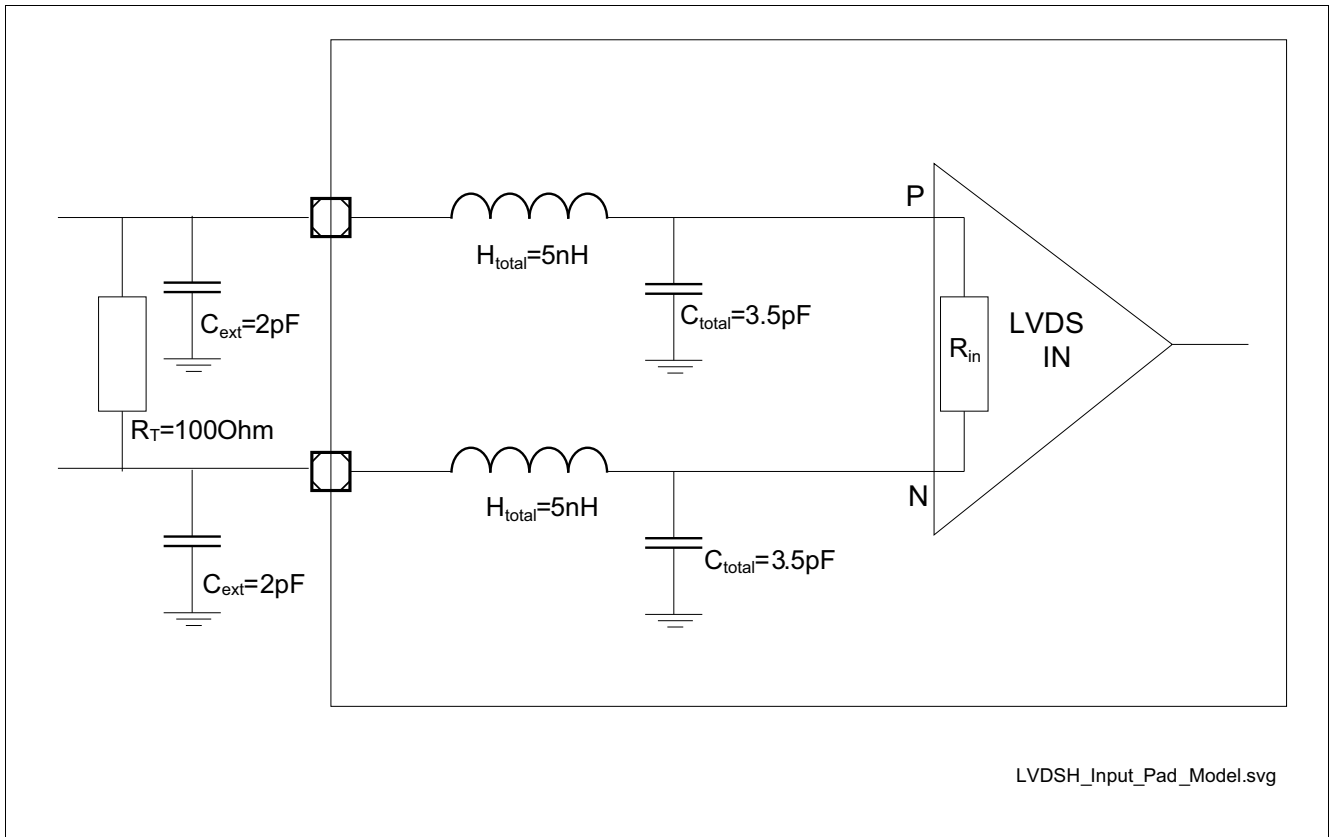


Figure 3-1 LVDS pad Input model

3.7 VADC Parameters

The accuracy of the converter results depends on the reference voltage range. The parameters in the table below are valid for a reference voltage range of $(V_{AREF} - V_{AGND}) \geq 4.5 \text{ V}$. If the reference voltage range is below 4.5 V by a factor of k (e.g. 3.3 V), the accuracy parameters increase by a factor of $1.1/k$ (e.g. $1.1 \times 4.5 / 3.3 = 1.5$).

Noise on supply voltage V_{DDM} influences the conversion. The accuracy (error) parameters are defined for a supply voltage ripple of below 20 mVpp up to 10 MHz (below 5 mVpp above 10 MHz).

Digital functions overlapping analog inputs influence accuracy.

The total unadjusted error (TUE) is defined without noise. The overall deviation depends on TUE and EN_{RMS} (depending on the noise distribution). Example: For a noise distribution of 4 sigma and $EN_{RMS} = 1.0$ the additional peak-peak noise error is $\pm(4 \times 1.0) = 8 \text{ LSB}_{12}$.

The noise reduction feature improves the result by adding additional conversion steps. The conversion times, therefore, increase accordingly ($4 \times t_{ADCI} + 3 \times t_{ADC}$ for each of 1, 3, or 7 steps).

Table 3-19 VADC 5V

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
EVADC IVR output voltage	$V_{DDK} \text{ CC}$	1.13	-	1.33	V	Measured at low temperature.
Deviation of IVR output voltage V_{DDK}	$dV_{DDK} \text{ CC}$	-2	-	2	%	Based on device-specific value
Analog reference voltage ¹⁾	$V_{AREF} \text{ SR}$	4.5	5.0	$V_{DDM} + 0.05$	V	$4.5 \text{ V} \leq V_{DDM} \leq 5.5 \text{ V}$
		2.97	3.3	$V_{DDM} + 0.05$	V	$2.97 \text{ V} \leq V_{DDM} < 4.5 \text{ V}$
Analog reference ground	$V_{AGND} \text{ SR}$	V_{SSM}	V_{SSM}	V_{SSM}	V	V_{SSM} and V_{AGND} are connected together
Analog input voltage range	$V_{AIN} \text{ SR}$	V_{AGND}	-	V_{AREF}	V	V_{AIN} is limited by the respective pad supply voltage; see pin configuration (buffer type)
Converter reference clock	$f_{ADCI} \text{ SR}$	16	40	53.33	MHz	$4.5 \text{ V} \leq V_{DDM} \leq 5.5 \text{ V}$
		16	20	26.67	MHz	$2.97 \text{ V} \leq V_{DDM} < 4.5 \text{ V}$
Total Unadjusted Error ^{2) 3)}	$TUE \text{ CC}$	-4	-	4	LSB	12-bit resolution for primary groups
INL Error ²⁾	$EA_{INL} \text{ CC}$	-3	-	3	LSB	
DNL error ²⁾⁴⁾	$EA_{DNL} \text{ CC}$	-1	-	3	LSB	
Gain Error ²⁾	$EA_{GAIN} \text{ CC}$	-3.5	-	3.5	LSB	
Offset Error ²⁾³⁾	$EA_{OFF} \text{ CC}$	-4	-	4	LSB	
RMS Noise ^{2)5) 6)}	$EN_{RMS} \text{ CC}$	-	0.5	0.8	LSB	Noise reduction level 3
		-	0.5	1.0	LSB	Standard conversion

Electrical Specification VADC Parameters

Table 3-19 VADC 5V (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Reference input charge consumption per conversion (from V_{AREF}) ^{7) 8) 9)}	Q_{CONV} CC	-	-	20	pC	$V_{AIN} = 0$ V (worst case), precharging disabled
		-	-	10	pC	$V_{AIN} = 0$ V (worst case), precharging enabled, $V_{DDM} - 5\% < V_{AREF} < V_{DDM} + 50$ mV
Switched capacitance of an analog input	C_{AINS} CC	-	2.5	3.4	pF	Input buffer disabled
Analog input charge consumption ¹⁰⁾	Q_{AINS} CC	-	-	3.5	pC	Primary groups; $V_{AIN} = V_{AREF}$; $V_{DDM} = 5.0$ V; input buffer enabled; $T_J \leq 150^\circ\text{C}$
		-	-	3.8	pC	Primary groups; $V_{AIN} = V_{AREF}$; $V_{DDM} = 5.0$ V; input buffer enabled; $T_J > 150^\circ\text{C}$
Sampling time	t_S SR	100	-	-	ns	Primary group channel, $4.5\text{ V} \leq V_{DDM} \leq 5.5\text{ V}$; input buffer disabled
		300	-	-	ns	Primary group channel, $4.5\text{ V} \leq V_{DDM} \leq 5.5\text{ V}$; input buffer enabled
		200	-	-	ns	Primary group channel, $2.97\text{ V} \leq V_{DDM} < 4.5\text{ V}$; input buffer disabled
		400	-	-	ns	Primary group channel, $2.97\text{ V} \leq V_{DDM} < 4.5\text{ V}$; input buffer enabled
Sampling time for calibration	t_{SCAL} SR	50	-	-	ns	$4.5\text{ V} \leq V_{DDM} \leq 5.5\text{ V}$
		100	-	-	ns	$2.97\text{ V} \leq V_{DDM} < 4.5\text{ V}$
Input buffer switch-on time	t_{BUF} CC	-	0.4	1	μs	
Wakeup time	t_{WU} CC	-	0.1	0.2	μs	Fast standby mode
		-	1.6	3	μs	Slow standby mode
Broken wire detection delay against V_{AREF}	t_{BWR} CC	-	100	-	cycles	Result above 80% of full scale range, analog input buffer disabled
Broken wire detection delay against V_{AGND}	t_{BWG} CC	-	100	-	cycles	Result below 10% of full scale range, analog input buffer disabled
Converter diagnostics unit resistance ¹¹⁾	R_{CSD} CC	45	-	75	kOhm	
Converter diagnostics voltage accuracy	dV_{CSD} CC	-10	-	10	%	Percentage refers to V_{DDM}

Electrical Specification VADC Parameters

Table 3-19 VADC 5V (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Resistance of the multiplexer diagnostics pull-up device	$R_{MDU\ CC}$	30	-	42	kOhm	$0\text{ V} \leq V_{IN} \leq 0.9 \cdot V_{DDM}$, Automotive Levels
		56	-	78	kOhm	$0\text{ V} \leq V_{IN} \leq 0.9 \cdot V_{DDM}$, TTL Levels
Resistance of the multiplexer diagnostics pull-down device	$R_{MDD\ CC}$	43	-	58	kOhm	$0.1 \cdot V_{DDM} \leq V_{IN} \leq V_{DDM}$, Automotive level
		18	-	25	kOhm	$0.1 \cdot V_{DDM} \leq V_{IN} \leq V_{DDM}$, TTL level
Resistance of the pull-down test device	$R_{PDD\ CC}$	-	-	0.3	kOhm	Measured at pad input voltage $V_{IN} = V_{DDM} / 2$.

- These limits apply to the standard reference input as well as to the alternate reference input.
- Parameter depends on reference voltage range and supply ripple, see introduction.
Resulting worst case combined error is arithmetic combination of TUE and EN_{RMS} .
Tests are done with postcalibration disabled, after completing the startup calibration.
- Analog inputs mapped to pads of the type SLOW influence accuracy. The values for this parameter increase by 3 LSB₁₂.
- Monotonic characteristic, no missing codes when calibrated.
- Parameter EN_{RMS} refers to a 1 sigma distribution.
- Analog inputs mapped to pads of the type SLOW the RMS noise (EN_{RMS}) can be up to 2 LSB₁₂ (soft switching for DC/DC enabled).
- For reduced reference voltages $V_{AREF} < 3.375\text{V}$, the consumed charge QCONV is reduced by the factor of $k_2 = V_{AREF} [\text{V}] / 3.375$. For reduced reference voltages $4.5\text{V} < V_{AREF} \leq 3.375\text{V}$, QCONV is not reduced.
- Maximum charge increases by 15 pC when BWD (Broken Wire Detection) is active.
- Fast compare channels only consume 1/3 of the charge for a primary/secondary group.
- For analog inputs with overlaid digital GPIOs or with PDD function this value increases by 1 pC.
- Use a sample time of at least 1.1 μs to enable proper settling of the test voltage.

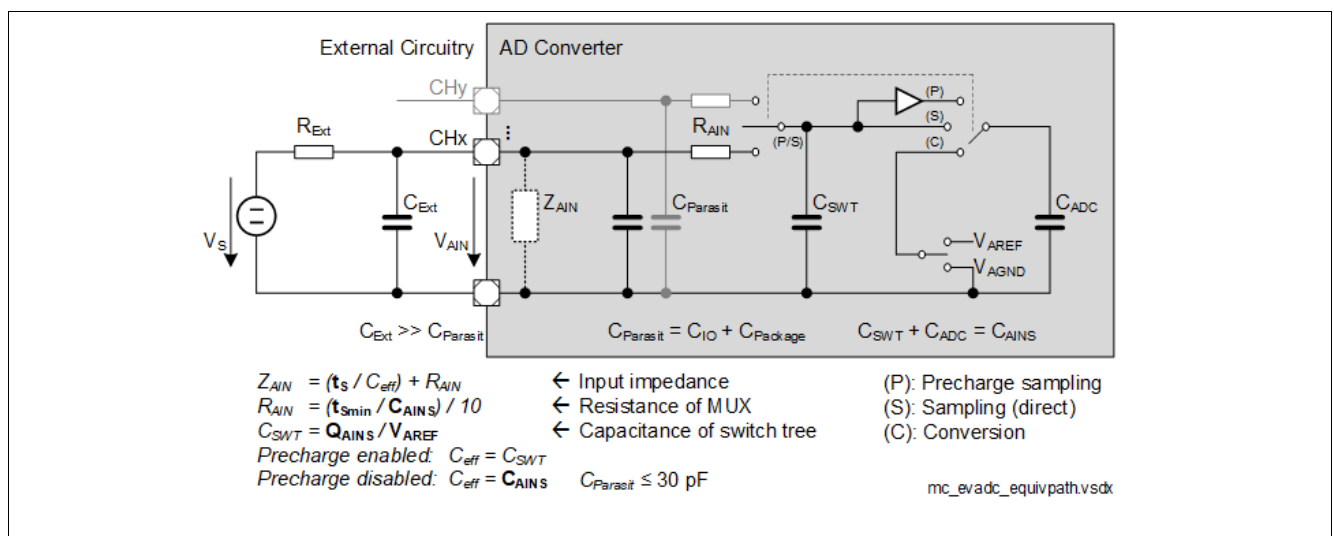


Figure 3-2 Equivalent Circuitry for Analog Inputs

3.8 MHz Oscillator

OSC_XTAL is used as accurate and exact clock source. OSC_XTAL supports 16 MHz to 40 MHz crystals external outside of the device. Support of ceramic resonators is also provided.

Table 3-20 OSC_XTAL

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Input current at XTAL1	I_{IX1} CC	-70	-	70	μA	$V_{IN} > 0\text{V}$; $V_{IN} < V_{EXT}$
Oscillator frequency	f_{OSC} SR	4	-	40	MHz	Direct Input Mode selected, if shaper is not bypassed
		16	-	40	MHz	External Crystal Mode selected
Oscillator start-up time	t_{OSCS} CC	-	-	3 ¹⁾	ms	$20\text{MHz} \leq f_{OSC}$ and 8pF load capacitance
Input voltage at XTAL1 ²⁾	V_{IX} SR	-0.7	-	$V_{EXT} + 0.5$	V	If shaper is not bypassed
Input amplitude (peak to peak) at XTAL1	V_{PPX} SR	$0.3 \cdot V_{EXT}$	-	$V_{EXT} + 1.0$	V	If shaper is not bypassed; $f_{OSC} > 25\text{MHz}$
		$0.35 \cdot V_{EXT}$	-	$V_{EXT} + 1.0$	V	If shaper is not bypassed; $f_{OSC} \leq 25\text{MHz}$
Internal load capacitor	C_{L0} CC	1.30	1.40	1.55	pF	enabled via bit OSCCON.CAP0EN
Internal load capacitor	C_{L1} CC	3.05	3.35	3.70	pF	enabled via bit OSCCON.CAP1EN
Internal load capacitor	C_{L2} CC	7.85	8.70	9.55	pF	enabled via bit OSCCON.CAP2EN
Internal load capacitor	C_{L3} CC	12.05	13.35	14.65	pF	enabled via bit OSCCON.CAP3EN
Internal load stray capacitor between XTAL1 and XTAL2	C_{XINTS} CC	1.15	1.20	1.25	pF	
Internal load stray capacitor between XTAL1 and ground	C_{XTAL1} CC	-	2.5	4	pF	
Duty cycle at XTAL1 ³⁾	DC_{X1} SR	35	-	65	%	$V_{XTAL1} = 0.5 \cdot V_{PPX}$
Absolute RMS jitter at XTAL1 ³⁾	J_{ABSX1} SR	-	-	28	ps	10 KHz to $f_{OSC}/2$
Slew rate at XTAL1 ³⁾	SR_{XTAL1} SR	0.3	-	-	V/ns	Maximum 30% difference between rising and falling slew rate

1) t_{OSCS} is defined from the moment when the Oscillator Mode is set to External Crystal Mode until the oscillations reach an amplitude at XTAL1 of $0.3 \cdot V_{EXT}$. This value depends on the frequency of the used external crystal. For faster crystal frequencies this value decrease.

2) For Supply ($V_{EXT} < 5.3\text{V}$ V_{IX}) min could be down to -0.9V. For XTAL1 an input level down to -0.9V will not cause a damage or a reliability problem operating with an external crystal.

3) Square wave input signal for XTAL1.

Note: It is strongly recommended to measure the oscillation allowance (negative resistance) in the final target system (layout) to determine the optimal parameters for the oscillator operation. Please refer to the limits specified by the crystal or ceramic resonator supplier.

3.9 Back-up Clock

The back-up clock provides an alternative clock source.

Table 3-21 Back-up Clock

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Back-up clock accuracy before trimming	f_{BACKUT} CC	70	100	130	MHz	$V_{\text{EXT}} \geq 2.97\text{V}$
Back-up clock accuracy after trimming ¹⁾	f_{BACKT} CC	98	100	102	MHz	$V_{\text{EXT}} \geq 2.97\text{V}$
Standby clock	f_{SB} CC	25	70	110	kHz	$V_{\text{EXT}} \geq 2.97\text{V}$

1) A short term trimming providing the accuracy required by LIN communication is possible by periodic trimming every 2 ms for temperature and voltage drifts up to temperatures of 125 celcius

3.10 Temperature Sensor

Table 3-22 DTS PMS

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Measurement time for each conversion ¹⁾	t_M CC	-	-	2.7	ms	Measured from cold power-on reset release
Calibration reference accuracy	T_{CALACC} CC	-1	-	1	°C	calibration points @ $T_J = -40^{\circ}\text{C}$ and $T_J = 127^{\circ}\text{C}$
Accuracy over temperature range	T_{NL} CC	-2	-	2	°C	T_{CALACC} has to be added in addition
DTS temperature range	T_{SR} SR	-40	-	170	°C	

1) After warm reset t_M is not restarted and is measured from last conversion.

Table 3-23 DTS Core

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Measurement time for each conversion ¹⁾	t_M CC	-	-	2.7	ms	Measured from cold power-on reset release
Temperature difference between on chip temperature sensors	ΔT CC	-3	-	3	°C	
Calibration reference accuracy	T_{CALACC} CC	-2	-	2	°C	calibration points @ $T_J = -40^{\circ}\text{C}$ and $T_J = 127^{\circ}\text{C}$
Accuracy over temperature range	T_{NL} CC	-2	-	2	°C	T_{CALACC} has to be added in addition
DTS temperature range	T_{SR} SR	-40	-	170	°C	

1) After warm reset t_M is not restarted and is measured from last conversion.

3.11 Power Supply Current

The total power supply current defined below consists of leakage and switching component.

Application relevant values are typically lower than those given in the following table and depend on the customer's system operating conditions (e.g. thermal connection or used application configurations).

The operating conditions for the parameters in the following table are:

The ADAS realistic (real) power pattern defines the following conditions:

- $T_J = 150\text{ °C}$
- $f_{SRI} = f_{CPUx} = 200\text{ MHz}$
- $f_{SPU} = 200\text{ MHz}$; (FFT length =512, complex windowing)
- $f_{SPB} = f_{STM} = f_{BAUD1} = f_{BAUD2} = f_{ASCLINx} = 100\text{ MHz}$
- $V_{DD} = 1.275\text{ V}$
- $V_{DDP3/EXT/FLEX/EVRSB} = 3.366\text{ V}$
- $V_{DDM} = 5.1\text{ V}$
- CPU0 (IPC=1.2) and CPU1 (IPC=0.6) cores are active including CPU0 lockstep core
- the following modules are inactive: HSM, GETH, FCE, and MTU

The ADAS maximum (max) power pattern defines the following conditions:

- $T_J = 150\text{ °C}$
- $f_{SRI} = f_{CPUx} = 200\text{ MHz}$
- $f_{SPU} = 200\text{ MHz}$; (FFT length =2048, complex windowing)
- $f_{SPB} = f_{STM} = f_{BAUD1} = f_{BAUD2} = f_{ASCLINx} = 100\text{ MHz}$
- $V_{DD} = 1.375\text{ V}$
- $V_{DDP3/EXT/FLEX/EVRSB} = 3.63\text{ V}$
- $V_{DDM} = 5.5\text{ V}$
- CPU0 (IPC=1.2) and CPU1 (IPC=1.2) cores are active including CPU0 lockstep core
- the following modules are inactive: GETH, FCE, and MTU

Table 3-24 Current Consumption

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Σ Sum of I_{DD} core and peripheral supply currents (incl. $I_{DDPORST} + \Sigma I_{DDC0} + \Sigma I_{DDCxx} + I_{DDGTM} + I_{DDSB}$)	$I_{DDRAIL\ CC}$	-	-	920	mA	ADAS max power pattern; valid for Feature Package DA and DZ products
		-	-	730	mA	ADAS real power pattern; valid for Feature Package DA and DZ products

Electrical Specification Power Supply Current

Table 3-24 Current Consumption (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
I_{DD} core current during active power-on reset (PORST pin held low). Leakage current of core domain. ¹⁾	$I_{DDPORST}$ CC	-	-	110	mA	$V_{DD} = 1.275V$; $T_J = 125^\circ C$; valid for Feature Package DA and DZ products
		-	-	192	mA	$V_{DD} = 1.275V$; $T_J = 150^\circ C$; valid for Feature Package DA and DZ products
		-	-	240	mA	$V_{DD} = 1.275V$; $T_J = 160^\circ C$; valid for Feature Package DA and DZ products
		-	-	260	mA	$V_{DD} = 1.275V$; $T_J = 165^\circ C$; valid for Feature Package DA and DZ products
Σ Sum of I_{DDP3} 3.3 V supply currents	$I_{DDP3RAIL}$ CC	-	-	35	mA	max power pattern incl. Flash read current and Dflash programming current.
		-	-	26 ²⁾	mA	real power pattern incl. Flash read current and Dflash programming current.
Σ Sum of external I_{EXT} supply currents (incl. $I_{EXTFLEX} + I_{EVRSB} + I_{EXTLVDS}$)	$I_{EXTRAIL}$ CC	-	-	44	mA	max power pattern; ADAS max power pattern
		-	-	39	mA	real power pattern; ADAS real power pattern
I_{EXT} and I_{FLEX} supply current	$I_{EXTFLEX}$ CC	-	-	11 ¹⁾³⁾	mA	real power pattern with port activity absent; PORST output inactive.
I_{EVRSB} supply current ¹⁾	I_{EVRSB} CC	-	-	8.5	mA	real power pattern; PMS/EVR module current considered without SCR and Standby RAM during RUN mode.
Σ Sum of external I_{DDM} supply currents (incl. $I_{DDMEVADC} + I_{DDMEDSADC}$)	I_{DDM} CC	-	-	6	mA	real power pattern; ADAS real power pattern

Electrical Specification Power Supply Current

Table 3-24 Current Consumption (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Σ Sum of all currents (incl. $I_{\text{EXTRAIL}} + I_{\text{DDMRIL}} + I_{\text{DDx3RAIL}} + I_{\text{DD}}$)	$I_{\text{DDTOT CC}}$	-	-	1011	mA	ADAS max power pattern; valid for Feature Package DA and DZ products
		-	-	801	mA	ADAS real power pattern; valid for Feature Package DA and DZ products
Σ Sum of all currents with DC-DC EVRC regulator active ⁴⁾	$I_{\text{DDTOTDC3 CC}}$	-	-	450	mA	ADAS real power pattern; valid for Feature Package DA and DZ products; EVRC LCD CDC reset settings with 72% efficiency; $V_{\text{EXT}} = 3.3\text{V}$
Σ Sum of all currents with DC-DC EVRC regulator active ⁴⁾	$I_{\text{DDTOTDC5 CC}}$	-	-	330	mA	ADAS real power pattern; valid for Feature Package DA and DZ products; EVRC LCD CDC reset settings with 72% efficiency; $V_{\text{EXT}} = 5\text{V}$
Σ Sum of all currents (SLEEP mode) ¹⁾	$I_{\text{SLEEP CC}}$	-	-	25	mA	All CPUs in idle, All peripherals in sleep, $f_{\text{SRI/SPB}} = 1\text{ MHz}$ via LPDIV divider; $T_{\text{J}} = 25^{\circ}\text{C}$
Σ Sum of all currents (STANDBY mode) drawn at V_{EVRSB} supply pin ⁵⁾	$I_{\text{STANDBY CC}}$	-	-	130 ⁶⁾	μA	32 kB Standby RAM block active. SCR inactive. Power to remaining domains switched off. $T_{\text{J}} = 25^{\circ}\text{C}$; $V_{\text{EVRSB}} = 5\text{V}$
Maximum power dissipation ⁷⁾	$PD\text{ SR}$	-	-	1720	mW	ADAS max power pattern; valid for Feature Package DA and DZ products
		-	-	1260	mW	ADAS real power pattern; valid for Feature Package DA and DZ products

1) Limits are defined for real power pattern ($V_{\text{DD}}=1.275\text{V}$). For max power pattern limit has to be multiplied by the factor 1.22.

Electrical Specification Power Supply Current

- 2) Realistic Pflash read pattern with 50% Pflash bandwidth utilization and a code mix of 50% 0s and 50% 1s. A common decoupling capacitor of at least 100nF for (V_{DDP3}) is used. Continuous Dflash programming in burst mode with 3.3 V supply and realistic Pflash read access in parallel. Erase currents of the corresponding flash modules are less than the respective programming currents at V_{DDP3} pin. Programming and erasing flash may generate transient current spikes of up to 45 mA / 20 ns which are handled by the decoupling and buffer capacitors. This parameter is relevant for external power supply dimensioning and not for thermal considerations.
- 3) The current consumption includes only minimal port activity.
- 4) The total current drawn from external regulator is estimated with 72% EVRC SMPS regulator efficiency. IDDTOTDCx is calculated from IDDTOT using the scaled core current $[(I_{DD} \times V_{DD}) / (V_{in} \times \text{Efficiency})]$ and constitutes all other rail currents and IDDM.
- 5) The same current limits apply also for the other power pattern.
- 6) Σ Sum of all currents during RUN mode at VEVRSB supply pin is less than (IEVRSB + 4 mA Standby RAM current + ISCRSB if SCR active). Σ It is recommended to have atleast 100 nF decoupling capacitor at this pin. 32kB of Standby SRAM contributes less than 10uA to ISTANDBY current.
- 7) The values are only valid if all supplies are applied from external and do not contain the power losses of EVR33 and EVRC.

Table 3-25 Module Current Consumption

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
I_{DDP3} supply current for programming of a Pflash or Dflash bank ¹⁾	$I_{DDP3PROG}$ CC	-	-	25	mA	Pflash 3.3V programming current adder when using external 3.3V supply.
		-	-	9 ²⁾	mA	Pflash 3.3V programming current adder when using external 5V supply.
I_{EXT} supply current added by LVDS pads in LVDS mode ¹⁾	$I_{EXTLVDS}$ CC	-	-	9 ³⁾	mA	real power pattern; 6 pairs of LVDS pins active with receive function
Σ Sum of external I_{DDM} supply currents (incl. $I_{DDMEVADC} + I_{DDMEDSADC}$)	I_{DDM} CC	-	-	6	mA	real power pattern; current for EVADC modules; 2 EVADC modules active.
		-	-	12 ⁴⁾	mA	max power pattern; current for EVADC modules only; 4 EVADC modules active.
I_{DDP3} supply current for erasing of a Pflash or Dflash bank	$I_{DDP3ERASE}$ CC	-	-	25	mA	Pflash 3.3V erasing current adder when using external 3.3V supply.

Electrical Specification Power Supply Current

Table 3-25 Module Current Consumption (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
SCR 8-bit Standby Controller current incl. PMS in STANDBY Mode drawn at $V_{EVR\text{SB}}$ supply pin	$I_{\text{SCR}\text{SB}}$ CC	-	-	7.0	mA	SCR power pattern incl. PMS current consumption with fback clock active; $f_{\text{SYS_SCR}} = 20\text{MHz}$; $T_J = 150^\circ\text{C}$
		-	0.150	-	mA	SCR power pattern incl. PMS current consumption with fback inactive; $f_{\text{SYS_SCR}} = 70\text{kHz}$; $T_J = 25^\circ\text{C}$
SCR 8-bit Standby Controller CPU in IDLE mode ⁵⁾	$I_{\text{SCR}\text{IDLE}}$ CC	-	-	3.5	mA	real power pattern. CPU set into idle mode.

- 1) The same current limits apply also for the other power pattern.
- 2) During Pflash programming at 5V, additional 2 mA is drawn at VEXT supply rail.
- 3) A single LVDS pair with receive function is limited to 1.5mA (t_{EXTLVDS}).
- 4) A single VADC unit consumes 1.3 mA.
- 5) Limits are defined for real power pattern ($V_{\text{DD}} = 1.275\text{V}$). For max power pattern limit has to be multiplied by the factor 1.22.

Table 3-26 Module Core Current Consumption

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
I_{DD} core current of CPUx main core with CPUx lockstep core inactive	$I_{\text{DDC}\text{x0}}$ CC	-	-	70	mA	max power pattern; $f_{\text{CPU}} = 300\text{MHz}$; $\text{IPC} = 1.2$
		-	-	45	mA	real power pattern; $f_{\text{CPU}} = 300\text{MHz}$; $\text{IPC} = 0.6$
I_{DD} core current of CPUx main core with CPUx lockstep core active	$I_{\text{DDC}\text{xx}}$ CC	-	-	$I_{\text{DDC}\text{x0}} + 50$	mA	max power pattern; $f_{\text{CPU}} = 300\text{MHz}$; $\text{IPC} = 1.2$
		-	-	$I_{\text{DDC}\text{x0}} + 40$	mA	real power pattern; $f_{\text{CPU}} = 300\text{MHz}$; $\text{IPC} = 0.6$
I_{DD} core current added by HSM	$I_{\text{DDH}\text{SM}}$ CC	-	-	20 ¹⁾	mA	max power pattern; HSM running at 100MHz.
I_{DD} core current added by SPU	$I_{\text{DDSPU}\text{1}}$ CC	-	-	360 ^{2) 3)}	mA	CTRL.DIV = 00; SPU @ 300 MHz; FFT length 2048; DATSRC=EMEM.

Electrical Specification Power Supply Current

Table 3-26 Module Core Current Consumption (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
I_{DD} core current added by SPU	I_{DDSPU2} CC	-	-	310 ²⁾³⁾	mA	CTRL.DIV = 00; SPU @ 300 MHz; FFT length 512; DATSRC=EMEM.
I_{DD} core dynamic current load jump during IDDSPU1 pattern.	$I_{DDSPULJ1}$ CC	-	-	390 ⁴⁾	mA	CTRL.DIV = 00; SPU @ 300 MHz; FFT length 2048; DATSRC=EMEM.
I_{DD} core dynamic current load jump during IDDSPU2 pattern.	$I_{DDSPULJ2}$ CC	-	-	310 ⁴⁾	mA	CTRL.DIV = 00; SPU @ 300 MHz; FFT length 512; DATSRC=EMEM.
I_{DD} core dynamic current added by LBIST	I_{DDLBI} CC	-	-	200 ⁵⁾	mA	LBIST Configuration A; $1.2V \leq V_{DD}$
I_{DD} core dynamic current added by MBIST	$I_{DDMBIST}$ CC	-	-	200	mA	Non Destructive Test (4N) on all RAMs sequentially; fMBIST = 300MHz; tMBIST < 6ms.

- 1) The current consumption includes basic HSM activity incl. AES module.
- 2) The current is estimated as the sum of the SPU base load current at clock activation and average current caused by SPU dynamic activity as defined in the conditions. Secondary Voltage Monitor over-voltage threshold shall be set to $V_{DD} + 10\%$ and under-voltage threshold shall be set to $V_{DD} - 9\%$ respectively.
During the SPU operational phase for I_{DDSPU1} usecase, the externally supplied V_{DD} voltage has to be equal or greater than 1.225V (V_{DD} nominal - 2%) for static accuracy part and the overall static and dynamic at the V_{DD} supply pin shall be limited to (V_{DD} nominal - 8%).
- 3) The current is estimated as the sum of the SPU base load current at clock activation and average current caused by SPU dynamic activity as defined in the conditions. Secondary Voltage Monitor over-voltage threshold shall be set to $V_{DD} + 10\%$ and under-voltage threshold shall be set to $V_{DD} - 9\%$ respectively.
- 4) The dynamic current load jump during SPU activity as defined by the conditions observed at the V_{DD} pin beyond a settling time duration of 20 us.
- 5) LBIST is executed either during start-up phase or can be triggered by application software. Secondary voltage monitors are inactive during the LBIST execution time (t_{LBIST}).
During the start-up phase externally supplied V_{DD} voltage has to be equal or greater than 1.2V (V_{DD} nominal - 4%) for static accuracy.
If V_{DD} is supplied internally by EVRC, EVRC takes care not to violate the V_{DD} 1.2V static under voltage limit.

3.11.1 Calculating the 1.3 V Current Consumption

The current consumption of the 1.3 V rail compose out of two parts:

- Static current consumption
- Dynamic current consumption

Electrical Specification Power Supply Infrastructure and Supply Start-up

The static current consumption is related to the device temperature T_J and the dynamic current consumption depends of the configured clocking frequencies and the software application executed. These two parts needs to be added in order to get the rail current consumption.

(3.1)

$$I_0 = 3,0289 \left[\frac{\text{mA}}{\text{C}} \right] \times e^{0,0228 \times T_J[\text{C}]}$$

(3.2)

$$I_0 = 6,0675 \left[\frac{\text{mA}}{\text{C}} \right] \times e^{0,0226 \times T_J[\text{C}]}$$

Equation (3.1) defines the typical static current consumption and **Equation (3.2)** defines the maximum static current consumption. Both functions are valid for $V_{DD} = 1.275 \text{ V}$.

3.12 Power Supply Infrastructure and Supply Start-up

3.12.1 Supply Ramp-up and Ramp-down Behavior

Start-up slew rates for supply rails shall comply to SR (see [Table 3-30](#) Supply Ramp).

3.12.1.1 Single Supply mode (a)

Electrical Specification Power Supply Infrastructure and Supply Start-up

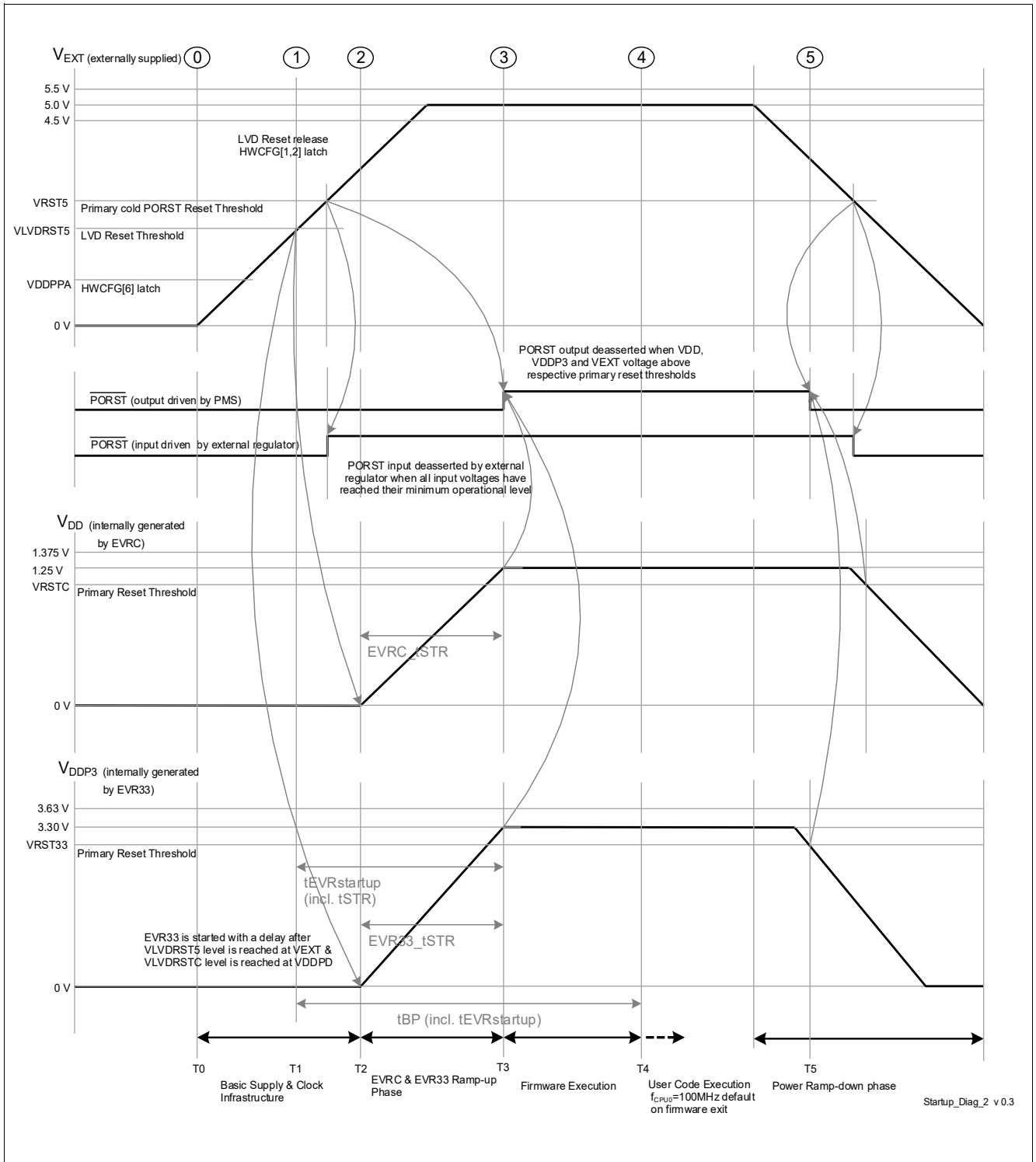


Figure 3-3 Single Supply mode (a) - V_{EXT} (5 V) single supply

$V_{EXT} = 5 V$ single supply mode. V_{DD} and V_{DDP3} are generated internally by the EVRC and EVR33 internal regulators.

- The rate at which current is drawn from the external regulator (dI_{EXT}/dt) is limited during the basic infrastructure and EVRx regulator start-up phase (T_0 up to T_2) to a maximum of 100 mA with 100 μs settling time. Start-up slew rates for supply rails shall comply to parameter SR. The slope is defined as the maximal tangential slope between 0% to 100% voltage level. Actual waveform may not represent the specification.

Electrical Specification Power Supply Infrastructure and Supply Start-up

- Furthermore it is also ensured that the current drawn from the regulator (dI_{DD}/dt) is limited during the Firmware start-up phase (T3 up to T4) to a maximum of 100 mA with 100 μ s settling time.
- PORST is active/asserted when either PORST (input) or PORST (output) is active/asserted.
- PORST (input) active means that the reset is held active by external agents by pulling the PORST pin low. It is recommended to keep the PORST (input) asserted until the external supply is above the respective primary reset threshold.
- PORST (output) active means that μ C asserts the reset internally and drives the PORST pin low thus propagating the reset to external devices. The PORST (output) is asserted by the μ C when at least one among the three supply domains (V_{DD} , V_{DDP3} or V_{EXT}) violate their primary under-voltage reset thresholds. The PORST (output) is de-asserted by the μ C when all supplies are above their primary reset thresholds and the basic supply and clock infrastructure is available. During reset release at T3, the load jump of up to 150 mA (dI_{DD}) is expected.
- The power sequence as shown in [Figure 3-3](#) is enumerated below
 - T1 up to T2 refers to the period in time when basic supply and clock infrastructure components are available as the external supply ramps up. The bandgap and internal clock sources are started. The supply mode is evaluated based on the HWCFG[1:2,4,5,6] and TESTMODE pins. These events are initiated after LVD reset release at T1 after V_{EXT} and VEVRSB supply rails have reached VLVD RST5 level and internal pre-regulator VDDPD voltage has reached VLVD RSTC level.
 - T2 refers to the point in time where consequently a soft start of EVRC and EVR33 regulators are initiated. PORST (input) does not have any affect on EVR33 or EVRC output and regulators continue to generate the respective voltages though PORST is asserted and the device is in reset state. The generated voltage follows a soft ramp-up over the tSTR time to avoid overshoots.
 - T3 refers to the point in time when all supplies are above their primary reset thresholds denoted by VRST5, VRST33 and VRSTC supply voltage levels. EVRC and EVR33 regulators have ramped up. (output) is de-asserted and HWCFG[3:5] pins are latched on PORST rising edge by SCU. Firmware execution is initiated. The time between T1 and T3 is documented as tEVRstartup.
 - T4 refers to the point in time when Firmware execution is completed and User code execution starts with CPU0 at a default frequency of 100 MHz. The time between T0 and T4 is documented as tBP.
 - T5 refers to the point in time during the ramp-down phase when at least one of the externally provided or generated supplies (V_{DD} , V_{DDP3} or V_{EXT}) drop below their respective primary under-voltage reset thresholds.

3.12.1.2 Single Supply mode (e)

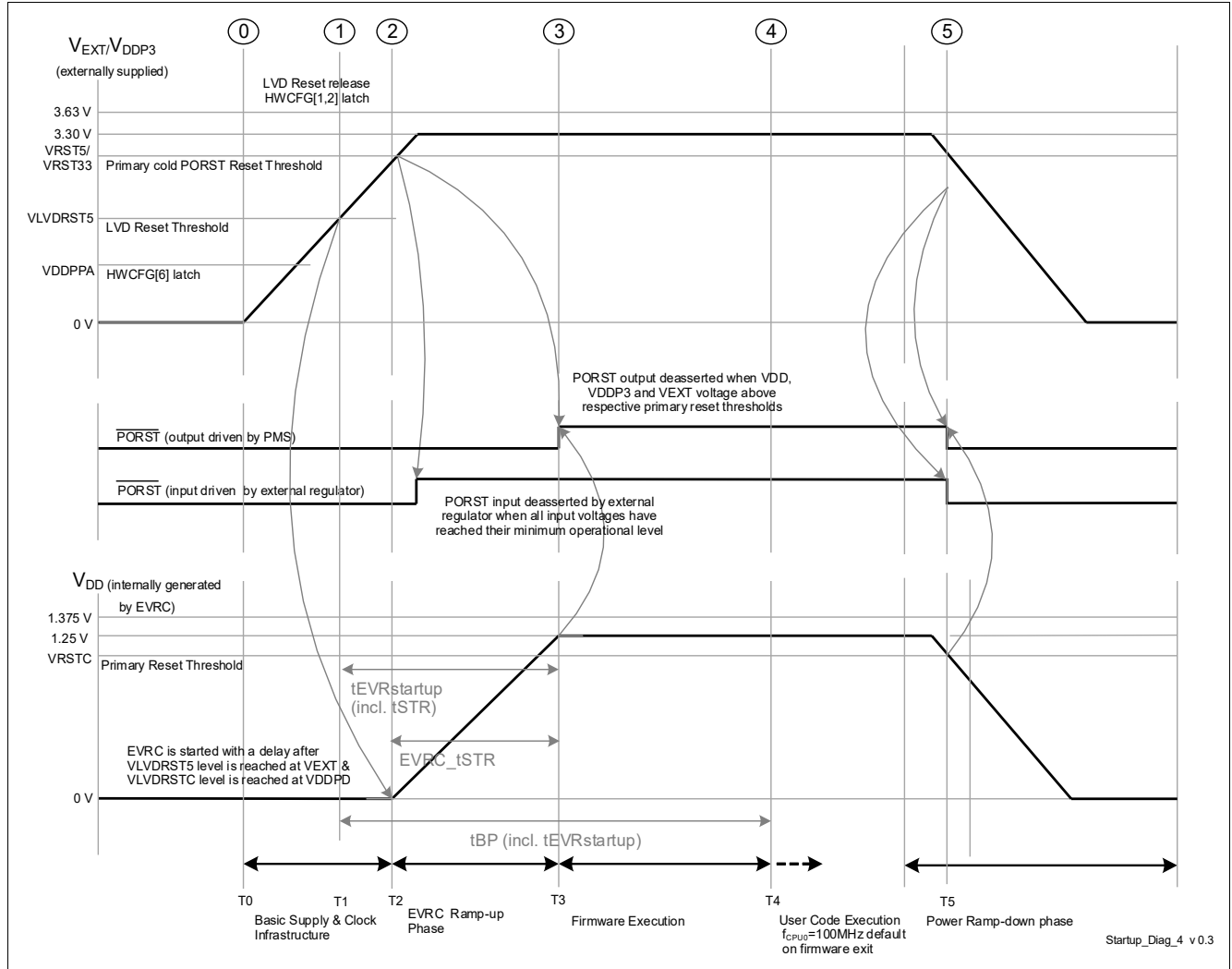


Figure 3-4 Single Supply mode (e) - (V_{EXT} & V_{DDP3}) 3.3 V single supply

$V_{EXT} = V_{DDP3} = 3.3\text{ V}$ single supply mode. V_{DD} is generated internally by the EVRC regulator.

- The rate at which current is drawn from the external regulator (dI_{EXT}/dt) is limited in the Start-up phase to a maximum of 100 mA with 100 μs settling time. Start-up slew rates for supply rails shall comply to SR. The slope is defined as the maximal tangential slope between 0% to 100% voltage level. Actual waveform may not represent the specification.
- $PORST$ is active/asserted when either $PORST$ (input) or $PORST$ (output) is active/asserted.
- $PORST$ (input) active means that the reset is held active by external agents by pulling the $PORST$ pin low. It is recommended to keep the $PORST$ (input) asserted until the external supply is above the respective primary reset threshold.
- $PORST$ (output) active means that μC asserts the reset internally and drives the $PORST$ pin low thus propagating the reset to external devices. The $PORST$ (output) is asserted by the μC when at least one among the three supply domains (V_{DD} , V_{DDP3} or V_{EXT}) violate their primary under-voltage reset thresholds. The $PORST$ (output) is de-asserted by the μC when all supplies are above their primary reset thresholds and the basic supply and clock infrastructure is available. During reset release at T3, the load jump of up to 150 mA (dI_{DD}) is expected.

Electrical Specification Power Supply Infrastructure and Supply Start-up

- The power sequence as shown in [Figure 3-4](#) is enumerated below
 - T1 up to T2 refers to the period in time when basic supply and clock infrastructure components are available as the external supply ramps up. The bandgap and internal clock sources are started. The supply mode is evaluated based on the HWCFG[1:2,4,5,6] and TESTMODE pins. These events are initiated after LVD reset release at T1 after V_{EXT} and VEVRSB supply rails have reached VLVD RST5 level and internal pre-regulator VDDPD voltage has reached VLVD RSTC level.
 - T2 refers to the point in time where consequently a soft start of EVRC regulator is initiated. PORST (input) does not have any affect on EVRC output and regulators continue to generate the respective voltages though PORST is asserted and the device is in reset state. The generated voltage follows a soft ramp-up over the tSTR time to avoid overshoots.
 - T3 refers to the point in time when all supplies are above their primary reset thresholds denoted by VRST5, VRST33 and VRSTC supply voltage levels. EVRC regulator has ramped up. PORST (output) is de-asserted and HWCFG[3:5] pins are latched on PORST rising edge by SCU. Firmware execution is initiated. The time between T1 and T3 is documented as tEVRstartup.
 - T4 refers to the point in time when Firmware execution is completed and User code execution starts with CPU0 at a default frequency of 100 MHz. The time between T0 and T4 is documented as tBP.
 - T5 refers to the point in time during the ramp-down phase when at least one of the externally provided or generated supplies (V_{DD} , V_{DDP3} or V_{EXT}) drop below their respective primary under-voltage reset thresholds.

3.12.1.3 External Supply mode (d)

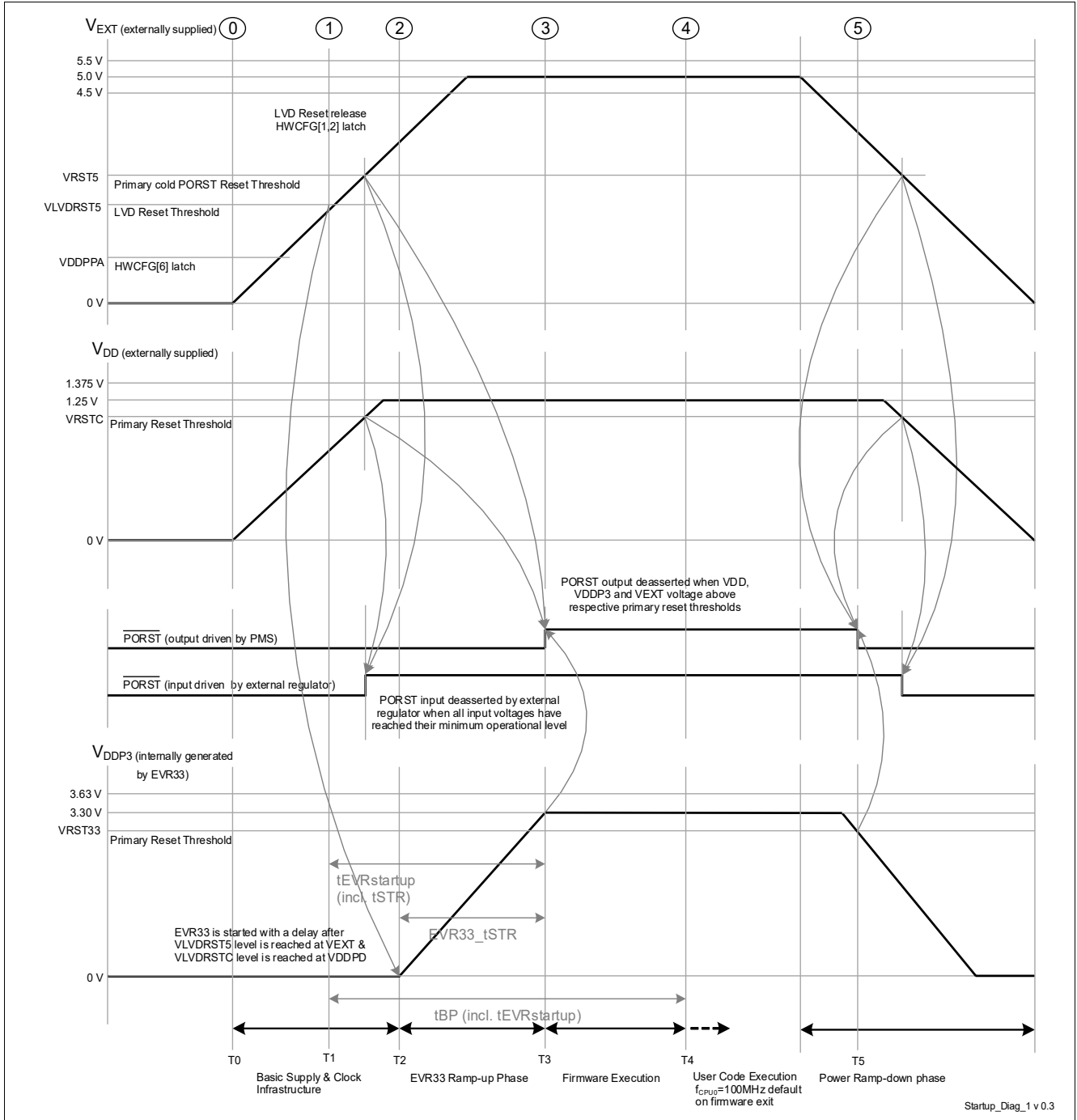


Figure 3-5 External Supply mode (d) - V_{EXT} and V_{DD} externally supplied

$V_{EXT} = 5\text{ V}$ and V_{DD} supplies are externally supplied. 3.3V is generated internally by the EVR33 regulator.

- External supplies V_{EXT} and V_{DD} may ramp-up or ramp-down independent of each other with regards to start, rise and fall time(s). Start-up slew rates for supply rails shall comply to SR. The slope is defined as the maximal tangential slope between 0% to 100% voltage level. Actual waveform may not represent the specification. It is expected that during start-up, V_{EXT} ramps up before V_{DD} rail. In case V_{DD} voltage rail is ramped up before V_{EXT} ; V_{DD} supply overshoots during start-up shall be limited within the operational voltage range.

Electrical Specification Power Supply Infrastructure and Supply Start-up

- The rate at which current is drawn from the external regulator (dI_{EXT}/dt or dI_{DD}/dt) is limited in the Start-up phase to a maximum of 100 mA with 100 μ s settling time.
- PORST is active/asserted when either PORST (input) or PORST (output) is active/asserted.
- PORST (input) active means that the reset is held active by external agents by pulling the PORST pin low. It is recommended to keep the PORST (input) asserted until all the external supplies are above their primary reset thresholds.
- PORST (output) active means that μ C asserts the reset internally and drives the PORST pin low thus propagating the reset to external devices. The PORST (output) is asserted by the μ C when at least one among the three supply domains (V_{DD} , V_{DDP3} or V_{EXT}) violate their primary under-voltage reset thresholds. The PORST (output) is de-asserted by the μ C when all supplies are above their primary reset thresholds and the basic supply and clock infrastructure is available. During reset release at T3, the load jump of up to 150 mA (dI_{DD}) is expected.
- The power sequence as shown in [Figure 3-5](#) is enumerated below
 - T1 up to T2 refers to the period in time when basic supply and clock infrastructure components are available as the external supply ramps up. The bandgap and internal clock sources are started. The supply mode is evaluated based on the HWCFG[1:2,4,5,6] and TESTMODE pins. These events are initiated after LVD reset release at T1 after V_{EXT} and VEVR33 supply rails have reached VLVDVRS5 level and internal pre-regulator VDDPD voltage has reached VLVDVRS5 level.
 - T2 refers to the point in time where consequently a soft start of EVR33 regulator is initiated. PORST (input) does not have any affect on EVR33 output and regulators continue to generate the respective voltages though PORST is asserted and the device is in reset state. The generated voltage follows a soft ramp-up over the tSTR time to avoid overshoots.
 - T3 refers to the point in time when all supplies are above their primary reset thresholds denoted by VRST5, VRST33 and VRSTC supply voltage levels. EVR33 regulators has ramped up. PORST (output) is de-asserted and HWCFG[3:5] pins are latched on PORST rising edge by SCU. Firmware execution is initiated. The time between T1 and T3 is documented as tEVRstartup.
 - T4 refers to the point in time when Firmware execution is completed and User code execution starts with CPU0 at a default frequency of 100 MHz. The time between T0 and T4 is documented as tBP.
 - T5 refers to the point in time during the ramp-down phase when at least one of the externally provided or generated supplies (V_{DD} , V_{DDP3} or V_{EXT}) drop below their respective primary under-voltage reset thresholds.

3.12.1.4 External Supply mode (h)

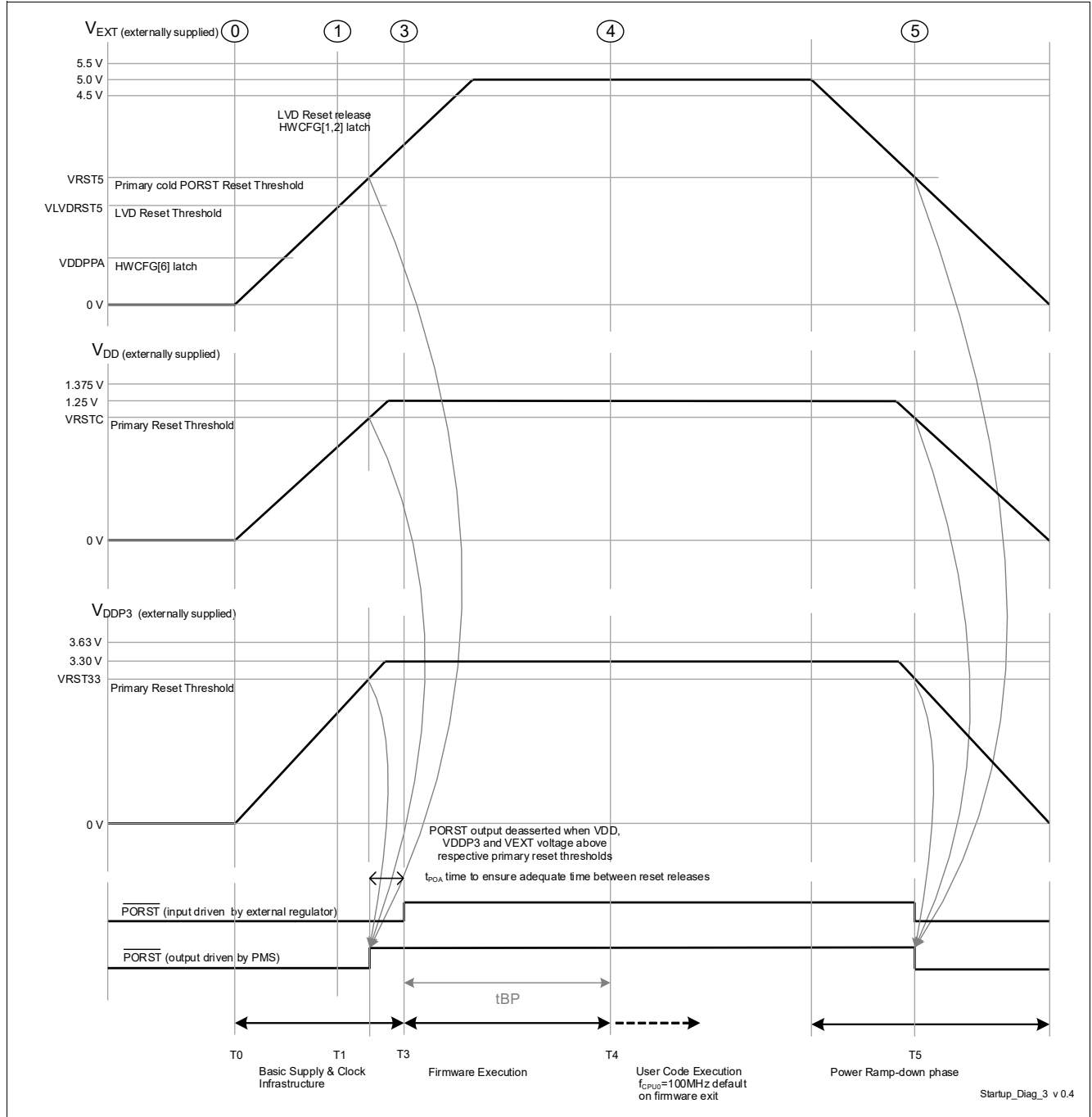


Figure 3-6 External Supply mode (h) - V_{EXT} , V_{DDP3} & V_{DD} externally supplied

All supplies, namely V_{EXT} , V_{DDP3} & V_{DD} are externally supplied.

- External supplies V_{EXT} , V_{DDP3} & V_{DD} may ramp-up or ramp-down independent of each other with regards to start, rise and fall time(s). Start-up slew rates for supply rails shall comply to SR. The slope is defined as the maximal tangential slope between 0% to 100% voltage level. Actual waveform may not represent the specification. It is expected that during start-up, V_{EXT} ramps up before V_{DDP3} and V_{DD} rails. Incase smaller voltage rails are ramped up before V_{EXT} , V_{DD} and V_{DDP3} supply overshoots during start-up shall be limited within the operational voltage ranges of the respective rails.

Electrical Specification Power Supply Infrastructure and Supply Start-up

- The rate at which current is drawn from the external regulator (dI_{EXT}/dt , dI_{DD}/dt or dI_{DDP3}/dt) is limited in the Start-up phase to a maximum of 100 mA with 100 μ s settling time.
- PORST is active/asserted when either PORST (input) or PORST (output) is active/asserted.
- PORST (input) active means that the reset is held active by external agents by pulling the PORST pin low. It is recommended to keep the PORST (input) asserted until all the external supplies are above their primary reset thresholds.
- PORST (output) active means that μ C asserts the reset internally and drives the PORST pin low thus propagating the reset to external devices. The PORST (output) is asserted by the μ C when at least one among the three supply domains (V_{DD} , V_{DDP3} or V_{EXT}) violate their primary under-voltage reset thresholds. The PORST (output) is de-asserted by the μ C when all supplies are above their primary reset thresholds and the basic supply and clock infrastructure is available. During reset release at T3, the load jump of up to 150 mA (dI_{DD}) is expected.
- The power sequence as shown in [Figure 3-6](#) is enumerated below
 - T1 up to T3 refers to the period in time when basic supply and clock infrastructure components are available as the external supply ramps up. The bandgap and internal clock sources are started. The supply mode is evaluated based on the HWCFG[1:2,4,5,6] and TESTMODE pins. These events are initiated after LVD reset release at T1 after V_{EXT} and V_{EVRSB} supply rails have reached $VLVDRST5$ level and internal pre-regulator $VDDPD$ voltage has reached $VLVDRSTC$ level.
 - T3 refers to the point in time when all supplies are above their primary reset thresholds denoted by $VRST5$, $VRST33$ and $VRSTC$ supply voltage levels. PORST (output) is de-asserted and HWCFG[3:5] pins are latched on PORST rising edge by SCU. Firmware execution is initiated.
 - T4 refers to the point in time when Firmware execution is completed and User code execution starts with CPU0 at a default frequency of 100 MHz. The time between T0 and T4 is documented as tBP.
 - T5 refers to the point in time during the ramp-down phase when at least one of the externally provided supplies (V_{DD} , V_{DDP3} or V_{EXT}) drop below their respective primary under-voltage reset thresholds.

3.13 Reset Timing

Table 3-27 Reset

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Application Reset Boot Time	t_B CC	-	-	400	μ s	operating with max. frequencies, with valid BMI header
System Reset Boot Time	t_{BS} CC	-	-	1.1	ms	RAM initialization and HSM boot time are not included, with valid BMI header
Cold Power on Reset Boot Time ¹⁾	t_{BP} CC	-	-	3.1	ms	$dVEXT/dT=1V/ms$. $VEXT>VLVD RST5$. Boot time after Cold PORST including EVR ramp-up and Firmware execution time; RAM initialization and HSM boot time are not included.
		-	-	1.6	ms	Firmware execution time after PORST release without EVR ramp-up; RAM initialization and HSM boot time is not included
Minimum cold PORST reset hold time in case of power fail event issued by EVR primary monitors	t_{EVRPOR} CC	$10^{2)}$	-	-	μ s	
PMS Infrastructure, EVRC and EVR33 overall start-up time till cold PORST reset release	$t_{EVRstartup}$ CC	-	-	1	ms	$dV/dT=1V/ms$. EVRC and EVR33 active
Minimum PORST active hold time externally after power supplies are stable at operating levels after start-up	t_{POA} SR	$1^{3)}$	-	-	ms	
Configurable PORST digital filter delay in addition to analog pad filter delay	$t_{PORSTDF}$ CC	600	-	1200	ns	
Warm Reset Sequencing Delay	$t_{WARMRSTSEQ}$ CC	-	-	180	μ s	
HWCFG pins hold time from ESR0 rising edge	t_{HDH} CC	$16 / f_{SPB}$	-	-	ns	

Electrical Specification Reset Timing

Table 3-27 Reset (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
HWCFG pins setup time to ESR0 rising edge	t_{HDS} CC	0	-	-	ns	
Ports inactive after ESR0 reset active	t_{PI} CC	$8000/f_{BAC}$ KT	-	$18000/f_{BA}$ CKT	s	
Ports inactive after PORST reset active	t_{PIP} CC	-	-	160	ns	
Hold time from PORST rising edge	t_{POH} SR	150	-	-	ns	
Setup time to PORST rising edge	t_{POS} SR	0	-	-	ns	
Warm PORST reset boot time	t_{BWP} CC	-	-	1.5	ms	without RAM initialization
LBIST execution time extending the boot time	t_{LBIST} CC	-	-	6	ms	LBIST Configuration A; $1.2V \leq V_{DD}$
SCR reset boot time	t_{SCR} CC	-	-	5	μs	User Mode 0
		-	-	16	μs	User Mode 1
		-	13.3	-	μs	WDT double bit ECC, soft reset
Minimum external supplies hold time after warm reset assertion	$t_{SUPHOLD}$ CC	-	-	250	μs	external supplies are V_{EVRSB} , V_{EXT} , V_{FLEX} , V_{DDM} , V_{DDP3} and V_{DD}

- 1) RAM initialization add 500μs in addition.
- 2) Cold PORST reset is driven by uC and maintained in an extended voltage range between VDDPPA limit and absolute maximum rating VEXT/VEVRSB voltage limits.
- 3) The reset release on supply ramp-up or supply restoration is delayed by a voltage hysteresis of 1.5% (default value) above the undervoltage reset limit implemented on VEXT, VDDP3 and VDD rails. This mechanism helps to avoid multiple consecutive cold PORST events during slow supply ramp-ups owing to voltage drop/current jumps when reset is released.

Electrical Specification Reset Timing

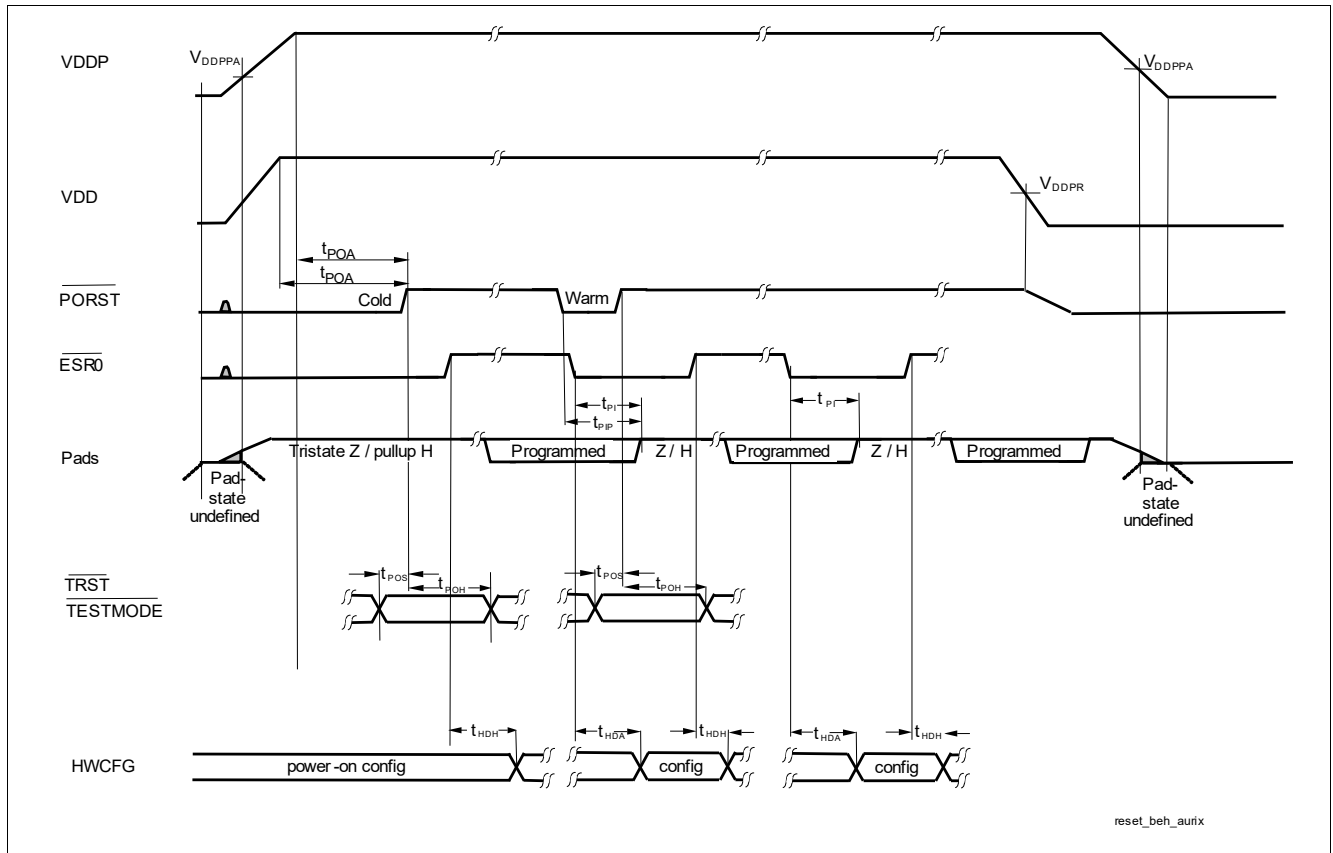


Figure 3-7 Power, Pad and Reset Timing

3.14 EVR

Table 3-28 EVR33 LDO

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Input voltage range	V_{IN} SR	3.60 ¹⁾	-	5.50	V	Normal RUN mode
		2.97 ²⁾	-	5.50	V	Low voltage cranking mode
Output voltage operational range including load/line regulation and aging ³⁾	V_{OUT} CC	2.97	3.3	3.63	V	Normal RUN mode
		2.60	3.3	3.63	V	Low voltage cranking mode; $I_{DDP3}=50mA$
Output V_{DDx3} static voltage accuracy after trimming and aging without dynamic load/line regulation.	V_{OUTT} CC	3.225	3.3	3.375	V	Normal RUN mode
		2.78	3.3	3.375	V	Low voltage cranking mode; $I_{DDP3}=50mA$
Output buffer capacitance on V_{OUT}	C_{OUT} SR	1.45	2.2	3	μF	
Output buffer capacitor ESR	C_{OUTESR} SR	-	-	100 ⁴⁾	mOhm	$f > 0.5MHz$; $f < 10MHz$
Maximum output current of the regulator	I_{MAX} CC	60 ⁵⁾	-	-	mA	Normal RUN mode
Startup time	t_{STR} CC	-	500	1000	μs	Normal RUN mode
External V_{IN} supply ramp ⁶⁾	dV_{in}/dt SR	-	1	-	V/ms	
Ripple on Output Voltage	ΔV_{OUTTC} CC	-	-	33	mV	$V_{EXT} \geq 2.97V$; $V_{EXT} \leq 5.5V$; $I_{OUTTC} \geq 10mA$; $I_{OUTTC} \leq 60mA$; ΔV_{OUTTC} = (peak to peak ripple / 2)
Load step response ⁷⁾	dV_{out}/dI_{out} CC	-165	-	-	mV	Normal RUN mode; $dI=10$ to $60mA$; $dt=20ns$; $T_{settle}=20us$
		-	-	165	mV	Normal RUN mode; $dI=60$ to $10mA$; $dt=20ns$; $T_{settle}=20us$
		-180	-	-	mV	Low voltage cranking mode; $dI=10$ to $50mA$; $dt=20ns$; $T_{settle}=20us$
		-	-	180	mV	Low voltage cranking mode; $dI=50$ to $10mA$; $dt=20ns$; $T_{settle}=20us$

Electrical Specification EVR

Table 3-28 EVR33 LDO (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Line step response	dV_{out}/dV_{in} CC	-	-	40	mV	$dV_{in}/dT=1V/ms$; $dV=3.6$ to $5V$; $I_{MAX}=60mA$
		-40	-	-	mV	$dV_{in}/dT=1V/ms$; $dV=5$ to $3.6V$; $I_{MAX}=60mA$
		-	-	280	mV	$dV_{in}/dT=50V/ms$; $dV=3.6$ to $5V$; $I_{MAX}=60mA$
		-165	-	-	mV	$dV_{in}/dT=50V/ms$; $dV=5$ to $3.6V$; $I_{MAX}=60mA$

- 1) A maximum pass device dropout voltage of 300mV is included in the minimum input voltage to ensure optimal pass device performance during normal operation.
- 2) VEXT Input voltage drop up to 2.97V leading to VDDP3 output voltage drop upto 2.6V can be tolerated if Flash is switched before to low performance mode.
- 3) No external inductive load permissible if EVR33 is used.
- 4) It is also recommended that the resistance of the supply trace from the pin to the EVR output capacitor is less than 100 mOhm. An additional decoupling capacitor of 100nF shall be located close to the pin before Cout.
- 5) In case EVR33 is not used, Injection current into 3.3V VDDP3 supply rail with active sink on 5V VEXT rail should be limited to 500 mA if during power sequencing 3.3V is supplied before 5V by external regulator.
- 6) EVR is robust against residual voltage ramp-up starting between 0 - 2.97 V. A VEXT voltage ramp range between 0.5V/min upto 120V/ms is covered in robustness validation. The generated voltage itself follows a soft ramp-up over the tSTR time to avoid overshoots.
- 7) Settling time is defined until output voltage is within $\pm 1\%$ of the mean(VOUTT) of the individual device.

Table 3-29 Supply Monitors

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Primary Undervoltage Reset threshold for V_{DDP3} before trimming ¹⁾	V_{RST33} CC	-	-	3.00	V	by reset release before EVR trimming on supply ramp-up
Primary undervoltage reset threshold for V_{DD} before trimming	V_{RSTC} CC	-	-	1.138	V	by reset release before trimming on supply ramp-up including 2 LSB voltage Hysteresis
V_{EXT} primary undervoltage monitor accuracy after trimming ²⁾	$V_{EXTPRIUV}$ CC	2.86	2.92	2.97	V	V_{EXT} = Undervoltage cold PORST Primary Monitor Threshold
V_{DDP3} primary undervoltage monitor accuracy after trimming ²⁾	$V_{DDP3PRIUV}$ CC	2.86 ³⁾	2.90	2.97	V	VDDP3 = Undervoltage cold PORST Primary Monitor Threshold
V_{DD} primary undervoltage monitor accuracy after trimming ²⁾	$V_{DDPRIUV}$ CC	1.08 ³⁾	1.105	1.125	V	VDD = Undervoltage cold PORST Primary Monitor Threshold

Electrical Specification EVR

Table 3-29 Supply Monitors (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
EVR primary monitor measurement latency for a new supply value	t_{PRIUV} CC	-	-	300	ns	The supply ramp / line jump slope is limited to 50V/ms for V_{EXT} , V_{DDP3} and V_{DD} rails.
V_{EXT} , V_{DDM} & V_{EVRSB} secondary supply monitor accuracy after trimming ^{4) 5)}	V_{EXTMON} CC	5.3	5.4	5.5	V	SWDxxVAL, VDDMxxVAL & SBxxVAL monitoring threshold=5.4V=EBh(UV)/ECh(OV). For BGA packages: EVRMONFILT.SWDFI L=1
		3.2	3.3	3.4	V	SWDxxVAL, VDDMxxVAL & SBxxVAL monitoring threshold=3.3V=90h(OV,UV). For BGA packages: EVRMONFILT.SWDFI L=1.
		4.5	4.6	4.7	V	SWDxxVAL, VDDMxxVAL & SBxxVAL monitoring threshold=4.6V=C8h(UV)/C9h(OV). For BGA packages: EVRMONFILT.SWDFI L=1
		4.9	5.0	5.1	V	SWDxxVAL, VDDMxxVAL & SBxxVAL monitoring threshold=5V=D9h(UV)/DAh(OV). For BGA packages: EVRMONFILT.SWDFI L=1

Electrical Specification EVR

Table 3-29 Supply Monitors (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
V_{DDP3} secondary supply monitor accuracy after trimming ⁵⁾	$V_{DDP3MON}$ CC	2.97	3.035	3.1	V	EVR33xxVAL monitoring threshold=3.035V=CBh (UV)/CCh(OV). EVRMONFILT.EVR33 FIL = 3.
		3.235	3.30	3.365	V	EVR33xxVAL monitoring threshold=3.3V=DDh(OV,UV). EVRMONFILT.EVR33 FIL = 3.
		3.5	3.565	3.63	V	EVR33xxVAL monitoring threshold=3.565V=EEh (UV)/EFh(OV). EVRMONFILT.EVR33 FIL = 3.
V_{DD} & V_{DDPD} secondary supply monitor accuracy after trimming ⁵⁾	V_{DDMON} CC	1.125	1.15	1.175	V	EVRCxxVAL & PRExxVAL monitoring threshold=1.15V=C7h (UV)/C8h(OV). EVRMONFILT.EVRCFIL = 1.
		1.225	1.25	1.275	V	EVRCxxVAL & PRExxVAL monitoring threshold=1.25V=D9h(OV,UV). EVRMONFILT.EVRCFIL = 1.
		1.325	1.35	1.375	V	EVRCxxVAL & PRExxVAL monitoring threshold=1.35V=EAh (UV)/EBh(OV). EVRMONFILT.EVRCFIL = 1.
V_{EXT} LVD Primary undervoltage reset Monitor threshold	$V_{LVDRST5}$ CC	2.3	-	2.72	V	Power-down
		2.4	-	2.75	V	Power-up
V_{EVR5B} LVD Primary undervoltage reset Monitor threshold	$V_{LVDRST5B}$ CC	2.18	-	2.47	V	Power-down
		2.21	-	2.5	V	Power-up
V_{EXT} and V_{EVR5B} PBIST primary overvoltage Monitor threshold	V_{PBIST5} CC	5.63	-	-	V	

Electrical Specification EVR

Table 3-29 Supply Monitors (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Primary undervoltage reset threshold for V_{EXT} before trimming	V_{RST5} CC	-	-	3.0	V	by last cold PORST release on supply ramp-up including voltage hysteresis.
EVR secondary monitor measurement latency for all 6 supply rails	t_{MON} CC	-	-	3.2	μ s	HPOSC and SHPBG bandgap trimmed. Filter inactive.

- 1) The reset release on supply ramp-up is delayed by a time duration 20-40 μ s after reaching undervoltage reset threshold and by a voltage hysteresis of 1.5% above the undervoltage reset limit. These mechanisms serve as hysteresis to avoid multiple consecutive cold PORST events during slow supply ramp-ups owing to voltage drop/current jumps when reset is released. The reset limit of 2,97V at pin is for the case with 3.3V generated internally from EVR33. In case the 3.3V supply is provided externally, the bondwire drop will cause a reset at a higher voltage of 3.0V at the VDDP3 pin.
- 2) The monitor tolerances constitute the inherent variation of the band gap and ADC over process, voltage and temperature operational ranges. The VxxPRIUV parameters are device individually tested in production with $\pm$ 1% tolerance about the VxxPRIUV limits. All voltages are measured on pins.
- 3) VRSTxx parameters are relevant only for the first cold PORST release. Later the reset levels are trimmed by the Firmware and reflected as VxxPRIUV parameters before device is used with full performance. The cold PORST is released with a voltage hysteresis on all the primary monitors to avoid consecutive PORST toggling behavior.
- 4) In case the application is using 3.3V single supply (Single Supply mode (e), i.e. VEXT and VDDP3 are shorted together), it is recommended to use secondary supply monitoring on channel VDDP3, because of the better accuracy of parameter VDDP3MON.
- 5) To monitor voltage level not provided in conditions the values for OV and UV thresholds can be generated by a linear interpolation or extrapolation based on the given points.

Table 3-30 Supply Ramp

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
External V_{EXT} & V_{EVRSB} supply ramp-up and ramp-down slope 1) 2) 3)	dV_{EXT}/dt SR	8.3E-6	1	100	V/ms	
External V_{DDP3} supply ramp-up and ramp-down slope 1)3)	dV_{DDP3}/dt SR	8.3E-6	1	100	V/ms	
External V_{DD} supply ramp-up and ramp-down slope 1)3)	dV_{DD}/dt SR	8.3E-6	1	100	V/ms	
External V_{DDM} supply ramp-up and ramp-down slope 1)3)	dV_{DDM}/dt SR	8.3E-6	1	100	V/ms	

- 1) The device is robust against residual voltage ramp-up starting between 0 - 2.97 V for VEXT, VEVR SB, VDDP3 and VDDM and 0-1 V for VDD. A voltage ramp range between 0.5V/min upto 120V/ms is covered in robustness validation.
- 2) Also valid in case EVR33 or EVRC is used. The generated voltage itself follows a soft ramp-up over the tSTR time to avoid overshoots.
- 3) The slope is defined as the maximal tangential slope between 0% to 100% voltage level. Actual waveform may not represent the specification.

Up to 1,000,000 power cycles, matching the limits defined in table 'Supply Ramp', are allowed for TC33x without any restriction to reliability.

Electrical Specification EVR

Table 3-31 EVRC SMPS

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Input V_{EXT} Voltage range	V_{IN} SR	2.97	-	5.5	V	Start-up V_{EXT} voltage > 2.6 V
SMPS regulator output voltage range including load/line regulation and aging	V_{DDDC} CC	1.125	-	1.375	V	$V_{EXT} \geq 2.97V$; $V_{EXT} \leq 5.5V$; $I_{DDDC} \geq I_{SLEEP}$; $I_{DDDC} \leq 300mA$; untrimmed
		1.125	-	1.375	V	LBIST Configuration A; SHPBG/HPOSC/primary monitors trimmed; controller/driver update not performed; $V_{EXT} \geq 2.97V$; $V_{EXT} \leq 5.5V$; $I_{DDDC} \geq I_{SLEEP}$; $I_{DDDC} \leq 300mA$
SMPS regulator static voltage output accuracy after trimming without dynamic load/line regulation.	V_{DDDC} CC	1.225	1.25	1.275	V	$V_{EXT} \geq 2.97V$; $V_{EXT} \leq 5.5V$; $I_{DDDC} \geq I_{SLEEP}$; $I_{DDDC} \leq I_{MAX}$
Programmable switching frequency	f_{DCDC} SR	1.6	1.82	2.0	MHz	Start-up frequency switches from 500 KHz in open loop operation to 1.7MHz during PWM untrimmed operation and is then changed by user to 1.8 MHz in closed loop operation.
		0.72	0.8	0.88	MHz	Start-up frequency switches from 500 KHz in open loop operation to 1.7MHz during PWM untrimmed operation and is then changed by user to 0.8 MHz in closed loop operation.
Startup time	t_{STRDC} CC	-	-	900	μs	SMPS Start-up Mode. It is defined between $V_{EXTPRIUV}$ reset threshold till PORST release, on condition that all other PORST requirements were released before. $I_{START} < 300mA$ (for both LE and HE components).

Electrical Specification EVR
Table 3-31 EVRC SMPS (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Switching frequency modulation spread	Δf_{DCSPR} CC	-	1.8%	-	MHz	
Maximum ripple at I_{MAX}	ΔV_{DDDC} CC	-	-	16	mV	$V_{EXT} \geq 2.97V$; $V_{EXT} \leq 5.5V$; $I_{DDDC} \geq 300mA$; $I_{DDDC} \leq I_{MAX}$; $\Delta V_{DDDC} =$ (Peak to Peak ripple / 2)
No load current consumption of SMPS regulator	I_{DCNL} CC	-	15	19	mA	$f_{DCDC}=1.82MHz$; $I_{DDDC}=I_{SLEEP}$; $V_{EXT} > 2.97 V$; $T_J=25^{\circ}C$
		-	5	-	mA	LPM mode; $I_{DDDC}=I_{SLEEP}$; $V_{EXT} > 2.97 V$; $T_J=25^{\circ}C$
SMPS regulator load transient response	dV_{DDDC} / dI_{OUT} CC	-50	-	75	mV	Low-End (LE) 1.8MHz and 0.8MHz components; $dI < -200mA$; $I_{DDDC}=230-700mA$; $t_f=0.1\mu s$; $t_r=0.1\mu s$; $V_{DDDC}=1.25V$; $T_{settle}=100 \mu s$
		-50	-	87	mV	Low-End (LE) 1.8MHz and 0.8MHz components; $dI < -350mA$; $I_{DDDC}=380-700mA$; $t_f=0.1\mu s$; $t_r=0.1\mu s$; $V_{DDDC}=1.25V$; $T_{settle}=100 \mu s$
		-100	-	145	mV	Low-End (LE) 1.8MHz and 0.8MHz components; $dI < -500mA$; $I_{DDDC}=530-700mA$; $t_f=0.1\mu s$; $t_r=0.1\mu s$; $V_{DDDC}=1.25V$; $T_{settle}=100 \mu s$
		-26	-	26	mV	Low-End (LE) 1.8MHz and 0.8MHz components; $dI < 100mA$; $I_{DDDC}=30-600mA$; $t_f=0.1\mu s$; $t_r=0.1\mu s$; $V_{DDDC}=1.25V$; $T_{settle}=20\mu s$;

Electrical Specification EVR

Table 3-31 EVRC SMPS (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
SMPS regulator load transient response (cont'd)	$dV_{DDDC} / dI_{OUT\ CC}$	-26	-	26	mV	Low-End (LE) 1.8MHz and 0.8MHz components; $dI < -100mA$; $I_{DDDC}=130-700mA$; $t_r=0.1\mu s$; $t_f=0.1\mu s$; $V_{DDDC}=1.25V$; $T_{settle}=20\mu s$;
		-60	-	90	mV	Low-End (LE) 1.8MHz and 0.8MHz components; $dI < I_{DDSPULJ1}$; Refer to SPU pattern for $I_{DDSPULJ1}$.
		-50	-	75	mV	High-End (HE) 1.8MHz and 0.8MHz components; $dI < -250mA$; $I_{DDDC}=280-1000mA$; $t_r=0.1\mu s$; $t_f=0.1\mu s$; $V_{DDDC}=1.25V$; $T_{settle}=100\mu s$
		-50	-	87	mV	High-End (HE) 1.8MHz and 0.8MHz components; $dI < -450mA$; $I_{DDDC}=480-1000mA$; $t_r=0.1\mu s$; $t_f=0.1\mu s$; $V_{DDDC}=1.25V$; $T_{settle}=100\mu s$
		-100	-	145	mV	High-End (HE) 1.8MHz and 0.8MHz components; $dI < -700mA$; $I_{DDDC}=730-1000mA$; $t_r=0.1\mu s$; $t_f=0.1\mu s$; $V_{DDDC}=1.25V$; $T_{settle}=100\mu s$
		-26	-	26	mV	High-End (HE) 1.8MHz and 0.8MHz components; $dI < 100mA$; $I_{DDDC}=30-900mA$; $t_r=0.1\mu s$; $t_f=0.1\mu s$; $V_{DDDC}=1.25V$; $T_{settle}=20\mu s$;

Electrical Specification EVR

Table 3-31 EVRC SMPS (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
SMPS regulator load transient response (cont'd)	$dV_{DDDC} / dI_{OUT} CC$	-26	-	26	mV	High-End (HE) 1.8MHz and 0.8MHz components; $dI < -100mA$; $I_{DDDC}=130-1000mA$; $t_r=0.1\mu s$; $t_f=0.1\mu s$; $V_{DDDC}=1.25V$; $T_{settle}=20\mu s$;
		-60	-	90	mV	High-End (HE) 1.8MHz and 0.8MHz components; $dI < I_{DDSPULJ1}$; Refer to SPU pattern for $I_{DDSPULJ1}$.
Maximum output current	$I_{MAX} CC$	100	-	-	mA	LPM mode. Typical current in LPM Mode = I_{SLEEP}
		1	-	-	A	Limited by thermal constraints and component choice, High-End (HE) components.
		700	-	-	mA	Limited by thermal constraints and component choice, Low-End (LE) components.
SMPS regulator line transient response	$dV_{DDDC} / dV_{IN} CC$	-75	-	75	mV	$dV/dT=120V/ms$; $dV < 2.97 - 5.5V$; $I_{DDDC}=50mA-I_{MAX}$;
		-12.5	-	12.5	mV	$dV/dT=1V/ms$; $dV < 2.97 - 5.5V$; $I_{DDDC}=50mA-I_{MAX}$;
SMPS regulator efficiency	$\eta_{DC} CC$	-	80	-	%	$V_{IN}=3.3V$; $I_{DDDC}=1500mA$; $f_{DCDC}=1.82MHz$
		-	75	-	%	$V_{IN}=5V$; $I_{DDDC}=1500mA$; $f_{DCDC}=1.82MHz$
Input Synchronisation frequency	$f_{DCDCSYNC} SR$	1.6	1.82	2.0	MHz	$f_{DCDC}=1.82MHz$
		0.66	0.8	0.94	MHz	$f_{DCDC}=0.8MHz$

Table 3-32 EVRC SMPS External components

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
External output capacitor value 1)	C_{OUT} SR	20.8	32	43.2	μF	$I_{DDDC}=I_{MAX}$; $f_{DDDC} = 0.8MHz$; HE components
		15.4	22	29.7	μF	$I_{DDDC}=I_{MAX}$; $f_{DDDC} = 1.82MHz$; HE components
		13	20	27	μF	$I_{DDDC}=I_{MAX}$; $f_{DDDC} = 0.8MHz$; LE components
		9.6	14.7	19.8	μF	$I_{DDDC}=I_{MAX}$; $f_{DDDC} = 1.82MHz$; LE components
External output capacitor ESR	C_{OUT_ESR} SR	-	-	50	mOhm	$f \geq 0.5MHz$; $f \leq 10MHz$
		-	-	100	Ohm	$f=10Hz$
External input capacitor value 1)	C_{IN} SR	6.5	10	13.5	μF	$I_{DDDC}=I_{MAX}$; HE components
		4.42	6.8	9.18	μF	$I_{DDDC}=I_{MAX}$; LE components
External input capacitor ESR	C_{IN_ESR} SR	-	-	50	mOhm	$f \geq 0.5MHz$; $f \leq 10MHz$
		-	-	100	Ohm	$f=100Hz$
External inductor value	L_{DC} SR	3.29	4.7	6.11	μH	$f_{DCDC}=0.8MHz$
		2.31	3.3	4.29	μH	$f_{DCDC}=1.82MHz$
External inductor DCR	L_{DC_DCR} SR	-	-	0.2	Ohm	
P + N-channel MOSFET logic level	V_{LL} SR	-	-	2.5	V	
P + N-channel MOSFET drain source breakdown voltage	$ V_{BR_DS} $ SR	+7	-	-	V	NMOS - $V_{GS} = 0$.
		-	-	-7	V	PMOS - $V_{GS} = 0$.
P + N-channel MOSFET drain source ON-state resistance	R_{ON} SR	-	150	-	mOhm	$I_{DDDC}=1.0A$; $ V_{GS} =2.5V$; $T_A=25^{\circ}C$
		-	200	-	mOhm	$I_{DDDC}=700mA$; $ V_{GS} =2.5V$; $T_A=25^{\circ}C$

Electrical Specification EVR

Table 3-32 EVRC SMPS External components (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
P + N-channel MOSFET Gate Charge	Q_G SR	-	-	8	nC	$I_{DDDC}=1.0A$; NMOS- $ V_{GS} =5V$; 1.5A pulsed drain current
		-8	-	-	nC	$I_{DDDC}=1.0A$; PMOS- $ V_{GS} =5V$; 1.5A pulsed drain current
		-	-	4	nC	$I_{DDDC}=700mA$; NMOS- $ V_{GS} =5V$; 0.5A pulsed drain current
		-4	-	-	nC	$I_{DDDC}=700mA$; PMOS- $ V_{GS} =5V$; 0.5A pulsed drain current
External Inductor Saturation Current Margin	ΔI_{SAT} SR	400	-	-	mA	The saturation current of the coil must be larger than $I_{DDDC} + \Delta I_{SAT}$
P + N-channel MOSFET Gate threshold voltage	V_{GSTH} SR	-	1	-	V	NMOS
		-	-1	-	V	PMOS
N-channel MOSFET reverse diode forward voltage	V_{RDN} SR	-	0.8	-	V	

1) Capacitor min-max range represent typical $\pm 35\%$ tolerance including DC bias effect. The trace resistance from the capacitor to the supply or ground rail should be limited to 25 mOhm.

3.15 System Phase Locked Loop (SYS_PLL)

Table 3-33 PLL System

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
DCO Input frequency range	f_{REF} CC	10	-	40	MHz	
Modulation Amplitude	MA CC	0	-	2	%	
Peak Period jitter	DP CC	-200	-	200	ps	without modulation (PLL output frequency)
Peak Accumulated Jitter	D_{PP} CC	-5	-	5	ns	without modulation
Total long term jitter	J_{TOT} CC	-	-	11.5	ns	including modulation; MA 1.25%; f_{REF} 20MHz
System frequency deviation	f_{SYSD} CC	-	-	0.01	%	with active modulation
DCO frequency range	f_{DCO} CC	400	-	800	MHz	
PLL lock-in time	t_L CC	4	-	100	μs	

Note: The specified PLL jitter values are valid if the capacitive load per pin does not exceed $C_L = 20$ pF with the maximum driver and sharp edge.

Note: The maximum peak-to-peak noise on the power supply voltage, is limited to a peak-to-peak voltage of $V_{PP} = 100$ mV for noise frequencies below 300 KHz and $V_{PP} = 40$ mV for noise frequencies above 300 KHz. These conditions can be achieved by appropriate blocking of the supply voltage as near as possible to the supply pins and using PCB supply and ground planes.

Electrical Specification Peripheral Phase Locked Loop (PER_PLL)

3.16 Peripheral Phase Locked Loop (PER_PLL)

Table 3-34 PLL Peripheral

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Peak Accumulated jitter at SYSCLK pin	D_{PP} CC	-1000	-	1000	ps	Peak only
Peak accumulated jitter	D_{PPI} CC	-700	-	700	ps	Peak only
RMS Accumulated jitter	D_{RMS} CC	-100	-	100	ps	measured over 1 μ s; $f_{REF} = 20$ MHz and $f_{DCO} = 640$ MHz or $f_{REF} = 25$ MHz and $f_{DCO} = 800$ MHz
Peak Period jitter	DP CC	-200	-	200	ps	$f_{DCO} = 640$ MHz or $f_{DCO} = 800$ MHz
Absolute RMS jitter (PLL out)	J_{ABS10} CC	-125	-	125	ps	$f_{REF} = 10$ MHz; $f_{DCO} = 640$ MHz
Absolute RMS jitter (PLL out)	J_{ABS20} CC	-85	-	85	ps	$f_{REF} = 20$ MHz; $f_{DCO} = 640$ MHz
Absolute RMS jitter (PLL out)	J_{ABS25} CC	-85	-	85	ps	$f_{REF} = 25$ MHz; $f_{DCO} = 800$ MHz
DCO frequency range	f_{DCO} CC	400	-	800	MHz	
DCO input frequency range	f_{REF} CC	10	-	40	MHz	
PLL lock-in time	t_L CC	4	-	100	μ s	

Note: The specified PLL jitter values are valid if the capacitive load per pin does not exceed $C_L = 20$ pF with the maximum driver and sharp edge.

Note: The maximum peak-to-peak noise on the power supply voltage, is limited to a peak-to-peak voltage of $V_{PP} = 100$ mV for noise frequencies below 300 KHz and $V_{PP} = 40$ mV for noise frequencies above 300 KHz. These conditions can be achieved by appropriate blocking of the supply voltage as near as possible to the supply pins and using PCB supply and ground planes.

3.17 AC Specifications

All AC parameters are specified for the complete operating range defined in [Chapter 3.4](#) unless otherwise noted in column Note / Test Condition.

Unless otherwise noted in the figures the timings are defined with the following guidelines:

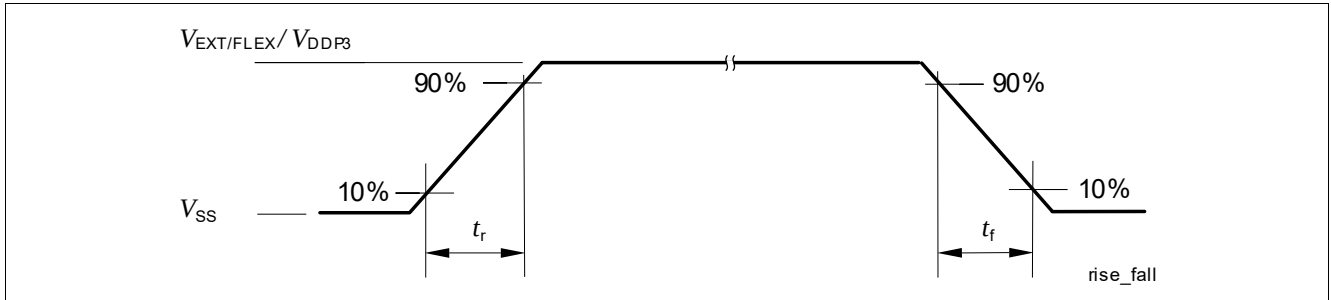


Figure 3-8 Definition of rise / fall times

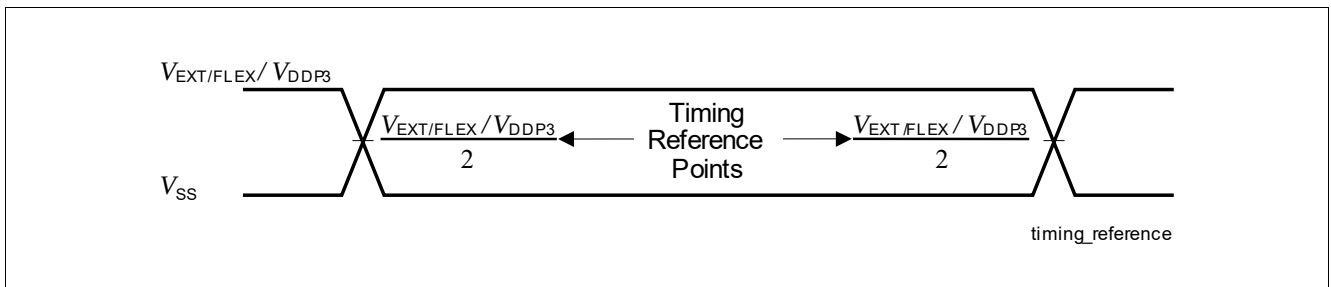


Figure 3-9 Time Reference Point Definition

3.18 JTAG Parameters

The following parameters are applicable for communication through the JTAG debug interface. The JTAG module is fully compliant with IEEE1149.1-2000.

Table 3-35 JTAG

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
TCK clock period	t_1 SR	50	-	-	ns	
TCK high time	t_2 SR	10	-	-	ns	
TCK low time	t_3 SR	10	-	-	ns	
TCK clock rise time	t_4 SR	-	-	4	ns	
TCK clock fall time	t_5 SR	-	-	4	ns	
TDI/TMS setup to TCK rising edge	t_6 SR	6.0	-	-	ns	
TDI/TMS hold after TCK rising edge	t_7 SR	6.0	-	-	ns	
TDO valid after TCK falling edge (propagation delay)	t_8 CC	3.0	-	-	ns	$C_L \leq 20\text{pF}$
		-	-	25	ns	$C_L \leq 50\text{pF}$
TDO hold after TCK falling edge	t_{18} CC	2	-	-	ns	
TDO high impedance to valid from TCK falling edge	t_9 CC	-	-	25	ns	$C_L \leq 50\text{pF}$
TDO valid output to high impedance from TCK falling edge	t_{10} CC	-	-	25	ns	$C_L \leq 50\text{pF}$

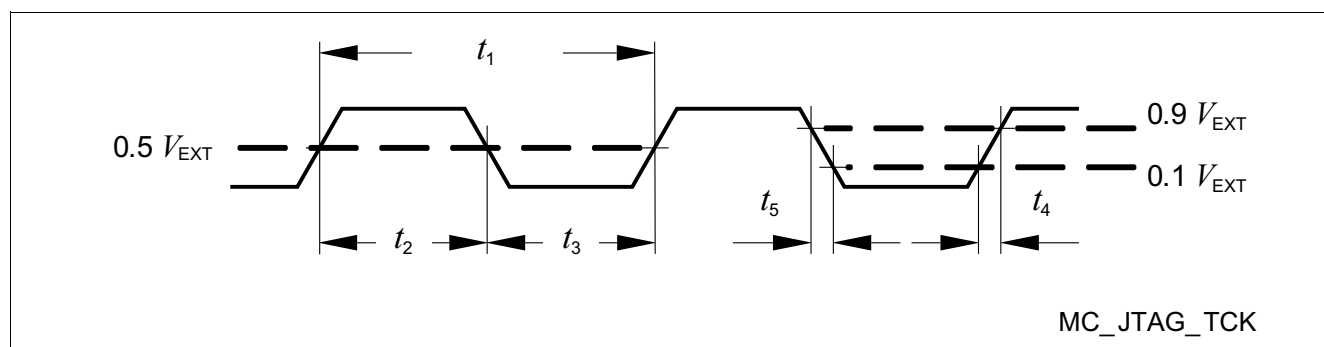


Figure 3-10 Test Clock Timing (TCK)

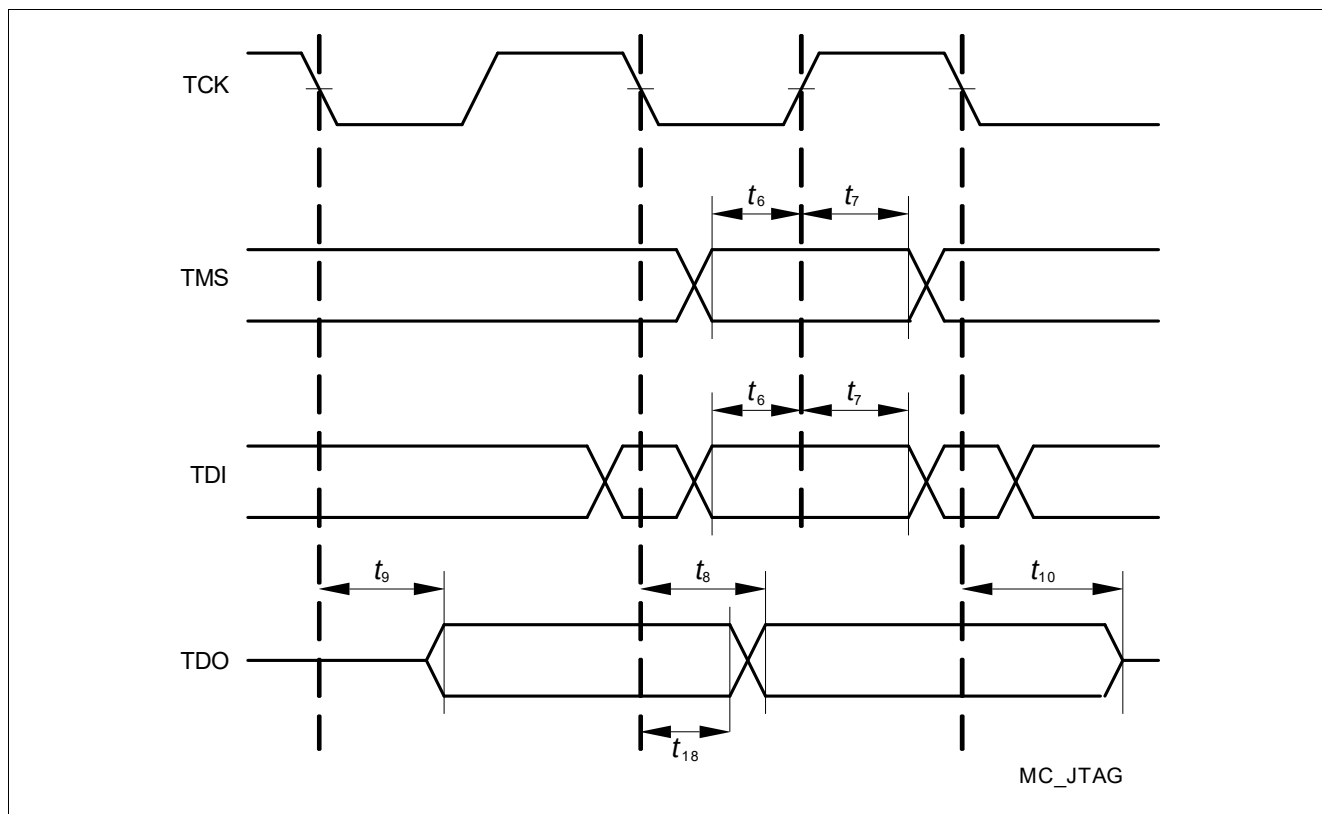


Figure 3-11 JTAG Timing

3.19 DAP Parameters

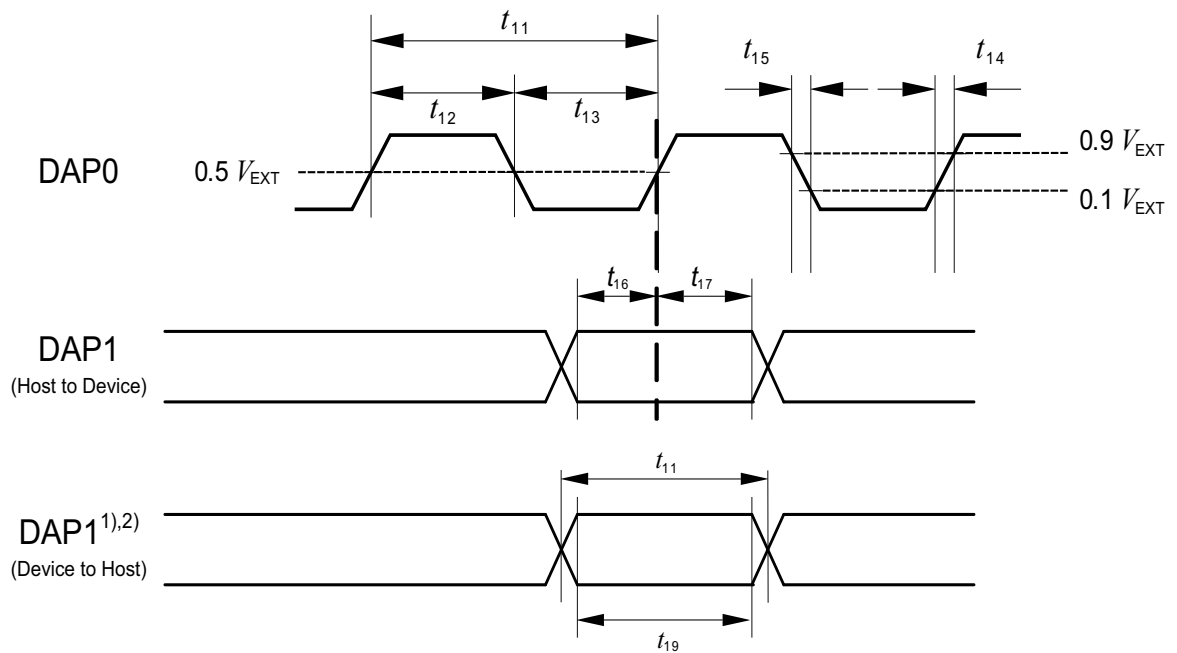
The following parameters are applicable for communication through the DAP debug interface.

Table 3-36 DAP

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
DAP0 clock rise time	t_{14} SR	-	-	1	ns	f=160MHz
		-	-	4	ns	f=40MHz
		-	-	2	ns	f=80MHz
DAP0 clock fall time	t_{15} SR	-	-	1	ns	f=160MHz
		-	-	4	ns	f=40MHz
		-	-	2	ns	f=80MHz
DAP1 setup to DAP0 rising edge	t_{16} SR	4	-	-	ns	
		5	-	-	ns	f=40MHz
DAP1 hold after DAP0 rising edge	t_{17} SR	2	-	-	ns	
DAP1 valid per DAP0 clock period	t_{19} CC	4	-	-	ns	$C_L=20\text{pF}$; f=160MHz
		8	-	-	ns	$C_L=20\text{pF}$; f=80MHz
		10	-	-	ns	$C_L=50\text{pF}$; f=40MHz
DAP0 high time	t_{12} SR	2	-	-	ns	
DAP0 low time	t_{13} SR	2	-	-	ns	
DAP0 clock period	t_{11} SR	6.25	-	-	ns	

Table 3-37 SCR DAP

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
DAP0 clock rise time	t_{14} SR	-	-	8	ns	f=20MHz
DAP0 clock fall time	t_{15} SR	-	-	8	ns	f=20MHz
DAP1 setup to DAP0 rising edge	t_{16} SR	10	-	-	ns	
DAP1 hold after DAP0 rising edge	t_{17} SR	10	-	-	ns	
DAP1 valid per DAP0 clock period	t_{19} CC	30	-	-	ns	$C_L=20\text{pF}$; f=20MHz
DAP0 high time	t_{12} SR	15	-	-	ns	
DAP0 low time	t_{13} SR	15	-	-	ns	
DAP0 clock period	t_{11} SR	50	-	-	ns	



- 1) The DAP1 and DAP2 device to host timing is individual for both pins.
There is no guaranteed max. signal skew.
- 2) No explicit setup and hold times are given for DAP1 for the direction Device to Host.
Only t_{11} and t_{19} are guaranteed and the tool may set the sample point freely.

Figure 3-12 DAP Timing

Note: The DAP1 and DAP2 device to host timing is individual for both pins. There is no guaranteed max. signal skew.

3.20 ASCLIN SPI Master Timing

This section defines the timings for the ASCLIN in the TC33xEXT.

Note: Pad asymmetry is already included in the following timings.

Table 3-38 Master Mode strong sharp (ss) output pads

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
ASCLKO clock period	t_{50} CC	20	-	-	ns	$C_L=25\text{pF}$
Deviation from ideal duty cycle	t_{500} CC	-2	-	2	ns	$C_L=25\text{pF}$
MTSR delay from ASCLKO shifting edge	t_{51} CC	-3.5	-	3.5	ns	$C_L=25\text{pF}$
ASLSON delay from the first ASCLKO edge	t_{510} CC	-3	-	3.5	ns	$C_L=25\text{pF}$
MRST setup to ASCLKO latching edge	t_{52} SR	25	-	-	ns	$C_L=25\text{pF}$
MRST hold from ASCLKO latching edge	t_{53} SR	-2	-	-	ns	$C_L=25\text{pF}$

Table 3-39 Master Mode strong medium (sm) output pads

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
ASCLKO clock period	t_{50} CC	50	-	-	ns	$C_L=50\text{pF}$
Deviation from ideal duty cycle	t_{500} CC	-5	-	5	ns	$C_L=50\text{pF}$
MTSR delay from ASCLKO shifting edge	t_{51} CC	-7	-	7	ns	$C_L=50\text{pF}$
ASLSON delay from the first ASCLKO edge	t_{510} CC	-7	-	7	ns	$C_L=50\text{pF}$
MRST setup to ASCLKO latching edge	t_{52} SR	35	-	-	ns	$C_L=50\text{pF}$
MRST hold from ASCLKO latching edge	t_{53} SR	-5	-	-	ns	$C_L=50\text{pF}$

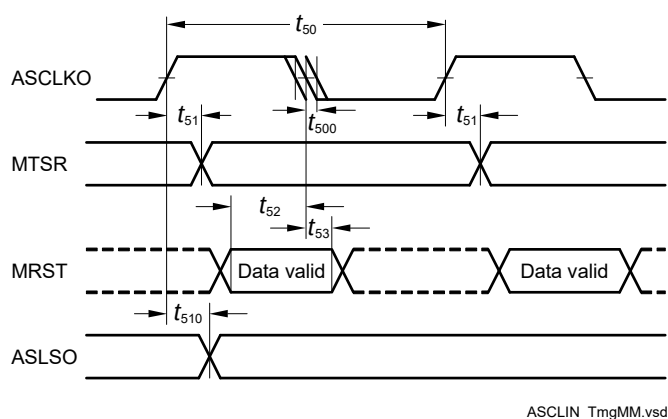
Table 3-40 Master Mode medium (m) output pads

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
ASCLKO clock period	t_{50} CC	160	-	-	ns	$C_L=50\text{pF}$
Deviation from ideal duty cycle	t_{500} CC	-10	-	10	ns	$C_L=50\text{pF}$
MTSR delay from ASCLKO shifting edge	t_{51} CC	-20	-	20	ns	$C_L=50\text{pF}$
ASLSON delay from the first ASCLKO edge	t_{510} CC	-20	-	20	ns	$C_L=50\text{pF}$

Electrical Specification ASCLIN SPI Master Timing

Table 3-40 Master Mode medium (m) output pads (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
MRST setup to ASCLKO latching edge	t_{52} SR	80	-	-	ns	$C_L=50\text{pF}$
MRST hold from ASCLKO latching edge	t_{53} SR	-15	-	-	ns	$C_L=50\text{pF}$



ASCLIN_TmgMM.vsd

Figure 3-13 ASCLIN SPI Master Timing

Electrical Specification QSPI Timings, Master and Slave Mode

3.21 QSPI Timings, Master and Slave Mode

This section defines the timings for the QSPI in the TC33xEXT.

It is assumed that SCLKO, MTSR, and SLSON pads have the same pad settings:

Note: Pad asymmetry is already included in the following timings.

Table 3-41 Master Mode Strong Sharp (ss) output pads

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
SCLKO clock period	t_{50} CC	50	-	-	ns	CL=25pF
Deviation from the ideal duty cycle	t_{500} CC	-2	-	2	ns	CL=25pF
MTSR delay from SCLKO shifting edge	t_{51} CC	-4	-	5	ns	CL=25pF
SLSON deviation from the ideal programmed position	t_{510} CC	-4	-	5	ns	CL=25pF
MRST setup to SCLK latching edge	t_{52} SR	25 ^{1) 2)}	-	-	ns	CL=25pF
MRST hold from SCLK latching edge	t_{53} SR	-2 ¹⁾²⁾	-	-	ns	CL=25pF

1) For compensation of the average on-chip delay the QSPI module provides the bit fields ECONz.A, B and C.

2) The setup and hold times are valid for both settings of the input pads thresholds: TTL and AL.

Table 3-42 Master Mode Strong Medium (sm) output pads

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
SCLKO clock period	t_{50} CC	50	-	-	ns	CL=50pF
Deviation from the ideal duty cycle	t_{500} CC	-5	-	5	ns	CL=50pF
MTSR delay from SCLKO shifting edge	t_{51} CC	-7	-	7	ns	CL=50pF
SLSON deviation from the ideal programmed position	t_{510} CC	-7	-	7	ns	CL=50pF
MRST setup to SCLK latching edge	t_{52} SR	35 ^{1) 2)}	-	-	ns	CL=50pF
MRST hold from SCLK latching edge	t_{53} SR	-5 ¹⁾²⁾	-	-	ns	CL=50pF

1) For compensation of the average on-chip delay the QSPI module provides the bit fields ECONz.A, B and C.

2) The setup and hold times are valid for both settings of the input pads thresholds: TTL and AL.

Electrical Specification QSPI Timings, Master and Slave Mode

Table 3-43 Master Mode Medium (m) output pads

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
SCLKO clock period	t_{50} CC	160	-	-	ns	CL=50pF
Deviation from the ideal duty cycle	t_{500} CC	-10	-	10	ns	CL=50pF
MTSR delay from SCLKO shifting edge	t_{51} CC	-20	-	20	ns	CL=50pF
SLSOn deviation from the ideal programmed position	t_{510} CC	-20	-	20	ns	CL=50pF
MRST setup to SCLK latching edge	t_{52} SR	80 ^{1) 2)}	-	-	ns	CL=50pF
MRST hold from SCLK latching edge	t_{53} SR	-15 ¹⁾²⁾	-	-	ns	CL=50pF
		-13 ¹⁾²⁾	-	-	ns	CL=50pF; SCR SSC

1) For compensation of the average on-chip delay the QSPI module provides the bit fields ECONZ.A, B and C.

2) The setup and hold times are valid for both settings of the input pads thresholds: TTL and AL.

Table 3-44 Slave mode timing

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
SCLK clock period	t_{54} SR	$4 \times T_{MAX}$	-	-	ns	
SCLK duty cycle	t_{55}/t_{54} SR	40	-	60	%	
MTSR setup to SCLK latching edge	t_{56} SR	6	-	-	ns	Input Level AL
		6	-	-	ns	Input Level TTL
MTSR hold from SCLK latching edge	t_{57} SR	4	-	-	ns	Input Level AL
		6	-	-	ns	Input Level TTL
SLSI setup to first SCLK shift edge	t_{58} SR	4	-	-	ns	Input Level AL
		6	-	-	ns	Input Level TTL
SLSI hold from last SCLK latching edge	t_{59} SR	3	-	-	ns	Input Level AL
		6	-	-	ns	Input Level TTL
MRST delay from SCLK shift edge	t_{60} CC	5	-	35	ns	driver = strong edge = medium ; $C_L=50pF$
		2	-	24	ns	driver = strong edge = sharp ; $C_L=50pF$
		15	-	80	ns	medium driver ; $C_L=50pF$
		14	-	-	ns	medium driver ; $C_L=50pF$; SCR SSC

Electrical Specification QSPI Timings, Master and Slave Mode

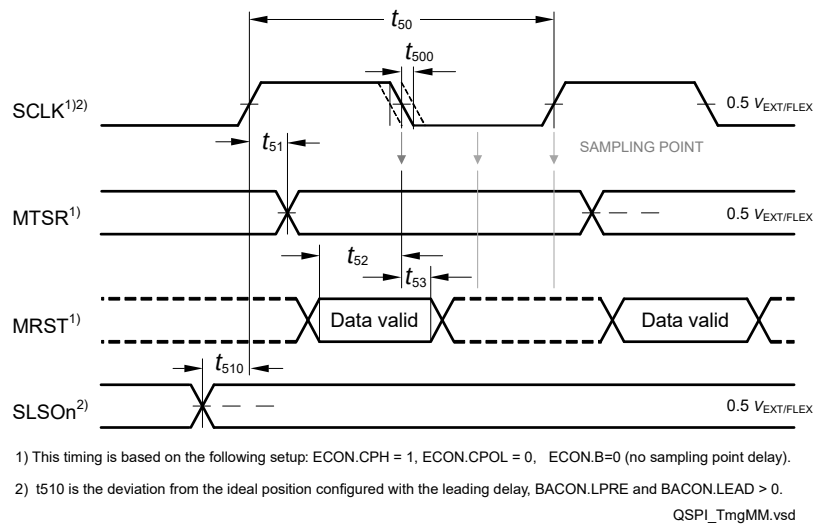


Figure 3-14 Master Mode Timing

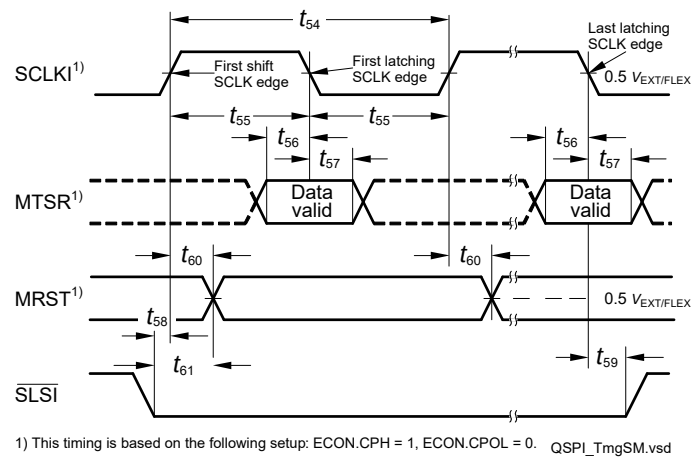


Figure 3-15 Slave Mode Timing

3.22 Ethernet Interface (ETH) Characteristics

3.22.1 ETH Measurement Reference Points

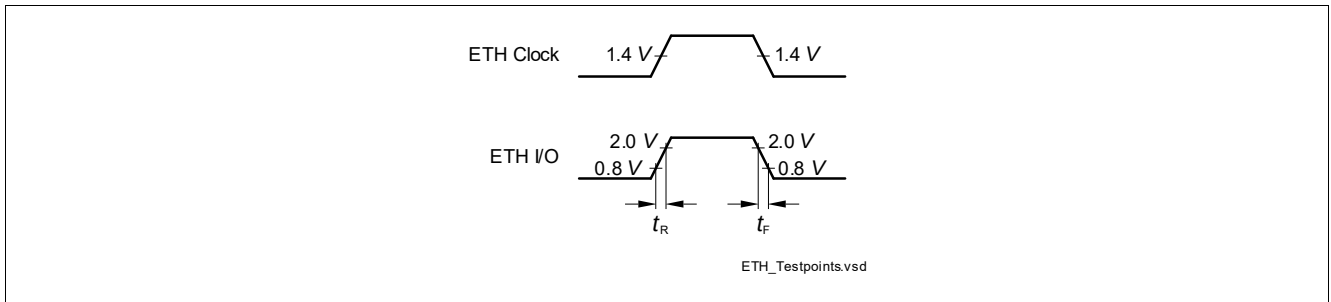


Figure 3-16 ETH Measurement Reference Points

3.22.2 ETH Management Signal Parameters (ETH_MDC, ETH_MDIO)

Table 3-45 ETH Management Signal Parameters valid for 3.3V

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
ETH_MDC period	t_1 CC	400	-	-	ns	CL=25pF
ETH_MDC high time	t_2 CC	160	-	-	ns	CL=25pF
ETH_MDC low time	t_3 CC	160	-	-	ns	CL=25pF
ETH_MDIO setup time (output)	t_4 CC	10	-	-	ns	CL=25pF
ETH_MDIO hold time (output)	t_5 CC	10	-	-	ns	CL=25pF
ETH_MDIO data valid (input)	t_6 SR	0	-	300	ns	CL=25pF

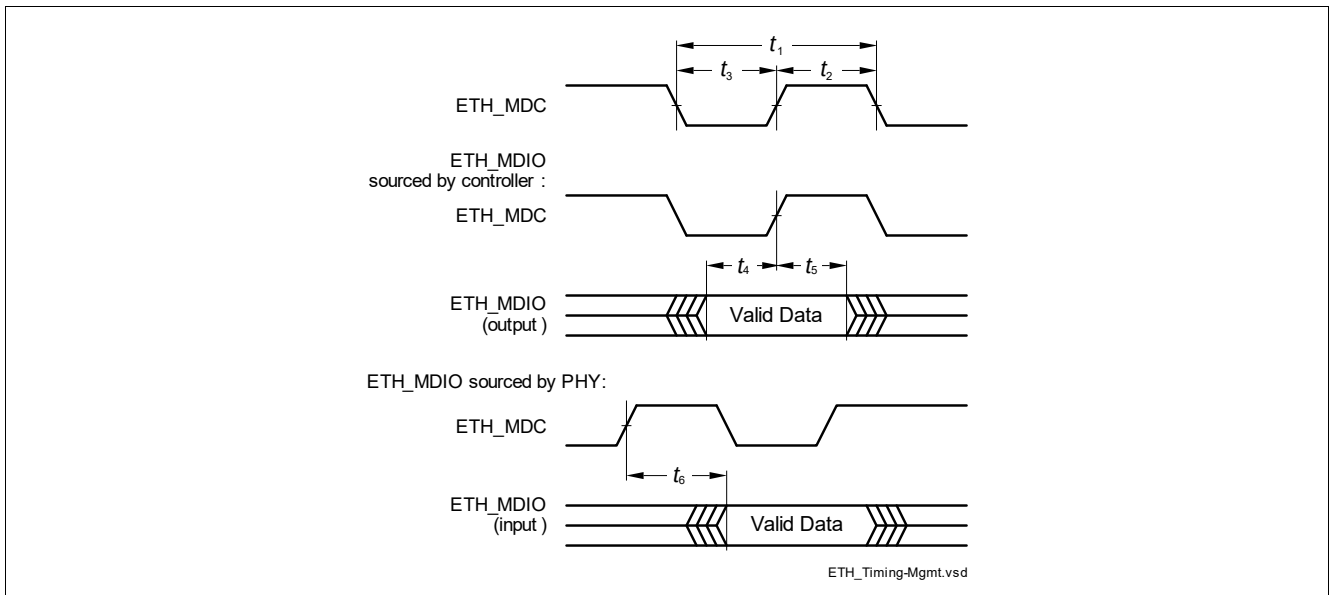


Figure 3-17 ETH Management Signal Timing

Electrical Specification Ethernet Interface (ETH) Characteristics

3.22.3 ETH MII Parameters

In the following, the parameters of the MII (Media Independent Interface) are described.

Table 3-46 ETH MII Signal Timing Parameters

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Clock period	t_7 SR	-	40	-	ns	CL=25pF ; baudrate=100Mbps
		-	400	-	ns	CL=25pF ; baudrate=10Mbps
Clock high time	t_8 SR	14	-	26	ns	CL=25pF ; baudrate=100Mbps
		140 ¹⁾	-	260 ²⁾	ns	CL=25pF ; baudrate=10Mbps
Clock low time	t_9 SR	14	-	26	ns	CL=25pF ; baudrate=100Mbps
		140 ¹⁾	-	260 ²⁾	ns	CL=25pF ; baudrate=10Mbps
Input setup time	t_{10} SR	10	-	-	ns	CL=25pF
Input hold time	t_{11} SR	10	-	-	ns	CL=25pF
Output valid time	t_{12} CC	0	-	25	ns	CL=25pF

1) Defined by 35% of clock period.

2) Defined by 65% of clock period.

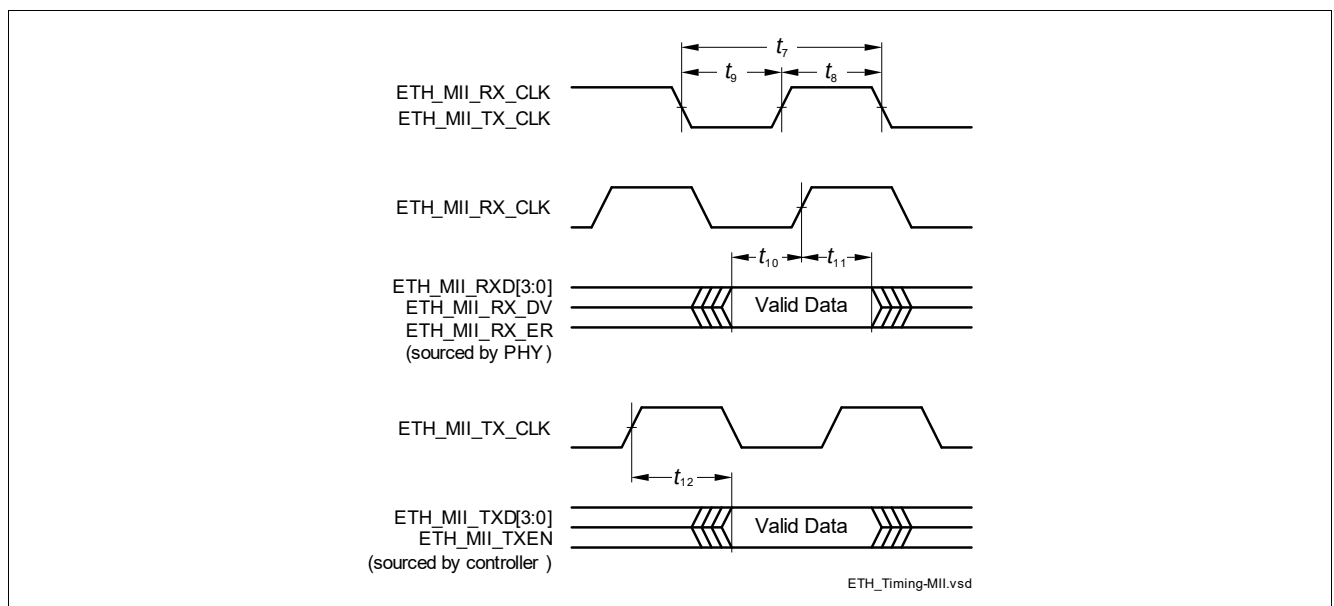


Figure 3-18 ETH MII Signal Timing

Electrical Specification Ethernet Interface (ETH) Characteristics

3.22.4 ETH RMII Parameters

In the following, the parameters of the RMII (Reduced Media Independent Interface) are described.

Table 3-47 ETH RMII Signal Timing Parameters valid for 3.3V

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
ETH_RMII_REF_CL clock period	t_{13} SR	-	20	-	ns	50ppm ; CL=25pF
ETH_RMII_REF_CL clock high time	t_{14} SR	7 ¹⁾	-	13 ²⁾	ns	CL=25pF
ETH_RMII_REF_CL clock low time	t_{15} SR	7 ¹⁾	-	13 ²⁾	ns	CL=25pF
ETHTXEN, ETHTXD[1:0], ETHRXD[1:0], ETHCRSDV; setup time ³⁾	t_{16} CC	4	-	-	ns	CL=25pF
ETHTXEN, ETHTXD[1:0], ETHRXD[1:0], ETHCRSDV; hold time ³⁾	t_{17} CC	2	-	-	ns	CL=25pF

1) Defined by 35% of clock period.

2) Defined by 65% of clock period.

3) For ETHRXD and ETHCRSDV signals this parameter is a SR.

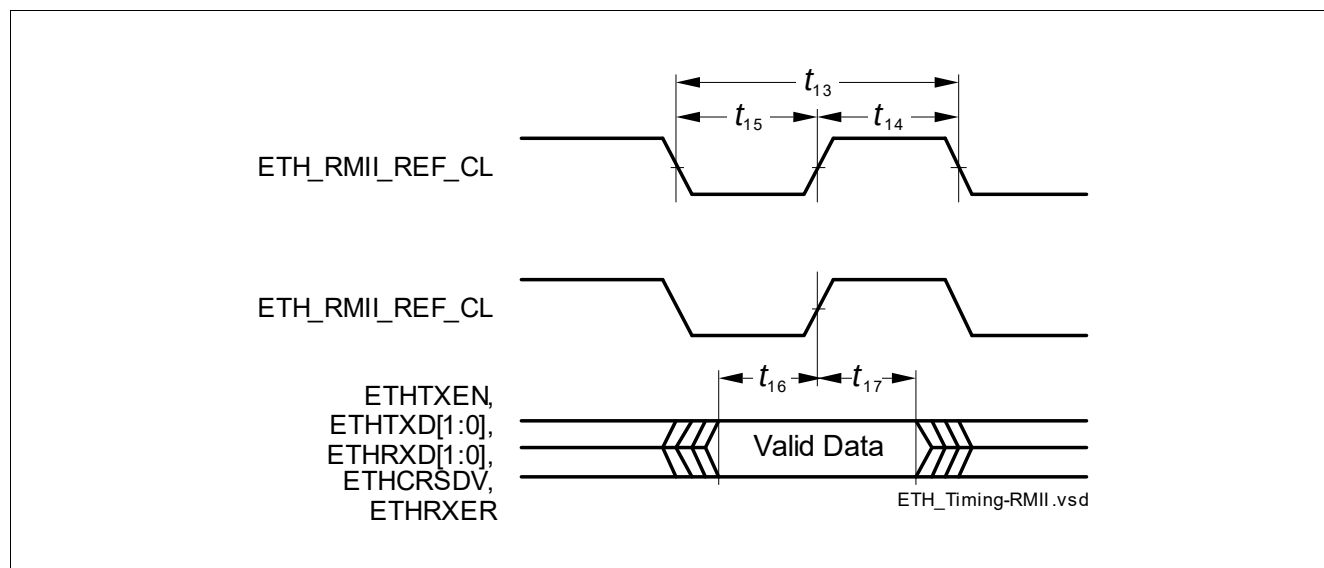


Figure 3-19 ETH RMII Signal Timing

Electrical Specification Ethernet Interface (ETH) Characteristics

3.22.5 ETH RGMII Parameters

In the following, the parameters of the RGMII are described.

Table 3-48 ETH RGMII Signal Timing Parameters valid for 3.3V

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
TX Clock period	t_{19} CC	36	40	44	ns	100Mbps
		360	400	440	ns	10Mbps
		7.2	8	8.8	ns	Gigabit
Data to Clock Output skew	t_{20} CC	-500	0	500	ps	
Data to Clock input skew (at receiver)	t_{21} SR	1	1.8	2.6	ns	
Clock duty cycle	t_{duty} CC	40	50	60	%	10/100Mbps
		45	50	55	%	Gigabit
GREFCLK duty cycle	t_{duty_in} SR	45	-	55	%	
GREFCLK Input accuracy	ACC SR	-0.005	-	0.005	%	

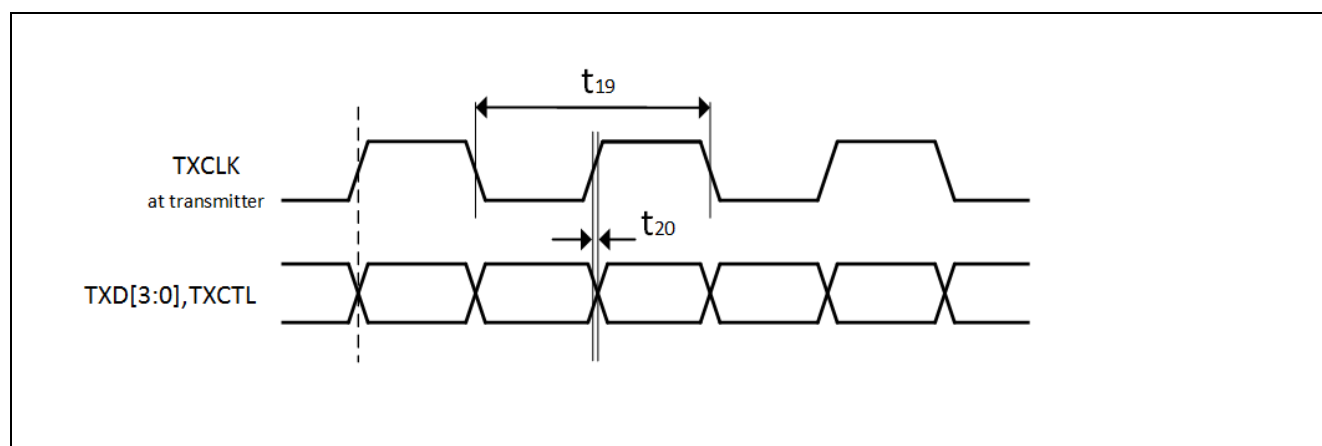


Figure 3-20 ETH RGMII TX Signal Timing (Delay on Destination (DoD))

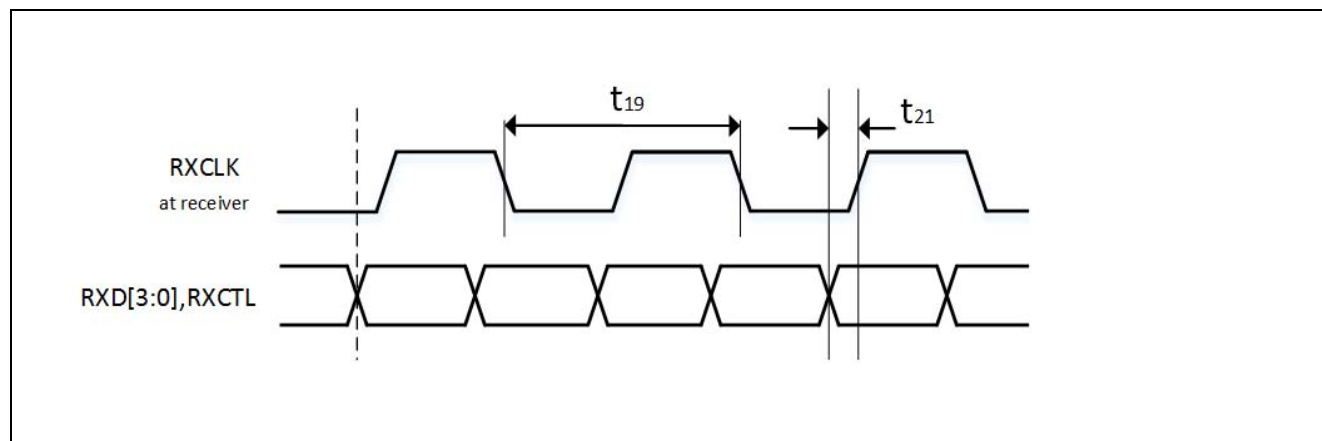


Figure 3-21 ETH RGMII RX Signal Timing (Delay on Source (DoS))

3.23 Radar Interface Timing

Table 3-49 Skew Calibration Related

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Bit time	t_{80} CC	2.5	-	-	ns	
Set-up time	t_{82} SR	0.8	-	-	ns	
Hold time	t_{83} SR	0.8	-	-	ns	
RAMP1 set-up time relative to the FRAME rising edge	t_{88} SR	30	-	-	ns	
RAMP1 hold time relative to the FRAME rising edge	t_{89} SR	30	-	-	ns	

3.24 SDMMC Interface Timing

Table 3-50 SDMMC

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Clock period Data Transfer Mode	t_1 CC	20	-	-	ns	push-pull, $C_L \leq 30\text{pF}$, $V_{EXT} = 3.3\text{V}$
Clock period Identification Mode	t_2 CC	-	-	2500	ns	open-drain, $C_L \leq 30\text{pF}$, $V_{EXT} = 3.3\text{V}$
Clock low time	t_3 CC	6,5	-	-	ns	$C_L \leq 30\text{pF}$, $V_{EXT} = 3.3\text{V}$
Clock high time	t_4 CC	6,5	-	-	ns	$C_L \leq 30\text{pF}$, $V_{EXT} = 3.3\text{V}$
Data output valid time before rising clock edge	t_5 CC	3	-	-	ns	$C_L \leq 30\text{pF}$, $V_{EXT} = 3.3\text{V}$
Data output valid time after rising clock edge	t_6 CC	3	-	-	ns	$C_L \leq 30\text{pF}$, $V_{EXT} = 3.3\text{V}$
Data input hold time	t_7 SR	2,5	-	-	ns	$C_L \leq 30\text{pF}$, $V_{EXT} = 3.3\text{V}$, TTL levels
Data Input delay time	t_8 SR	-	-	13,7	ns	$C_L \leq 30\text{pF}$, $V_{EXT} = 3.3\text{V}$, TTL levels
Data Input setup time	t_9 SR	5,2	-	-	ns	$C_L \leq 30\text{pF}$, $V_{EXT} = 3.3\text{V}$, TTL levels

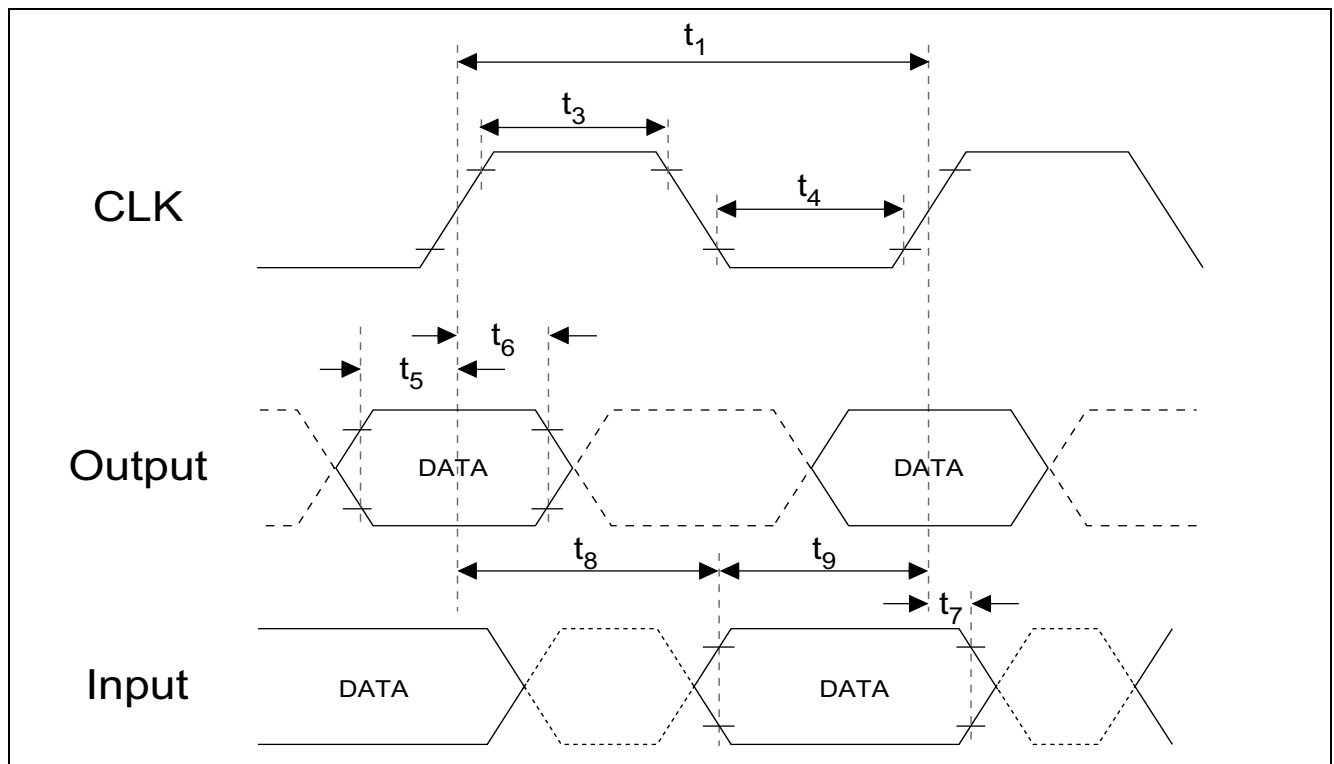


Figure 3-22 SDMMC Timing

3.25 FSP Parameter

Table 3-51 Safety

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Skew between FSP0 and FSP1	$t_{\text{FSPSKEW CC}}$	-8	-	9	ns	$C_L=50\text{pF}$, driver strength m
		-5	-	6	ns	$C_L=50\text{pF}$, driver strength sm
		-4	-	5	ns	$C_L=50\text{pF}$, driver strength ss

3.26 Flash Target Parameters

Table 3-52 Flash

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Program Flash Erase Time per logical sector ¹⁾	t_{ERP} CC	-	-	0.5	s	cycle count < 1000
Program Flash Erase Time per Multi-Sector Command ¹⁾	t_{MERP} CC	-	-	0.5	s	For consecutive logical sectors in a physical sector with total range ≤ 512 kByte; cycle count < 1000
Program Flash program time per page in 5 V mode ¹⁾	t_{PRP5} CC	-	-	80	μs	32 Byte
Program Flash program time per page in 3.3 V mode ¹⁾	t_{PRP3} CC	-	-	115	μs	32 Byte
Program Flash program time per burst in 5 V mode ¹⁾	t_{PRPB5} CC	-	-	220	μs	256 Byte
Program Flash program time per burst in 3.3 V mode ¹⁾	t_{PRPB3} CC	-	-	530	μs	256 Byte
Program Flash program time for 1 MByte with burst programming in 3.3 V mode excluding communication ¹⁾	t_{PRPB3_1MB} CC	-	-	2.2	s	Derived value for documentation purpose
Program Flash program time for 1 MByte with burst programming in 5 V mode excluding communication ¹⁾	t_{PRPB5_1MB} CC	-	-	1	s	Derived value for documentation purpose
Program Flash program time for complete PFlash with burst programming in 5 V mode excluding communication ¹⁾	t_{PRPB5_PF} CC	-	-	2	s	Derived value for documentation purpose
Write Page Once adder ¹⁾	t_{ADD} CC	-	-	20	μs	Adder to Program Time when using Write Page Once
Program Flash suspend to read latency ¹⁾	t_{SPNDP} CC	-	-	120	μs	For Write Burst, Verify Erased and for multi-(logical) sector erase commands
Data Flash Erase Disturb Limit (single ended sensing mode)	N_{DFD} CC	-	-	50	cycles	
Data Flash Erase Disturb Limit (complement sensing mode)	N_{DFDC} CC	-	-	500	cycles	
UCB Erase Disturb Limit	N_{UCBD} CC	-	-	500	cycles	

Electrical Specification Flash Target Parameters

Table 3-52 Flash (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Program time data flash per page ¹⁾²⁾	t_{PRD} CC	-	-	75	μ s	8 Byte
Complete Device Flash Erase Time PFlash and DFlash ^{1)3) 4) 5)}	t_{ER_Dev} CC	-	1.8	3	s	Valid for less than 1000 cycles, w/o UCB. Derived value for documentation purpose.
Data Flash program time per burst ¹⁾²⁾	t_{PRDB} CC	-	-	140	μ s	32 Byte
Data Flash suspend to read latency ¹⁾	t_{SPNDD} CC	-	-	120	μ s	
Wait time after margin change	$t_{FL_MarginDel}$ CC	-	-	2	μ s	
Program Flash Endurance per Logical Sector	N_{E_P} CC	-	-	1000	cycles	Replace logical sector command shall be used if a sector fails during erase or program
Number of erase operations per physical sector in program flash	N_{ERP} CC	-	-	16000	cycles	
Program Flash Retention Time, Sector	t_{RET} CC	20	-	-	years	Max. 1000 erase/program cycles
UCB Retention Time	t_{RTU} CC	20	-	-	years	Max. 100 erase/program cycles per UCB, max 500 erase/program cycles for all UCBs together
Data Flash access delay	t_{DF} CC	-	-	100	ns	see RFLASH of DMU register HF_DWAIT
Data Flash ECC Delay	t_{DFECC} CC	-	-	20	ns	see RECC of DMU register HF_DWAIT
Program Flash access delay	t_{PF} CC	-	-	30	ns	see RFLASH of DMU register HF_PWAIT
Program Flash ECC delay	t_{PFECC} CC	-	-	10	ns	see RECC and CECC of DMU register HF_PWAIT
Number of erase operations on DF0 over lifetime (complement sensing mode) ⁶⁾	N_{ERD0C} CC	-	-	4000000	cycles	
Number of erase operations on DF0 over lifetime (single ended sensing mode) ⁷⁾	N_{ERD0S} CC	-	-	750000	cycles	

Electrical Specification Flash Target Parameters

Table 3-52 Flash (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Number of erase operations on DF1 over lifetime (complement sensing mode) ⁶⁾	N_{ERD1C} CC	-	-	2000000	cycles	
Number of erase operations on DF1 over lifetime (single ended sensing mode) ⁷⁾	N_{ERD1S} CC	-	-	500000	cycles	
Data Flash Endurance per EEPROMx sector (complement sensing mode) ⁸⁾	$N_{\text{E_EEP10C}}$ CC	-	-	500000	cycles	Max. data retention time 10 years
DataFlash Endurance per EEPROMx sector (single ended sensing mode) ⁸⁾	$N_{\text{E_EEP10S}}$ CC	-	-	125000	cycles	Retention time and Tj according below example temperature profile
		-	-	125000	cycles	max data retention time 20y, Tj=110°C
		-	-	125000	cycles	max data retention time 8.2y, Tj=125°C
Data Flash Endurance per HSMx sector (complement sensing mode) ⁸⁾	$N_{\text{E_HSMC}}$ CC	-	-	250000	cycles	Max. data retention time 10 years
Data Flash Endurance per HSMx sector (single ended sensing mode) ⁸⁾	$N_{\text{E_HSMS}}$ CC	-	-	125000	cycles	Retention time and Tj according below example temperature profile
		-	-	125000	cycles	max data retention time 20y, Tj=110°C
		-	-	125000	cycles	max data retention time 8.2y, Tj=125°C
Junction temperature limit for PFlash program/erase operations	T_{JPFlash} SR	-	-	150	°C	
Data Flash Erase Time per Sector ¹⁾³⁾⁵⁾	t_{ERD1} CC	-	-	0.5	s	Max. 1000 erase/program cycles
Data Flash Erase Time per Sector ¹⁾³⁾⁵⁾	t_{ERDM} CC	-	-	1.5	s	Max allowed cycles, see NE_EEP10 and NE_HSM parameters
DataFlash Adder on Erase Time per 32kByte erase size when using complement sensing mode ¹⁾	$t_{\text{ER_ADDC32C}}$ CC	-	-	50	ms	Adder per 32 kByte on erase time; applicable only when using complement mode

Electrical Specification Flash Target Parameters
Table 3-52 Flash (cont'd)

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Data Flash Erase Time per Multi-Sector Command ¹⁾³⁾⁵⁾	t_{MERD1} CC	-	-	0.5	s	Max 1000 erase/program cycles; For consecutive logical sectors ≤ 256 KBytes
Data Flash Erase Time per Multi-Sector Command ¹⁾³⁾⁵⁾	t_{MERDM} CC	-	-	1.5	s	Max allowed cycles, see NE_EEP10x and NE_HSMx Parameters; For consecutive logical sectors ≤ 256 kByte
Program Flash Access Delay at reduced VDDP3 voltage supply during cranking	$t_{PF_low_VDDP3}$ CC	-	-	60	ns	see register DMU_HF_PWAIT.CFLASH
Data Flash Erase Verify time per page (Complement Sensing) ²⁾	$t_{VER_PAGE_DC}$ CC	-	-	10	μ s	Time per 8 Byte page for Verify Erased Page command
Data Flash Erase Verify time per page (Single Ended Sensing) ¹⁾	$t_{VER_PAGE_DS}$ CC	-	-	10	μ s	Time per 8 Byte page for Verify Erased Page command
Program Flash Erase Verify time per page ¹⁾	$t_{VER_PAGE_P}$ CC	-	-	10	μ s	Time per 32 Byte page for Verify Erased Page command
Data Flash Erase Verify time per sector (Complement Sensing) ¹⁾	$t_{VER_SEC_DC}$ CC	-	-	200	μ s	Time per 2 KB sector for Verify Erased Logical Sector Range command
Data Flash Erase Verify time per sector (Single Ended Sensing) ¹⁾	$t_{VER_SEC_DS}$ CC	-	-	360	μ s	Time per 4 KB sector for Verify Erased Logical Sector Range command
Program Flash Erase Verify time per sector ¹⁾	$t_{VER_SEC_P}$ CC	-	-	360	μ s	Time per 16KB sector for Verify Erased Logical Sector Range command
Data Flash Erase Verify time per wordline (Complement Sensing) ¹⁾	$t_{VER_WL_DC}$ CC	-	-	30	μ s	
Data Flash Erase Verify time per wordline (Single Ended Sensing) ¹⁾	$t_{VER_WL_DS}$ CC	-	-	50	μ s	
Program Flash Erase Verify time per wordline ¹⁾	$t_{VER_WL_P}$ CC	-	-	30	μ s	

1) Only valid for $f_{FSI} = 100$ MHz.

2) Time is not dependent on program mode (5V or 3.3V).

Electrical Specification Flash Target Parameters

- 3) Under out-of-spec conditions (e.g. over-cycling) or in case of activation of WL oriented defects, the duration of erase processes may be increased by up to 50%.
- 4) Using 512 kByte / 256 kByte erase commands (PFlash / DFlash).
- 5) If the DataFlash is operated in Complement Sensing Mode the erase time is increased by $\text{erase_size} / 32\text{kByte} \times t_{\text{ER_ADDC32C}}$
- 6) Allows segmentation of addressable memory into 8 logical sectors; round robin cycling must still be done to consider erase disturb limit N_{DFD} .
- 7) Allows segmentation of addressable memory into 6 logical sectors; round robin cycling must still be done to consider erase disturb limit N_{DFD} .
- 8) Only valid when a robust EEPROM emulation algorithm is used. For more details see the Users Manual.

3.27 Quality Declarations

Table 3-53 Quality Parameters

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Moisture Sensitivity Level	MSL CC	-	-	3		Conforming to Jedec J-STD--020C for 240C
ESD susceptibility according to Charged Device Model (CDM)	V_{CDM} SR	-	-	500 ¹⁾	V	for all other balls/pins; conforming to JESD22-C101-C
		-	-	750	V	for corner balls/pins; conforming to JESD22-C101-C
ESD susceptibility according to Human Body Model (HBM)	V_{HBM} SR	-	-	2000 ²⁾	V	Conforming to JESD22-A114-B
ESD susceptibility of the LVDS pins according to Human Body Model (HBM)	V_{HBM1} SR	-	-	2000	V	
Operation Lifetime	t_{OP} CC	-	-	24500	hour	see below temperature profile as an example

1) Pads of the AGBT interface are limited to a maximum value of 250V.

2) Pads of the AGBT interface are limited to a maximum value of 1000V.

Example Temperature Profile

The following temperature profile is an example. Application specific temperature profiles need to be aligned and approved by Infineon Technologies for the fulfillment of quality and reliability targets.

Table 3-54 Example Temperature Profile

$T_J =$	Duration [h]	Comment
$\leq 170^{\circ}\text{C}$	≤ 30	
$\leq 160^{\circ}\text{C}$	≤ 120	
$\leq 150^{\circ}\text{C}$	≤ 220	
$\leq 140^{\circ}\text{C}$	≤ 350	
$\leq 130^{\circ}\text{C}$	≤ 780	
$\leq 120^{\circ}\text{C}$	≤ 1600	
$\leq 110^{\circ}\text{C}$	≤ 3000	
$\leq 100^{\circ}\text{C}$	≤ 7000	
$\leq 90^{\circ}\text{C}$	≤ 8000	
$\leq 80^{\circ}\text{C}$	≤ 2400	
$\leq 70^{\circ}\text{C}$	≤ 1000	
	≤ 24500	total time

Electrical Specification Quality Declarations

Table 3-55 Example Inactive Lifetime Temperature Profile

T_J =	Duration [h]	Comment
$\leq 55^{\circ}\text{C}$	≤ 150700	

3.28 Package Outline

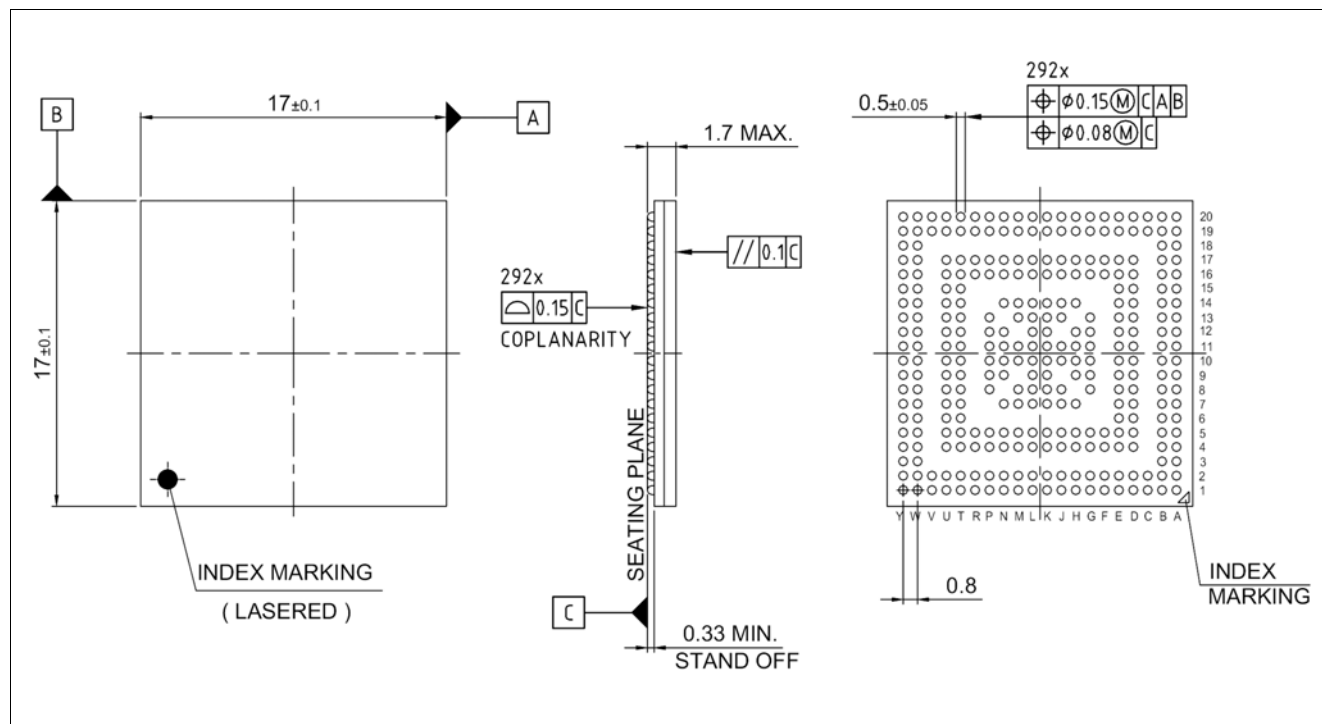


Figure 3-23 Package Outlines LFBGA-292

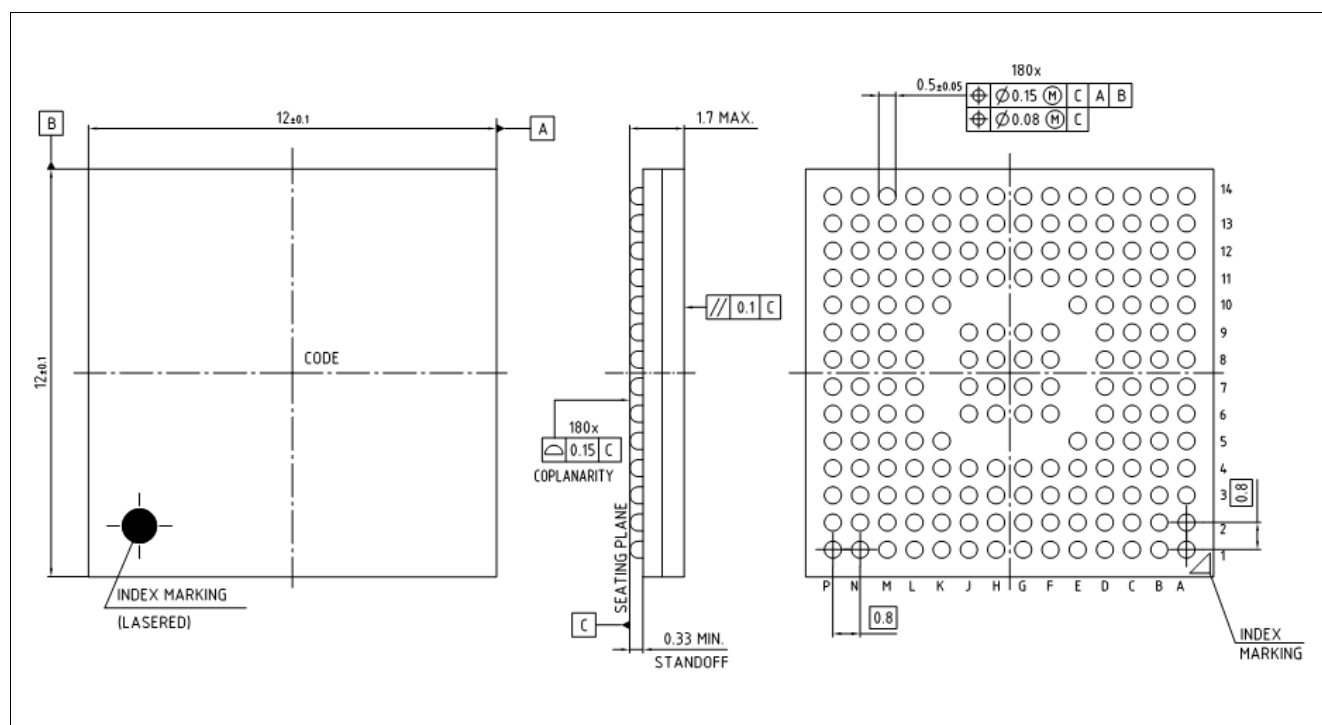


Figure 3-24 Package Outlines LFBGA-180

You can find all of our packages, sorts of packing and others in our Infineon Internet Page "Products": <http://www.infineon.com/products>.

3.28.1 Package Parameters

Table 3-56 Package Parameters

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Thermal resistance (junction to ambient) ¹⁾	RTH_JA CC	-	-	24	K/W	LFBGA180; Bottom Cooling
		-	-	16	K/W	LFBGA180; Top Cooling
		-	-	23	K/W	LFBGA292; Bottom Cooling
		-	-	14	K/W	LFBGA292; Top Cooling
		-	-	20	K/W	TQFP100
		-	-	18	K/W	TQFP144
Thermal resistance (junction to case bottom) ¹⁾	RTH_JCB CC	-	-	4.5	K/W	LFBGA180
		-	-	4.5	K/W	LFBGA292
		-	-	2	K/W	TQFP100
		-	-	2	K/W	TQFP144
Thermal resistance (junction to case top) ¹⁾	RTH_JCT CC	-	-	6	K/W	LFBGA180
		-	-	5	K/W	LFBGA292
		-	-	10	K/W	TQFP100
		-	-	10	K/W	TQFP144

1) The top and bottom thermal resistances between the case and the ambient (RTH_CTA, RTH_CBA) are to be combined with the thermal resistances between the junction and the case given above (RTH_JCT, RTH_JCB), in order to calculate the total thermal resistance between the junction and the ambient (RTH_JA). The thermal resistances between the case and the ambient (RTH_CTA, RTH_CBA) depend on the external system (PCB, case) characteristics and are under user responsibility. The junction temperature can be calculated using the following equation: $T_J = T_A + RTH_JA * P_D$, where the RTH_JA is the total thermal resistance between the junction and the ambient.

Thermal resistances as measured by the 'cold plate method' (MIL SPEC-883 Method 1012.1).

4 History

Version 0.4 is the first version of this document.

4.1 Changes from Version 0.4 to Version 0.6

Changes in chapter “Summary of Features”

- Changes in table “Platform Feature Overview” for feature values of SRAM, Data Flash, DMA, EVADC, and GTM

Changes in chapter “TC33x Pin Definition and Functions”

- Changes in sub-chapter “LFBGA-292 Package Variant Pin Configuration of TC33x for feature package LP”
 - Changes in “Port 22 Functions” table; added QSPI functions at P22.0
 - Changes in “Port 22 Functions” table; added QSPI and IOM functions at P22.1
 - Changes in “Port 22 Functions” table; added QSPI functions at P22.2
 - Changes in “Port 22 Functions” table; added QSPI functions at P22.3
 - Changes in table “Analog Inputs”, added functions at ball T9, Y9, T8, U8, W8, U7, Y8, W7
 - Changes in table “System I/O”, deleted functions at ball W17, Y17
 - Changes in table “Supply”, changed functions at ball N14, P13
- Changes in sub-chapter “LFBGA-292 Package Variant Pin Configuration of TC33x for feature package DA, DH, DT, DZ”
 - Changes in “Port 14 Functions” table; added CAN functions at P14.7
 - Changes in “Port 14 Functions” table; added CAN functions at P14.9
 - Changes in “Port 20 Functions” table; added CAN functions at P20.7
 - Changes in table “System I/O”, changed buffer type at ball L7, K7, P10, P11
- Changes in sub-chapter “LFBGA-180 Package Variant Pin Configuration of TC33x for feature package L and LP”
 - Changes in “Port 22 Functions” table; added QSPI functions at P22.0
 - Changes in “Port 22 Functions” table; added QSPI and IOM functions at P22.1
 - Changes in “Port 22 Functions” table; added QSPI functions at P22.2
 - Changes in “Port 22 Functions” table; added QSPI functions at P22.3
 - Changes in table “Analog Inputs”, added EVADC functions at ball N6, L5, M5, P5, N5, N3, M4, N4
 - Changes in table “System I/O”, deleted functions at ball N12, P12
 - Changes in table “Supply”, changed function at ball J8
- Changes in sub-chapter “LFBGA-180 Package Variant Pin Configuration of TC33x for feature package DA”
 - Changes in “Port 20 Functions” table; added CAN function at P20.7
 - Changes in table “Analog Inputs”, changed function at pin K3
 - Changes in table “Analog Inputs”, added function at pin J4
 - Changes in table “Supply”, deleted reserved functionality at pin J4
 - Changes in table “Supply”, added reserved functionality at pin L3
- Changes in sub-chapter “TQFP-144 Package Variant Pin Configuration of TC33x for feature package L and LP”
 - Changes in “Port 22 Functions” table; added QSPI functions at P22.0
 - Changes in “Port 22 Functions” table; added QSPI and IOM functions at P22.1

History Changes from Version 0.4 to Version 0.6

- Changes in “Port 22 Functions” table; added QSPI functions at P22.2
- Changes in “Port 22 Functions” table; added QSPI functions at P22.3
- Changes in “Port 34 Functions” table; changed pin number for P34.1
- Changes in “Port 34 Functions” table; changed pin number for P34.2
- Changes in “Port 34 Functions” table; deleted all functionality of P34.3
- Changes in table “Analog Inputs”, added functions at pins 47, 46, 45, 40, 39, 38, 37, 36
- Changes in table “System I/O”, deleted functionality at pin 70 and 71
- Changes in table “Supply”, added function at pin 53
- Changes in sub-chapter “TQFP-100 Package Variant Pin Configuration of TC33x for feature package L and LP”
 - Changes in table “Analog Inputs”, changed functions at pins 38, 37, 36, 35, 34, 33
 - Changes in table “Analog Inputs”, added function at pin 34, 33, 32, 29, 28, 27, 26, 25
 - Changes in table “System I/O”, deleted functionality at pin 48 and 49
 - Changes in table “Supply”, added function at pin 39
 - Changes in table “Supply”, changed function at pin 40, 31
 - Changes in table “Supply”, deleted function at pin 32
- Changes in sub-chapter “TQFP-80 Package Variant Pin Configuration of TC33x for feature package L and LP”
 - Changes in table “Analog Inputs”, deleted functions at pin 28
 - Changes in table “Analog Inputs”, added function at pins 27, 26, 23, 22, 21, 20
 - Changes in table “System I/O”, deleted functionality at pin 38 and 39
 - Changes in table “Supply”, changed function at pin 25, 29
 - Changes in table “Supply”, added function at pin 28
 - Changes in table “Supply”, deleted function at pin 26
- Changes in sub-chapter “Sequence of Pads in Pad Frame for feature package L and LP”
 - General changes in table “Pad List” concerning “X” and “Y” coordinates, numbers, pad names, pad types, and comments
- Changes in sub-chapter “Sequence of Pads in Pad Frame for feature package DA, DH, DT, and DZ”
 - General changes in table “Pad List” concerning “X” and “Y” coordinates, numbers, pad names, pad types, and comments
 - Added explanation following table “Pad List”
- Changes in sub-chapter “Legend”
 - Changed IO-Spirit versions

Changes in chapter “Electrical Specification”

- Changes in sub-chapter “Pin Reliability in Overload”
 - Changed table numbers in chapter description referring to Temperature Profile
 - Changed wording in test condition for parameter K_{OVAP}
- Changes in sub-chapter “5V/3.3V switchable Pads”
 - Changes in table Slow 5V GPIO for parameter I_{OZ} condition
 - Changes in table Slow 3.3V GPIO for parameter I_{OZ} condition
- Changes in table “LVDS - IEEE standard LVDS general purpose link (GPL)”
 - Changed value of parameter V_I
 - Changed test condition of parameter V_{idth}

History Changes from Version 0.6 to Version 0.7

- Added values and test condition of parameter V_{idth}
- Changed test condition of parameter R_{in}
- Added footnotes
- Changes in table “Current Consumption”
 - Changes in footnote 6)
- Changes in table “Reset”
 - Changed value of parameter t_{PIP}
 - Changed value of parameter t_{BWP}
- Changes in table “Supply Ramp”
 - Added conditions for values of table “Supply Ramp”
- Changes in table “EVR33 LDO”
 - Added values for parameter ΔV_{OUTTC}
- Changes in table “EVR13 SMPS”
 - Changed test condition of parameter V_{DDDC}
 - Changed value of parameter t_{STRDC}
- Changes in table “EVR13 SMPS External components”
 - Changed value of parameter C_{OUT}
- Changes in table “PLL System”
 - Changed value of parameter f_{REF}
- Changes in sub-chapter “ETH RGMII Parameters”
 - Added figure for “ETH RGMII TX Signal Timing (Delay on Destination (DoD))”
 - Added figure for “ETH RGMII RX Signal Timing (Delay on Source (DoS))”
- Changes in sub-chapter “SDMMC Interface Timing”
 - Added figure “SDMMC Timing”
- Changes in table “Flash”
 - Changed value of parameter t_{PRPB5_PF}
 - Changed value of parameter t_{ER_Dev}
- Changes in sub-chapter “Package Outline”
 - Changed figure numbering
- Changes in sub-chapter “Package Parameters”
 - Added values and notes of parameter RTH_JA in table “Package Parameters”

4.2 Changes from Version 0.6 to Version 0.7

General changes in Data Sheet TC33x: Data Sheet splitted and renamed to TC33xEXT for feature package DA/DZ and TC33xPD for feature package L/LP

- Changed Data Sheet version 0.6 to 0.7
- **Changes in chapter “Summary of Features”**
 - Changes in chapter “Summary of Features”
 - Changed wording of "DLMU" in table "Platform Feature Overview"
 - Changed value of "Data Flash" in table "Platform Feature Overview"

History Changes from Version 0.6 to Version 0.7

- Changed values of "EVADC" in table "Platform Feature Overview"
- Changed wording of "EVADC" in table "Platform Feature Overview"
- Changed values of "GTM" in table "Platform Feature Overview"
- Deleted wording of "GTM" in chapter "Summary of Features"
- Changed values of "FlexRay" in table "Platform Feature Overview"
- Deleted wording of "FlexRay" in chapter "Summary of Features"
- Changed values of "CAN" in table "Platform Feature Overview"
- Changed value of "CAN" in chapter "Summary of Features"
- Changed value of "ASCLIN" in chapter and table "Platform Feature Overview"
- Deleted wording of "IOM" in chapter "Summary of Features"
- Deleted value of "IOM" in table "Platform Feature Overview"
- Deleted packages TQFP144, TQFP-100 and TQFP-80 in table "Platform Feature Overview"
- Change in table "Platform Feature Overview" for T_{ambient} temperature range
- **Changes in chapter "TC33xEXT Pin Definition and Functions"**
 - Deleted Package Pinning for LFBGA-292, TC33x DH, DT, and LP
 - Deleted Package Pinning for LFBGA-180, TC33x L and LP
 - Deleted Package Pinning for TQFP-144, TC33x L and LP
 - Deleted Package Pinning for TQFP-100, TC33x L and LP
 - Deleted Package Pinning for TQFP-80, TC33x L and LP
 - Deleted Pad List in chapter "Sequence of Pads in Pad Frame for feature package L and LP"
- **Changes in sub-chapter "Legend"**
 - Spirit version for feature package TC33xpd has been deleted
 - Deleted redundant sentence regarding neighbor pads
 - Changed table numbering scheme due to deleted chapters in whole data sheet

Changes in chapter "Electrical Specification"

- Changes in sub-chapter "Parameter Interpretation"
 - Changed wording in sub-chapter "Parameter Interpretation"
- Changes in sub-chapter "Absolute Maximum Ratings"
 - Added footnote for parameter I_{IN}
 - Changed wording for parameter ΣI_{IN} Added footnote 5)
- Changes in sub-chapter "Pin Reliability in Overload"
 - Changed table numbers
 - Deleted value for parameter K_{OVDN} in table "Overload Parameters"
 - Deleted value for parameter K_{OVDP} in table "Overload Parameters"
- Changes in sub-chapter "Operating Conditions"
 - Changed wording in sub-chapter "Operating Conditions"
 - Changed value for parameter f_{GETH} in table "Operating Conditions"
 - Deleted value for parameter f_{ERAY} in table "Operating Conditions"
- Changes in sub-chapter "5V/3.3V switchable Pads"
 - Deleted values for parameter I_{OZ} in table "Fast 5V GPIO"
 - Deleted values for parameter I_{OZ} in table "Fast 3.3V GPIO"

- Changes in table “High performance LVDS Pads”
 - Changed wording in sub-chapter “High performance LVDS Pads”
 - Deleted parameter R_0 in table "LVDS - IEEE standard LVDS general purpose link (GPL)"
 - Deleted parameter t_{fall20} in table "LVDS - IEEE standard LVDS general purpose link (GPL)"
 - Deleted parameter V_{OD} in table "LVDS - IEEE standard LVDS general purpose link (GPL)"
 - Deleted parameter V_{OH} in table "LVDS - IEEE standard LVDS general purpose link (GPL)"
 - Deleted parameter V_{OL} in table "LVDS - IEEE standard LVDS general purpose link (GPL)"
 - Deleted parameter V_{OS} in table "LVDS - IEEE standard LVDS general purpose link (GPL)"
 - Deleted parameter V_{ODSM} in table "LVDS - IEEE standard LVDS general purpose link (GPL)"
 - Deleted parameter dR_0 in table "LVDS - IEEE standard LVDS general purpose link (GPL)"
 - Deleted parameter dV_{OS} in table "LVDS - IEEE standard LVDS general purpose link (GPL)"
 - Deleted parameter dV_{od} in table "LVDS - IEEE standard LVDS general purpose link (GPL)"
 - Deleted parameter t_{SET_LVDS} in table "LVDS - IEEE standard LVDS general purpose link (GPL)"
 - Deleted footnotes 1), 2), 3), 4)
- Changes in sub-chapter “VADC Parameters”
 - Changed value for parameter V_{AREF} in table "VADC 5V"
 - Deleted sentence “Fast compare....” in sub-chapter "VADC Parameters"
 - Added note for parameter V_{AREF} in table "VADC 5V"
 - Changed note for parameter T_{UE} in table "VADC 5V"
 - Changed notes for parameter Q_{AINS} in table "VADC 5V"
 - Deleted values for parameter Q_{AINS} in table "VADC 5V"
 - Deleted values for parameter t_s in table "VADC 5V"
 - Changed wording in notes for parameter t_s in table "VADC 5V"
 - Added note for parameter R_{PDD} in table "VADC 5V"
 - Changed wording of footnote 7)
 - Changed figure at table "VADC 5V"
- Changes in sub-chapter “MHz Oscillator”
 - Changed wording in footnote 1)
 - Changed wording in footnote 2)
- Changes in sub-chapter “Power Supply Current”
 - Deleted values for parameter I_{DDRAIL} in table "Current Consumption"
 - Deleted values for parameter $I_{DDPORST}$ in table "Current Consumption"
 - Changed value for parameter I_{EVRB} in table "Current Consumption"
 - Deleted values for parameter I_{DDTOT} in table "Current Consumption"
 - Deleted values for parameter $I_{DDTOTDC5}$ in table "Current Consumption"
 - Changed value for parameter $I_{STANDBY}$ in table "Current Consumption"
 - Deleted values for parameter $I_{STANDBY}$ in table "Current Consumption"
 - Deleted values for parameter PD in table "Current Consumption"
 - Added equations and details for "Calculating the 1.3 V Current Consumption"
 - Deleted value for parameter $I_{EXTLVDS}$ in table "Module Current Consumption"
 - Changed value for parameter I_{SCRSB} in table "Module Current Consumption"
 - Deleted parameter I_{DDGTM} in table "Module Core Current Consumption"

History Changes from Version 0.7 to Version 1.0

- Changes in sub-chapter “Reset Timing”
 - Changed values for parameter t_{PI} in table "Reset"
- Changes in sub-chapter “EVR”
 - Changed values and notes for parameter V_{EXTMON} in table "Supply Monitors"
 - Changed notes for parameter $V_{DDP3MON}$ in table "Supply Monitors"
 - Changed notes for parameter V_{DDMON} in table "Supply Monitors"
 - Changed note for parameter t_{MON} in table "Supply Monitors"
 - Deleted table "EVR13 SMPS" - causing changed numbering of all following tables
- Changes in sub-chapter “ASCLIN SPI Master Timing”
 - Changed wording in sub-chapter “ASCLIN SPI Master Timing”
- Changes in sub-chapter “QSPI Timings, Master and Slave Mode”
 - Changed wording in sub-chapter “QSPI Timings, Master and Slave Mode”
 - Deleted table "Master Mode Timing, LVDS output pads for data and clock"
- Changes in sub-chapter “ETH RGMII Parameters”
 - Changed note for parameter t_{21} in table "ETH RGMII Signal Timing Parameters valid for 3.3V"
- Deleted sub-chapter “Parameters Specific to the Emulation Part”
- Changes in sub-chapter “Quality Declarations”
 - Added footnote 1) to parameter V_{CDM} in table "Quality Parameters"
 - Added footnote 2) to parameter V_{HBM} in table "Quality Parameters"
 - Added footnotes 1) and 2)
- Changes in sub-chapter “Package Outline”
 - Deleted figure for Package Outlines TQFP-144
 - Deleted figure for Package Outlines TQFP-100
 - Deleted figure for Package Outlines TQFP-80
 - Deleted values for parameter "RTH_JA" in table "Package Parameters"
 - Deleted values for parameter "RTH_JCB" in table "Package Parameters"
 - Deleted values for parameter "RTH_JCT" in table "Package Parameters"

4.3 Changes from Version 0.7 to Version 1.0

- **Changes in chapter “Summary of Features”**
 - Deleted “IOM” entry
 - Changed wording in table "Platform Feature Overview" for SRAM feature
 - Changed wording in table "Platform Feature Overview" for Timer feature
 - Changed wording in table "Platform Feature Overview" for Low Power Features
- **Changes in chapter “Electrical Specification”**
 - Changes in sub-chapter “Absolute Maximum Ratings”
 - Changed wording and value in footnote 5) for table “Absolute Maximum Ratings”
- Changes in sub-chapter "Power Supply Current"
 - Deleted value for parameter " f_{GTM} "
- Changes in sub-chapter "Reset Timing"

- Changed unit for parameter " t_{PI} " in table "Reset"

4.4 Changes from Version 1.0 to Version 1.1

- **Changes in chapter "Summary of Features"**

- Changed wording for "DFLASH"
- Added description for "AEC-Q100"
- Added description for "ISO 26262 Safety Element"
- Added description for Data Flash in table "Platform Feature Overview"
- Added footnote 1) for parameter Debug/ AGBT in table "Platform Feature Overview"
- Added footnote 1) to table "Platform Feature Overview"

- **Changes in chapter "TC33xEXT Pin Definition and Functions"**

- Added notes to table "System I/O" for "LFBGA-292 Package Variant"
- Added notes to table "System I/O" for "LFBGA-180 Package Variant"
- Changed comments for pad names VDDP3 (34), VDDP3 (35) in Pad List of sub-chapter "Sequence of Pads in Pad Frame"

- **Changes in chapter "Electrical Specification"**

- Changed value in footnote 5) for sub-chapter "Absolute Maximum Ratings"
- Changed value for parameter GETH frequency in table "Operating Conditions"
- Changed note for parameter t_{TX_ASYM} in table "Fast 5V GPIO" of sub-chapter "5V/3.3V switchable Pads"
- Changed note for parameter t_{TX_ASYM} in table "Fast 3.3V GPIO" of sub-chapter "5V/3.3V switchable Pads"
- Changed note for parameter t_{TX_ASYM} in table "Slow 5V GPIO" of sub-chapter "5V/3.3V switchable Pads"
- Changed note for parameter t_{TX_ASYM} in table "Slow 3.3V GPIO" of sub-chapter "5V/3.3V switchable Pads"
- Changed note for parameter t_{TX_ASYM} in table "RFast 5V GPIO" of sub-chapter "5V/3.3V switchable Pads"
- Changed note for parameter t_{TX_ASYM} in table "RFast 3.3V pad" of sub-chapter "5V/3.3V switchable Pads"
- Typo corrected in footnotes for table "LVDS – IEEE standard LVDS general purpose link (GPL)" in sub-chapter "High performance LVDS Pads"
- Added footnote sentence for table "LVDS – IEEE standard LVDS general purpose link (GPL)" in sub-chapter "High performance LVDS Pads"
- Added footnote to parameter Q_{CONV} in table "VADC 5V" in sub-chapter "VADC Parameters"
- Changed footnote numbering for parameters Q_{AINS} , R_{CSD} in table "VADC 5V" in sub-chapter "VADC Parameters"
- Changed footnote 3) and 6) of table "VADC 5V" in sub-chapter "VADC Parameters"
- Added footnote 9) to table "VADC 5V" in sub-chapter "VADC Parameters"
- Thus changing following footnote numbers of table "VADC 5V" in sub-chapter "VADC Parameters"
- Changed and deleted parts of text for sub-chapter "Power Supply Current"
- Added footnote to parameter PD in table "Current Consumption" for sub-chapter "Power Supply Current"
- Changed footnote 2) for table "Current Consumption" in sub-chapter "Power Supply Current"
- Added footnote 7) for table "Current Consumption" in sub-chapter "Power Supply Current"
- Added sentence to sub-chapter "Supply Ramp-up and Ramp-down Behavior"

History Changes from Version 1.0 to Version 1.1

- Changed value (from Max. to Typ.) for parameter R_{ON} in table “EVRC SMPS External components” for sub-chapter “EVR”
- Changed values (from Min. to Typ.) for parameter t_7 in table “ETH MII Signal Timing Parameters” for sub-chapter “ETH MII Parameters”
- Changed symbols for parameters t_{13} , t_{14} , t_{15} in table “ETH RMII Signal Timing Parameters valid for 3.3V” in sub-chapter “ETH RMII Parameters”
- Changed value (from Min. to Typ.) for parameter t_{13} in table “ETH RMII Signal Timing Parameters valid for 3.3V” in sub-chapter “ETH RMII Parameters”
- Added footnote 3) to parameters t_{16} , t_{17} in table “ETH RMII Signal Timing Parameters valid for 3.3V” in sub-chapter “ETH RMII Parameters”
- Added footnote 3) for table “ETH RMII Signal Timing Parameters valid for 3.3V” in sub-chapter “ETH RMII Parameters”
- Deleted parameter t_{81} in table “Skew Calibration Related” in sub-chapter “Radar Interface Timing”
- Changed values for parameters t_{82} and t_{83} in table “Skew Calibration Related” in sub-chapter “Radar Interface Timing”
- Deleted notes for parameters t_{82} and t_{83} in table “Skew Calibration Related” in sub-chapter “Radar Interface Timing”
- Added values and notes for parameters RTH_JA, RTH_JCB, RTH_JCT to table “Package Parameters” in sub-chapter “Package Parameters”

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